

INTRODUCTION AND BASIC CONCEPTS

The science of thermodynamics deals with the *amount* of heat transfer as a system undergoes a process from one equilibrium state to another, and it makes no reference to *how long* the process will take. But in engineering, we are often interested in the *rate* of heat transfer, which is the topic of the science of *heat transfer*.

We start this chapter with a review of the fundamental concepts of thermodynamics that form the framework for heat transfer. We first present the relation of heat to other forms of energy and review the energy balance. We then present the three basic mechanisms of heat transfer, which are conduction, convection, and radiation, and we discuss thermal conductivity. *Conduction* is the transfer of energy from the more energetic particles of a substance to the adjacent, less energetic ones as a result of interactions between the particles. *Convection* is the mode of heat transfer between a solid surface and the adjacent liquid or gas that is in motion, and it involves the combined effects of conduction and fluid motion. *Radiation* is the energy emitted by matter in the form of electromagnetic waves (or photons) as a result of the changes in the electronic configurations of the atoms or molecules. We close this chapter with a discussion of simultaneous heat transfer.



OBJECTIVES

When you finish studying this chapter, you should be able to:

- Understand how thermodynamics and heat transfer are related to each other,
- Distinguish thermal energy from other forms of energy and heat transfer from other forms of energy transfer,
- Perform general energy balances as well as surface energy balances,
- Understand the basic mechanisms of heat transfer, which are conduction, convection, and radiation, and Fourier's law of heat conduction, Newton's law of cooling, and the Stefan–Boltzmann law of radiation,
- Identify the mechanisms of heat transfer that occur simultaneously in practice,
- Develop an awareness of the cost associated with heat losses, and
- Solve various heat transfer problems encountered in practice.

1-1 ■ THERMODYNAMICS AND HEAT TRANSFER

We all know from experience that a cold canned drink left in a room warms up and a warm canned drink left in a refrigerator cools down. This is accomplished by the transfer of *energy* from the warm medium to the cold one. The energy transfer is always from the higher temperature medium to the lower temperature one, and the energy transfer stops when the two media reach the same temperature.

You will recall from thermodynamics that energy exists in various forms. In this text we are primarily interested in **heat**, which is *the form of energy that can be transferred from one system to another as a result of temperature difference*. The science that deals with the determination of the *rates* of such energy transfers is **heat transfer**.

You may be wondering why we need to undertake a detailed study on heat transfer. After all, we can determine the amount of heat transfer for any system undergoing any process using a thermodynamic analysis alone. The reason is that thermodynamics is concerned with the *amount* of heat transfer as a system undergoes a process from one equilibrium state to another, and it gives no indication about *how long* the process will take. A thermodynamic analysis simply tells us how much heat must be transferred to realize a specified change of state to satisfy the conservation of energy principle.

In practice we are more concerned about the rate of heat transfer (heat transfer per unit time) than we are with the amount of it. For example, we can determine the amount of heat transferred from a thermos bottle as the hot coffee inside cools from 90°C to 80°C by a thermodynamic analysis alone. But a typical user or designer of a thermos bottle is primarily interested in *how long* it will be before the hot coffee inside cools to 80°C , and a thermodynamic analysis cannot answer this question. Determining the rates of heat transfer to or from a system and thus the times of heating or cooling, as well as the variation of the temperature, is the subject of heat transfer (Fig. 1-1).

Thermodynamics deals with equilibrium states and changes from one equilibrium state to another. Heat transfer, on the other hand, deals with systems that lack thermal equilibrium, and thus it is a *nonequilibrium* phenomenon. Therefore, the study of heat transfer cannot be based on the principles of thermodynamics alone. However, the laws of thermodynamics lay the framework for the science of heat transfer. The *first law* requires that the rate of energy transfer into a system be equal to the rate of increase of the energy of that system. The *second law* requires that heat be transferred in the direction of decreasing temperature (Fig. 1-2). This is like saying that a car parked on an inclined road must go downhill in the direction of decreasing elevation when its brakes are released. It is also analogous to the electric current flowing in the direction of decreasing voltage or the fluid flowing in the direction of decreasing total pressure.

The basic requirement for heat transfer is the presence of a *temperature difference*. There can be no net heat transfer between two bodies that are at the same temperature. The temperature difference is the *driving force* for heat transfer, just as the *voltage difference* is the driving force for electric current flow and *pressure difference* is the driving force for fluid flow. The rate of heat transfer in a certain direction depends on the magnitude of the *temperature gradient* (the temperature difference per unit length or the rate of change of temperature) in that direction. The larger the temperature gradient, the higher the rate of heat transfer.

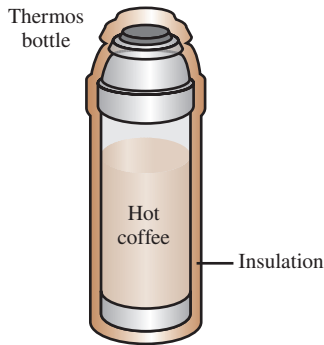


FIGURE 1-1

We are normally interested in how long it takes for the hot coffee in a thermos bottle to cool to a certain temperature, which cannot be determined from a thermodynamic analysis alone.

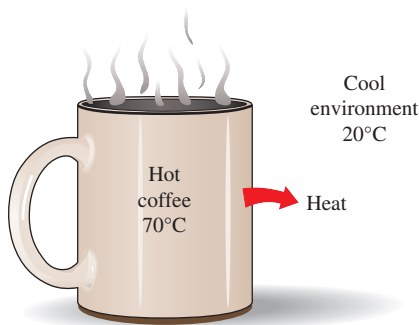


FIGURE 1-2

Heat flows in the direction of decreasing temperature.

Application Areas of Heat Transfer

Heat transfer is commonly encountered in engineering systems and other aspects of life, and one does not need to go very far to see some application areas of heat transfer. In fact, one does not need to go anywhere. The human body is constantly rejecting heat to its surroundings, and human comfort is closely tied to the rate of this heat rejection. We try to control this heat transfer rate by adjusting our clothing to the environmental conditions.

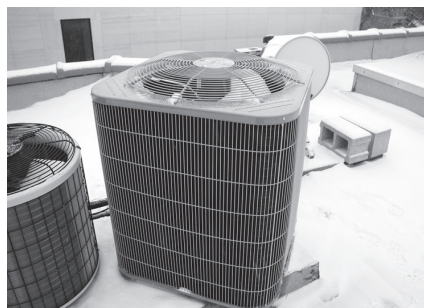
Many ordinary household appliances are designed, in whole or in part, by using the principles of heat transfer. Some examples include the electric or gas range, the heating and air-conditioning system, the refrigerator and freezer, the water heater, the iron, and even the computer, the TV, and the DVD player. Of course, energy-efficient homes are designed on the basis of minimizing heat loss in winter and heat gain in summer. Heat transfer plays a major role in the design of many other devices, such as car radiators, solar collectors, various components of power plants, and even spacecraft (Fig. 1–3). The optimal insulation thickness in the walls and roofs of houses, on hot water or steam pipes, or on water heaters is again determined on the basis of a heat transfer analysis with economic considerations.

Historical Background

Heat has always been perceived to be something that produces in us a sensation of warmth, and one would think that the nature of heat was one of the first things understood by humankind. But it was only in the middle of the



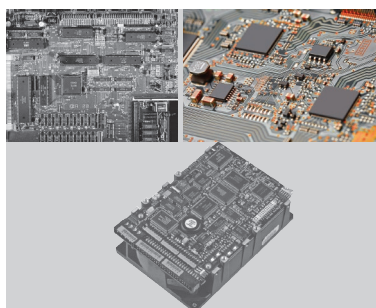
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Air-conditioning systems
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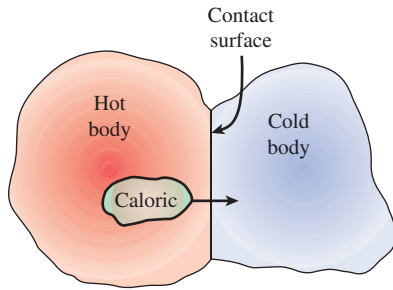
Power plants
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Refrigeration systems
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FIGURE 1–3

Some application areas of heat transfer.

**FIGURE 1–4**

In the early 19th century, heat was thought to be an invisible fluid called *caloric* that flowed from warmer bodies to cooler ones.

19th century that we had a true physical understanding of the nature of heat, thanks to the development at that time of the **kinetic theory**, which treats molecules as tiny balls that are in motion and thus possess kinetic energy. Heat is then defined as the energy associated with the random motion of atoms and molecules. Although it was suggested in the 18th and early 19th centuries that heat is the manifestation of motion at the molecular level (called the *live force*), the prevailing view of heat until the middle of the 19th century was based on the **caloric theory** proposed by the French chemist Antoine Lavoisier (1743–1794) in 1789. The caloric theory asserts that heat is a fluid-like substance called **caloric** that is a massless, colorless, odorless, and tasteless substance that can be poured from one body into another (Fig. 1–4). When caloric was added to a body, its temperature increased; and when caloric was removed from a body, its temperature decreased. When a body could not contain any more caloric, much the same way as when a glass of water could not dissolve any more salt or sugar, the body was said to be saturated with caloric. This interpretation gave rise to the terms *saturated liquid* and *saturated vapor* that are still in use today.

The caloric theory came under attack soon after its introduction. It maintained that heat was a substance that could not be created or destroyed. Yet it was known that heat could be generated indefinitely by rubbing one's hands together or rubbing two pieces of wood together. In 1798, the American Benjamin Thompson (Count Rumford) (1753–1814) showed in his papers that heat can be generated continuously through friction. The validity of the caloric theory was also challenged by several others. But it was the careful experiments of the Englishman James P. Joule (Fig. 1–5) published in 1843 that finally convinced the skeptics that heat was not a substance after all, thus putting the caloric theory to rest. Although the caloric theory was totally abandoned in the middle of the 19th century, it contributed greatly to the development of thermodynamics and heat transfer.

1–2 ■ ENGINEERING HEAT TRANSFER

Heat transfer equipment such as heat exchangers, boilers, condensers, radiators, heaters, furnaces, refrigerators, and solar collectors are designed primarily on the basis of heat transfer analysis. The heat transfer problems encountered in practice can be considered in two groups: (1) *rating* and (2) *sizing* problems. The rating problems deal with the determination of the heat transfer rate for an existing system at a specified temperature difference. The sizing problems deal with the determination of the size of a system in order to transfer heat at a specified rate for a specified temperature difference.

An engineering device or process can be studied either *experimentally* (testing and taking measurements) or *analytically* (by analysis or calculations). The experimental approach has the advantage that we deal with the actual physical system, and the desired quantity is determined by measurement, within the limits of experimental error. However, this approach is expensive, time-consuming, and often impractical. Besides, the system we are analyzing may not even exist. For example, the heating and plumbing systems of a building must usually be sized *before* the building is actually built on the basis of the specifications given. The analytical approach (including the numerical approach) has the advantage that it is fast and inexpensive, but the results

obtained are subject to the accuracy of the assumptions, approximations, and idealizations made in the analysis. In engineering studies, often a good compromise is reached by reducing the choices to just a few by analysis, and then verifying the findings experimentally.

Modeling in Engineering

The descriptions of most scientific problems involve equations that relate the changes in some key variables to each other. Usually the smaller the increment chosen in the changing variables, the more general and accurate the description. In the limiting case of infinitesimal or differential changes in variables, we obtain differential equations that provide precise mathematical formulations for the physical principles and laws by representing the rates of change as derivatives. Therefore, differential equations are used to investigate a wide variety of problems in sciences and engineering (Fig. 1–6). However, many problems encountered in practice can be solved without resorting to differential equations and the complications associated with them.

The study of physical phenomena involves two important steps. In the first step, all the variables that affect the phenomena are identified, reasonable assumptions and approximations are made, and the interdependence of these variables is studied. The relevant physical laws and principles are invoked, and the problem is formulated mathematically. The equation itself is very instructive because it shows the degree of dependence of some variables on others and the relative importance of various terms. In the second step, the problem is solved using an appropriate approach, and the results are interpreted.

Many processes that seem to occur in nature randomly and without any order are, in fact, being governed by some visible or not-so-visible physical laws. Whether we notice them or not, these laws are there, governing consistently and predictably what seem to be ordinary events. Most of these laws are well defined and well understood by scientists. This makes it possible to predict the course of an event before it actually occurs, or to study various aspects of an event mathematically without actually running expensive and time-consuming experiments. This is where the power of analysis lies. Very accurate results to meaningful practical problems can be obtained with relatively little effort by using a suitable and realistic mathematical model. The preparation of such models requires an adequate knowledge of the natural phenomena involved and the relevant laws, as well as sound judgment. An unrealistic model will obviously give inaccurate and thus unacceptable results.

An analyst working on an engineering problem often finds himself or herself in a position to make a choice between a very accurate but complex model and a simple but not-so-accurate model. The right choice depends on the situation at hand. The right choice is usually the simplest model that yields adequate results. For example, the process of baking potatoes or roasting a round chunk of beef in an oven can be studied analytically in a simple way by modeling the potato or the roast as a spherical solid ball that has the properties of water (Fig. 1–7). The model is quite simple, but the results obtained are sufficiently accurate for most practical purposes. As another example, when we analyze the heat losses from a building in order to select the right size for a heater, we determine the heat losses under anticipated worst conditions and select a

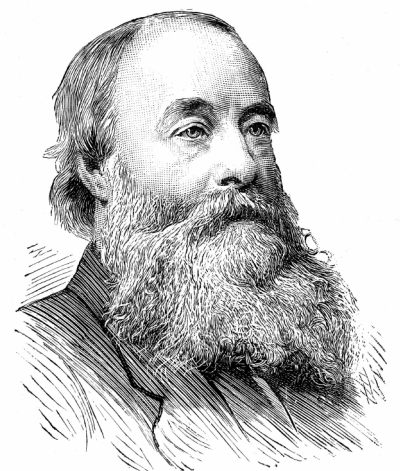
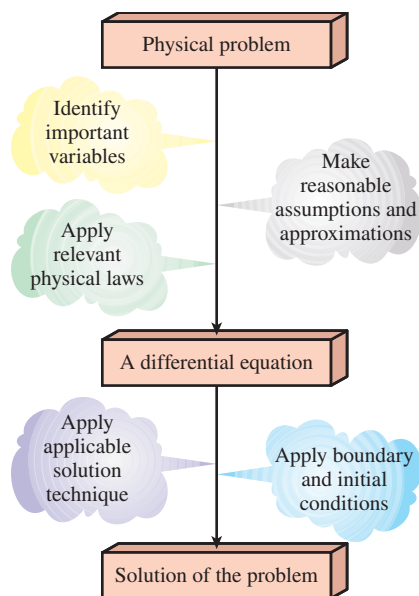


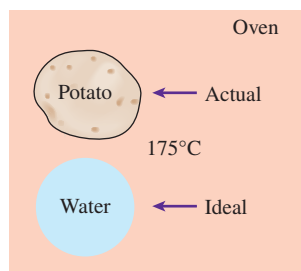
FIGURE 1–5

James Prescott Joule (1818–1889) was a British physicist born in Salford, Lancashire, England. Joule is best known for his work on the conversion of electrical and mechanical energy into heat and the first law of thermodynamics. The energy unit joule (J) is named after him. The Joule's law of electric heating that he formulated states that the rate of heat production in a conducting wire is proportional to the product of the resistance of the wire and the square of the electric current. Through his experiments, Joule demonstrated the mechanical equivalence of heat—the conversion of mechanical energy into an equivalent amount of thermal energy—which laid the foundation for the conservation of energy principle. Joule, together with William Thomson (later Lord Kelvin), discovered the temperature drop of a substance during free expansion, a phenomenon known as the Joule-Thomson effect, which provides the basis of operation for modern-day vapor-compression refrigeration and air-conditioning systems.

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**FIGURE 1-6**

Mathematical modeling of physical problems.

**FIGURE 1-7**

Modeling is a powerful engineering tool that provides great insight and simplicity at the expense of some accuracy.

furnace that will provide sufficient energy to make up for those losses. Often we tend to choose a larger furnace in anticipation of some future expansion or just to provide a factor of safety. A very simple analysis is adequate in this case.

When selecting heat transfer equipment, it is important to consider the actual operating conditions. For example, when purchasing a heat exchanger that will handle hard water, we must consider that some calcium deposits will form on the heat transfer surfaces over time, causing fouling and thus a gradual decline in performance. The heat exchanger must be selected on the basis of operation under these adverse conditions instead of under new conditions.

Preparing very accurate but complex models is usually not so difficult. But such models are not much use to an analyst if they are very difficult and time-consuming to solve. At the minimum, the model should reflect the essential features of the physical problem it represents. There are many significant real-world problems that can be analyzed with a simple model. But it should always be kept in mind that the results obtained from an analysis are as accurate as the assumptions made in simplifying the problem. Therefore, the solution obtained should not be applied to situations for which the original assumptions do not hold.

A solution that is not quite consistent with the observed nature of the problem indicates that the mathematical model used is too crude. In that case, a more realistic model should be prepared by eliminating one or more of the questionable assumptions. This will result in a more complex problem that, of course, is more difficult to solve. Thus any solution to a problem should be interpreted within the context of its formulation.

1-3 ■ HEAT AND OTHER FORMS OF ENERGY

Energy can exist in numerous forms such as thermal, mechanical, kinetic, potential, electrical, magnetic, chemical, and nuclear, and their sum constitutes the **total energy** E (or e on a unit mass basis) of a system. The forms of energy related to the molecular structure of a system and the degree of the molecular activity are referred to as the *microscopic energy*. The sum of all microscopic forms of energy is called the **internal energy** of a system and is denoted by U (or u on a unit mass basis).

The international unit of energy is *joule* (J) or *kilojoule* (1 kJ = 1000 J). In the English system, the unit of energy is the *British thermal unit* (Btu), which is defined as the energy needed to raise the temperature of 1 lbm of water at 60°F by 1°F. The magnitudes of kJ and Btu are almost identical (1 Btu = 1.055056 kJ). Another well-known unit of energy is the *calorie* (1 cal = 4.1868 J), which is defined as the energy needed to raise the temperature of 1 gram of water at 14.5°C by 1°C.

Internal energy may be viewed as the sum of the kinetic and potential energies of the molecules. The portion of the internal energy of a system associated with the kinetic energy of the molecules is called **sensible energy** or **sensible heat**. The average velocity and the degree of activity of the molecules are proportional to the temperature. Thus, at higher temperatures the molecules possess higher kinetic energy, and as a result, the system has a higher internal energy.

The internal energy is also associated with the intermolecular forces between the molecules of a system. These are the forces that bind the molecules to

each other, and, as one would expect, they are strongest in solids and weakest in gases. If sufficient energy is added to the molecules of a solid or liquid, they will overcome these molecular forces and simply break away, turning the system to a gas. This is a *phase change* process, and because of this added energy, a system in the gas phase is at a higher internal energy level than it is in the solid or the liquid phase. The internal energy associated with the phase of a system is called **latent energy** or **latent heat**.

The changes mentioned above can occur without a change in the chemical composition of a system. Most heat transfer problems fall into this category, and one does not need to pay any attention to the forces binding the atoms in a molecule together. The internal energy associated with the atomic bonds in a molecule is called **chemical** (or **bond**) **energy**, whereas the internal energy associated with the bonds within the nucleus of the atom itself is called **nuclear energy**. The chemical and nuclear energies are absorbed or released during chemical or nuclear reactions, respectively.

In the analysis of systems that involve fluid flow, we often encounter the combination of properties u and PV . For the sake of simplicity and convenience, this combination is defined as **enthalpy** h . That is, $h = u + PV$ where the term PV represents the *flow energy* of the fluid (also called the *flow work*), which is the energy needed to push a fluid and to maintain flow. In the energy analysis of flowing fluids, it is convenient to treat the flow energy as part of the energy of the fluid and to represent the microscopic energy of a fluid stream by enthalpy h (Fig. 1–8).

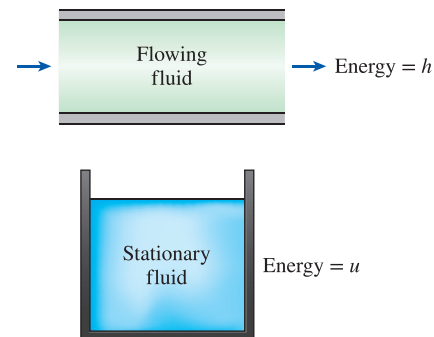


FIGURE 1–8

The *internal energy* u represents the microscopic energy of a nonflowing fluid, whereas *enthalpy* h represents the microscopic energy of a flowing fluid.

Specific Heats of Gases, Liquids, and Solids

You may recall that an **ideal gas** is defined as a gas that obeys the relation

$$PV = RT \text{ or } P = \rho RT \quad (1-1)$$

where P is the absolute pressure, v is the specific volume, T is the thermodynamic (or absolute) temperature, ρ is the density, and R is the gas constant. It has been experimentally observed that the ideal gas relation given above closely approximates the P - v - T behavior of real gases at low densities. At low pressures and high temperatures, the density of a gas decreases, and the gas behaves like an ideal gas. In the range of practical interest, many familiar gases such as air, nitrogen, oxygen, hydrogen, helium, argon, neon, and carbon dioxide and even heavier gases such as krypton can be treated as ideal gases with negligible error (often less than 1 percent). Dense gases such as water vapor in steam power plants and refrigerant vapor in refrigerators, however, should not always be treated as ideal gases since they usually exist at a state near saturation.

You may also recall that **specific heat** is defined as *the energy required to raise the temperature of a unit mass of a substance by one degree* (Fig. 1–9). In general, this energy depends on how the process is executed. We are usually interested in two kinds of specific heats: specific heat at constant volume c_v and specific heat at constant pressure c_p . The **specific heat at constant volume** c_v can be viewed as the energy required to raise the temperature of a unit mass of a substance by one degree as the volume is held constant. The energy required to do the same as the pressure is held constant is the

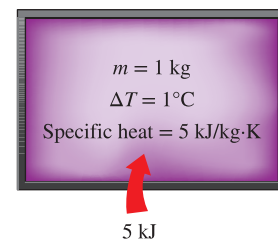
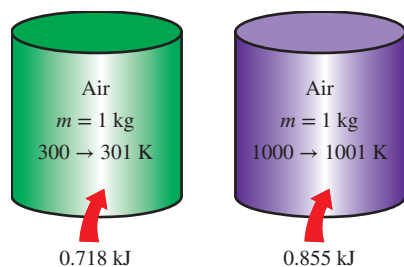


FIGURE 1–9

Specific heat is the energy required to raise the temperature of a unit mass of a substance by one degree in a specified way.

**FIGURE 1-10**

The specific heat of a substance changes with temperature.

specific heat at constant pressure c_p . The specific heat at constant pressure c_p is greater than c_v because at constant pressure the system is allowed to expand, and the energy for this expansion work must also be supplied to the system. For ideal gases, these two specific heats are related to each other by $c_p = c_v + R$.

A common unit for specific heats is $\text{kJ/kg}\cdot^\circ\text{C}$ or $\text{kJ/kg}\cdot\text{K}$. Notice that these two units are *identical* since $\Delta T(^{\circ}\text{C}) = \Delta T(\text{K})$, and a 1°C change in temperature is equivalent to a change of 1 K. Also,

$$1 \text{ kJ/kg}\cdot^\circ\text{C} \equiv 1 \text{ J/g}\cdot^\circ\text{C} \equiv 1 \text{ kJ/kg}\cdot\text{K} \equiv 1 \text{ J/g}\cdot\text{K}$$

The specific heats of a substance, in general, depend on two independent properties such as temperature and pressure. For an *ideal gas*, however, they depend on *temperature* only (Fig. 1–10). At low pressures all real gases approach ideal gas behavior, and therefore their specific heats depend on temperature only.

The differential changes in the internal energy u and enthalpy h of an ideal gas can be expressed in terms of the specific heats as

$$du = c_v dT \quad \text{and} \quad dh = c_p dT \quad (1-2)$$

The finite changes in the internal energy and enthalpy of an ideal gas during a process can be expressed approximately by using specific heat values at the average temperature as

$$\Delta u = c_{v, \text{avg}} \Delta T \quad \text{and} \quad \Delta h = c_{p, \text{avg}} \Delta T \quad (\text{J/g}) \quad (1-3)$$

or

$$\Delta U = mc_{v, \text{avg}} \Delta T \quad \text{and} \quad \Delta H = mc_{p, \text{avg}} \Delta T \quad (\text{J}) \quad (1-4)$$

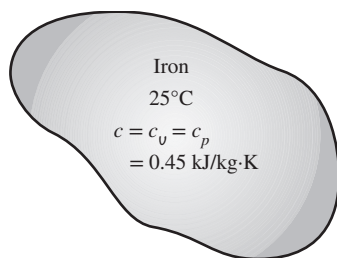
where m is the mass of the system.

A substance whose specific volume (or density) does not change with temperature or pressure is called an **incompressible substance**. The specific volumes of solids and liquids essentially remain constant during a process, and thus they can be approximated as incompressible substances without sacrificing much in accuracy.

The constant-volume and constant-pressure specific heats are identical for incompressible substances (Fig. 1–11). Therefore, for solids and liquids the subscripts on c_v and c_p can be dropped and both specific heats can be represented by a single symbol, c . That is, $c_p \cong c_v \cong c$. This result could also be deduced from the physical definitions of constant-volume and constant-pressure specific heats. Specific heats of several common gases, liquids, and solids are given in the Appendix.

The specific heats of incompressible substances depend on temperature only. Therefore, the change in the internal energy of solids and liquids can be expressed as

$$\Delta U = mc_{\text{avg}} \Delta T \quad (\text{J}) \quad (1-5)$$

**FIGURE 1-11**

The c_v and c_p values of incompressible substances are identical and are denoted by c .

where c_{avg} is the average specific heat evaluated at the average temperature. Note that the internal energy change of the systems that remain in a single phase (liquid, solid, or gas) during the process can be determined very easily using average specific heats.

Energy Transfer

Energy can be transferred to or from a given mass by two mechanisms: *heat transfer* Q and *work* W . An energy interaction is heat transfer if its driving force is a temperature difference. Otherwise, it is work. A rising piston, a rotating shaft, and an electrical wire crossing the system boundaries are all associated with work interactions. Work done *per unit time* is called **power** and is denoted by $\dot{W}$. The unit of power is W or hp (1 hp = 746 W). Car engines and hydraulic, steam, and gas turbines produce work; compressors, pumps, and mixers consume work. Notice that the energy of a system decreases as it does work and increases as work is done on it.

In daily life, we often refer to the sensible and latent forms of internal energy as **heat**, and we talk about the heat content of bodies (Fig. 1–12). In thermodynamics, however, those forms of energy are usually referred to as **thermal energy** to prevent any confusion with *heat transfer*.

The term *heat* and the associated phrases such as *heat flow*, *heat addition*, *heat rejection*, *heat absorption*, *heat gain*, *heat loss*, *heat storage*, *heat generation*, *electrical heating*, *latent heat*, *body heat*, and *heat source* are in common use today, and the attempt to replace *heat* in these phrases with *thermal energy* has had only limited success. These phrases are deeply rooted in our vocabulary, and they are used by both ordinary people and scientists without causing any misunderstanding. For example, the phrase *body heat* is understood to mean the *thermal energy content* of a body. Likewise, *heat flow* is understood to mean the *transfer of thermal energy*, not the flow of a fluid-like substance called *heat*, although the latter incorrect interpretation, based on the caloric theory, is the origin of this phrase. Also, the transfer of heat into a system is frequently referred to as *heat addition* and the transfer of heat out of a system as *heat rejection*.

Keeping in line with current practice, we will refer to the thermal energy as *heat* and the transfer of thermal energy as *heat transfer*. The amount of heat transferred during the process is denoted by Q . The amount of heat transferred per unit time is called the **heat transfer rate** and is denoted by $\dot{Q}$. The overdot stands for the time derivative, or “per unit time.” The heat transfer rate $\dot{Q}$ has the unit J/s, which is equivalent to W.

When the *rate* of heat transfer $\dot{Q}$ is available, then the total amount of heat transfer Q during a time interval Δt can be determined from

$$Q = \int_0^{\Delta t} \dot{Q} dt \quad (\text{J}) \quad (1-6)$$

provided that the variation of $\dot{Q}$ with time is known. For the special case of $\dot{Q} = \text{constant}$, Eq. 1–6 reduces to

$$Q = \dot{Q} \Delta t \quad (\text{J}) \quad (1-7)$$

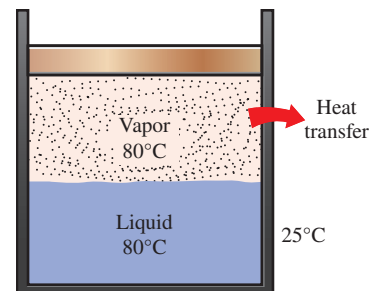
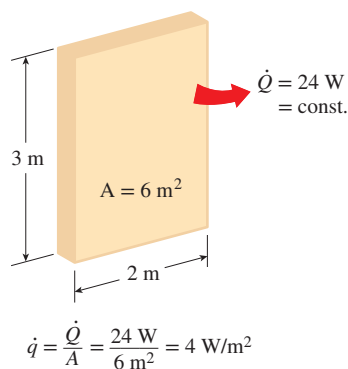
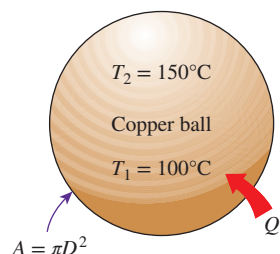


FIGURE 1–12

The sensible and latent forms of internal energy can be transferred as a result of a temperature difference, and they are referred to as *heat* or *thermal energy*.

**FIGURE 1-13**

Heat flux is heat transfer *per unit time* and *per unit area*, and is equal to $\dot{q} = \dot{Q}/A$ when $\dot{Q}$ is uniform over the area A .

**FIGURE 1-14**

Schematic for Example 1-1.

The rate of heat transfer per unit area normal to the direction of heat transfer is called **heat flux**, and the average heat flux is expressed as (Fig. 1-13)

$$\dot{q} = \frac{\dot{Q}}{A} \quad (\text{W/m}^2) \quad (1-8)$$

where A is the heat transfer area. The unit of heat flux in English units is $\text{Btu/h} \cdot \text{ft}^2$. Note that heat flux may vary with time as well as position on a surface.

EXAMPLE 1-1 Heating of a Copper Ball

A 10-cm-diameter copper ball is to be heated from 100°C to an average temperature of 150°C in 30 min (Fig. 1-14). Taking the average density and specific heat of copper in this temperature range to be $\rho = 8950 \text{ kg/m}^3$ and $c_p = 0.395 \text{ kJ/kg} \cdot ^\circ\text{C}$, respectively, determine (a) the total amount of heat transfer to the copper ball, (b) the average rate of heat transfer to the ball, and (c) the average heat flux.

SOLUTION The copper ball is to be heated from 100°C to 150°C . The total heat transfer, the average rate of heat transfer, and the average heat flux are to be determined.

Assumptions Constant properties can be used for copper at the average temperature.

Properties The average density and specific heat of copper are given to be $\rho = 8950 \text{ kg/m}^3$ and $c_p = 0.395 \text{ kJ/kg} \cdot ^\circ\text{C}$.

Analysis (a) The amount of heat transferred to the copper ball is simply the change in its internal energy and is determined from

$$\begin{aligned} \text{Energy transfer to the system} &= \text{Energy increase of the system} \\ Q &= \Delta U = mc_{\text{avg}}(T_2 - T_1) \end{aligned}$$

where

$$m = \rho V = \frac{\pi}{6} \rho D^3 = \frac{\pi}{6} (8950 \text{ kg/m}^3)(0.1 \text{ m})^3 = 4.686 \text{ kg}$$

Substituting,

$$Q = (4.686 \text{ kg})(0.395 \text{ kJ/kg} \cdot ^\circ\text{C})(150 - 100)^\circ\text{C} = \mathbf{92.6 \text{ kJ}}$$

Therefore, 92.6 kJ of heat needs to be transferred to the copper ball to heat it from 100°C to 150°C .

(b) The rate of heat transfer normally changes during a process with time. However, we can determine the *average* rate of heat transfer by dividing the total amount of heat transfer by the time interval. Therefore,

$$\dot{Q}_{\text{avg}} = \frac{Q}{\Delta t} = \frac{92.6 \text{ kJ}}{1800 \text{ s}} = 0.0514 \text{ kJ/s} = \mathbf{51.4 \text{ W}}$$

(c) Heat flux is defined as the heat transfer per unit time per unit area, or the rate of heat transfer per unit area. Therefore, the average heat flux in this case is

$$\dot{q}_{\text{avg}} = \frac{\dot{Q}_{\text{avg}}}{A} = \frac{\dot{Q}_{\text{avg}}}{\pi D^2} = \frac{51.4 \text{ W}}{\pi (0.1 \text{ m})^2} = \mathbf{1636 \text{ W/m}^2}$$

Discussion Note that heat flux may vary with location on a surface. The value calculated here is the *average* heat flux over the entire surface of the ball.

1-4 ■ THE FIRST LAW OF THERMODYNAMICS

The **first law of thermodynamics**, also known as the **conservation of energy principle**, states that *energy can neither be created nor destroyed during a process; it can only change forms*. Therefore, every bit of energy must be accounted for during a process. The conservation of energy principle (or the energy balance) for *any system* undergoing *any process* may be expressed as follows: *The net change (increase or decrease) in the total energy of the system during a process is equal to the difference between the total energy entering and the total energy leaving the system during that process*. That is,

$$\left(\begin{array}{c} \text{Total energy} \\ \text{entering the} \\ \text{system} \end{array} \right) - \left(\begin{array}{c} \text{Total energy} \\ \text{leaving the} \\ \text{system} \end{array} \right) = \left(\begin{array}{c} \text{Change in the} \\ \text{total energy of} \\ \text{the system} \end{array} \right) \quad (1-9)$$

Noting that energy can be transferred to or from a system by *heat*, *work*, and *mass flow*, and that the total energy of a simple compressible system consists of internal, kinetic, and potential energies, the **energy balance** for any system undergoing any process can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc., energies}}} \quad (\text{J}) \quad (1-10)$$

or, in the **rate form**, as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\substack{\text{Rate of net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{dE_{\text{system}}/dt}_{\substack{\text{Rate of change in internal} \\ \text{kinetic, potential, etc., energies}}} \quad (\text{W}) \quad (1-11)$$

Energy is a property, and the value of a property does not change unless the state of the system changes. Therefore, the energy change of a system is zero ($\Delta E_{\text{system}} = 0$) if the state of the system does not change during the process, that is, the process is steady. The energy balance in this case reduces to (Fig. 1-15)

Steady, rate form:

$$\underbrace{\dot{E}_{\text{in}}}_{\substack{\text{Rate of net energy transfer in} \\ \text{by heat, work, and mass}}} = \underbrace{\dot{E}_{\text{out}}}_{\substack{\text{Rate of net energy transfer out} \\ \text{by heat, work, and mass}}} \quad (1-12)$$

In the absence of significant electric, magnetic, motion, gravity, and surface tension effects (i.e., for stationary simple compressible systems), the change in the *total energy* of a system during a process is simply the change in its *internal energy*. That is, $\Delta E_{\text{system}} = \Delta U_{\text{system}}$.

In heat transfer analysis, we are usually interested only in the forms of energy that can be transferred as a result of a temperature difference, that is, heat or thermal energy. In such cases it is convenient to write a **heat balance** and to treat the conversion of nuclear, chemical, mechanical, and electrical energies into thermal energy as *heat generation*. The *energy balance* in that case can be expressed as

$$\underbrace{Q_{\text{in}} - Q_{\text{out}}}_{\substack{\text{Net heat} \\ \text{transfer}}} + \underbrace{E_{\text{gen}}}_{\substack{\text{Heat} \\ \text{generation}}} = \underbrace{\Delta E_{\text{thermal, system}}}_{\substack{\text{Change in thermal} \\ \text{energy of the system}}} \quad (\text{J}) \quad (1-13)$$

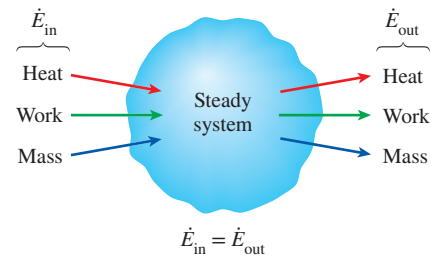
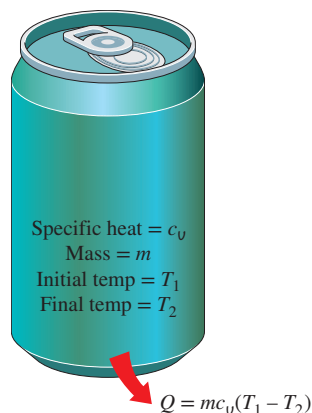


FIGURE 1-15

In steady operation, the rate of energy transfer to a system is equal to the rate of energy transfer from the system.

**FIGURE 1-16**

In the absence of any work interactions, the change in the energy content of a closed system is equal to the net heat transfer.

Energy Balance for Closed Systems (Fixed Mass)

A closed system consists of a *fixed mass*. The total energy E for most systems encountered in practice consists of the internal energy U . This is especially the case for stationary systems since they don't involve any changes in their velocity or elevation during a process. The energy balance relation in that case reduces to

$$\text{Stationary closed system:} \quad E_{\text{in}} - E_{\text{out}} = \Delta U = m c_v \Delta T \quad (\text{J}) \quad (1-14)$$

where we expressed the internal energy change in terms of mass m , the specific heat at constant volume c_v , and the temperature change ΔT of the system. When the system involves heat transfer only and no work interactions across its boundary, the energy balance relation further reduces to (Fig. 1-16)

$$\text{Stationary closed system, no work:} \quad Q = m c_v \Delta T \quad (\text{J}) \quad (1-15)$$

where Q is the net amount of heat transfer to or from the system. This is the form of the energy balance relation we will use most often when dealing with a fixed mass.

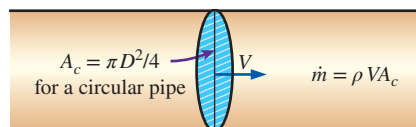
Energy Balance for Steady-Flow Systems

A large number of engineering devices such as water heaters and car radiators involve mass flow in and out of a system, and they are modeled as *control volumes*. Most control volumes are analyzed under steady operating conditions. The term *steady* means *no change with time* at a specified location. The opposite of steady is *unsteady* or *transient*. Also, the term *uniform* implies *no change with position* throughout a surface or region at a specified time. These meanings are consistent with their everyday usage (steady girlfriend, uniform distribution, etc.). The total energy content of a control volume during a *steady-flow process* remains constant ($E_{\text{CV}} = \text{constant}$). That is, the change in the total energy of the control volume during such a process is zero ($\Delta E_{\text{CV}} = 0$). Thus the amount of energy entering a control volume in all forms (heat, work, mass transfer) for a steady-flow process must be equal to the amount of energy leaving it.

The amount of mass flowing through a cross section of a flow device per unit time is called the **mass flow rate**, and it is denoted by $\dot{m}$. A fluid may flow in and out of a control volume through pipes or ducts. The mass flow rate of a fluid flowing in a pipe or duct is proportional to the cross-sectional area A_c of the pipe or duct, the density ρ , and the velocity V of the fluid. The mass flow rate through a differential area dA_c can be expressed as $\Delta \dot{m} = \rho V_n dA_c$ where V_n is the velocity component normal to dA_c . The mass flow rate through the entire cross-sectional area is obtained by integration over A_c .

The flow of a fluid through a pipe or duct can often be approximated to be *one-dimensional*. That is, the properties can be assumed to vary in one direction only (the direction of flow). As a result, all properties are assumed to be uniform at any cross section normal to the flow direction, and the properties are assumed to have *bulk average values* over the entire cross section. Under the one-dimensional flow approximation, the mass flow rate of a fluid flowing in a pipe or duct can be expressed as (Fig. 1-17)

$$\dot{m} = \rho V A_c \quad (\text{kg/s}) \quad (1-16)$$

**FIGURE 1-17**

The mass flow rate of a fluid at a cross section is equal to the product of the fluid density, average fluid velocity, and the cross-sectional area.

where ρ is the fluid density, V is the average fluid velocity in the flow direction, and A_c is the cross-sectional area of the pipe or duct.

The volume of a fluid flowing through a pipe or duct per unit time is called the **volume flow rate** $\dot{V}$ and is expressed as

$$\dot{V} = VA_c = \frac{\dot{m}}{\rho} \quad (\text{m}^3/\text{s}) \quad (1-17)$$

Note that the mass flow rate of a fluid through a pipe or duct remains constant during steady flow. This is not the case for the volume flow rate, however, unless the density of the fluid remains constant.

For a steady-flow system with one inlet and one exit, the rate of mass flow into the control volume must be equal to the rate of mass flow out of it. That is, $\dot{m}_{\text{in}} = \dot{m}_{\text{out}} = \dot{m}$. When the changes in kinetic and potential energies are negligible, which is usually the case, and there is no work interaction, the energy balance for such a steady-flow system reduces to (Fig. 1–18)

$$\dot{Q} = \dot{m} \Delta h = \dot{m} c_p \Delta T \quad (\text{kJ/s}) \quad (1-18)$$

where $\dot{Q}$ is the rate of net heat transfer into or out of the control volume. This is the form of the energy balance relation that we will use most often for steady-flow systems.

Surface Energy Balance

As mentioned in the chapter opener, heat is transferred by the mechanisms of conduction, convection, and radiation, and heat often changes vehicles as it is transferred from one medium to another. For example, the heat conducted to the outer surface of the wall of a house in winter is convected away by the cold outdoor air while being radiated to the cold surroundings. In such cases, it may be necessary to keep track of the energy interactions at the surface, and this is done by applying the conservation of energy principle to the surface.

A surface contains no volume or mass and thus no energy. Therefore, a surface can be viewed as a fictitious system whose energy content remains constant during a process (just like a steady-state or steady-flow system). Then the energy balance for a surface can be expressed as

Surface energy balance:
$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}} \quad (1-19)$$

This relation is valid for both steady and transient conditions, and the surface energy balance does not involve heat generation since a surface does not have a volume. The energy balance for the outer surface of the wall in Fig. 1–19, for example, can be expressed as

$$\dot{Q}_1 = \dot{Q}_2 + \dot{Q}_3 \quad (1-20)$$

where $\dot{Q}_1$ is conduction through the wall to the surface, $\dot{Q}_2$ is convection from the surface to the outdoor air, and $\dot{Q}_3$ is net radiation from the surface to the surroundings.

When the directions of interactions are not known, all energy interactions can be assumed to be toward the surface, and the surface energy balance can be expressed as $\sum \dot{E}_{\text{in}} = 0$. Note that the interactions in the opposite direction will end up having negative values, and balance this equation.

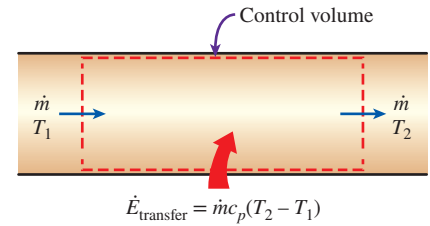


FIGURE 1–18

Under steady conditions, the net rate of energy transfer to a fluid in a control volume is equal to the rate of increase in the energy of the fluid stream flowing through the control volume.

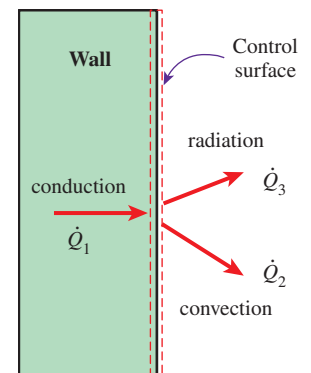


FIGURE 1–19

Energy interactions at the outer wall surface of a house.

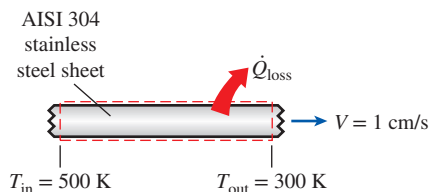


FIGURE 1–20

Schematic for Example 1–2.

EXAMPLE 1–2 Cooling of Stainless Steel Sheets

A heated continuous AISI 304 stainless steel sheet is being conveyed at a constant speed of 1 cm/s into a chamber to be cooled (Fig. 1–20). The stainless steel sheet is 5 mm thick and 2 m wide, and it enters and exits the chamber at 500 K and 300 K, respectively. Determine the rate of heat loss from the stainless steel sheet inside the chamber.

SOLUTION The rate of heat loss from a stainless steel sheet being conveyed inside a chamber is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The stainless steel sheet has constant properties. 3 Changes in potential and kinetic energy are negligible.

Properties The constant-pressure specific heat of AISI 304 stainless steel at the average temperature of $(500 + 300)/2 = 400 \text{ K}$ is $515 \text{ J/kg}\cdot\text{K}$. The density of AISI 304 stainless steel is 7900 kg/m^3 (Table A–3).

Analysis The mass of the stainless steel sheet being conveyed enters and exits the chamber at a rate of

$$\begin{aligned}\dot{m} &= \rho V w t \\ &= (7900 \text{ kg/m}^3)(0.01 \text{ m/s})(2 \text{ m})(0.005 \text{ m}) \\ &= 0.79 \text{ kg/s}\end{aligned}$$

The rate of heat loss from the stainless steel sheet in the chamber can be determined as follows:

$$\begin{aligned}\dot{Q}_{\text{loss}} &= \dot{m} c_p (T_{\text{in}} - T_{\text{out}}) \\ &= (0.79 \text{ kg/s})(515 \text{ J/kg}\cdot\text{K})(500 - 300) \text{ K} = 81,370 \text{ J/s} \\ &= \mathbf{81.4 \text{ kW}}\end{aligned}$$

Discussion The stainless steel sheet being conveyed in and out of the chamber is treated as a control volume.

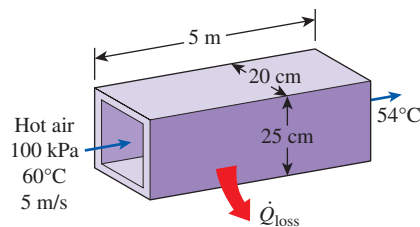


FIGURE 1–21

Schematic for Example 1–3.

EXAMPLE 1–3 Heat Loss from Heating Ducts in a Basement

A 5-m-long section of an air heating system of a house passes through an unheated space in the basement (Fig. 1–21). The cross section of the rectangular duct of the heating system is $20 \text{ cm} \times 25 \text{ cm}$. Hot air enters the duct at 100 kPa and 60°C at an average velocity of 5 m/s. The temperature of the air in the duct drops to 54°C as a result of heat loss to the cool space in the basement. Determine the rate of heat loss from the air in the duct to the basement under steady conditions. Also, determine the cost of this heat loss per hour if the house is heated by a natural gas furnace that has an efficiency of 80 percent and the cost of the natural gas in that area is $\$1.60/\text{therm}$ (1 therm = 100,000 Btu = 105,500 kJ).

SOLUTION The temperature of the air in the heating duct of a house drops as a result of heat loss to the cool space in the basement. The rate of heat loss from the hot air and its cost are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Air can be treated as an ideal gas with constant properties at room temperature.

Properties The constant-pressure specific heat of air at the average temperature of $(54 + 60)/2 = 57^\circ\text{C}$ is $1.007 \text{ kJ/kg}\cdot\text{K}$ (Table A–15).

Analysis We take the basement section of the heating system as our system, which is a steady-flow system. The rate of heat loss from the air in the duct can be determined from

$$\dot{Q} = \dot{m} c_p \Delta T$$

where $\dot{m}$ is the mass flow rate and ΔT is the temperature drop. The density of air at the inlet conditions is

$$\rho = \frac{P}{RT} = \frac{100 \text{ kPa}}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(60 + 273) \text{ K}} = 1.046 \text{ kg/m}^3$$

The cross-sectional area of the duct is

$$A_c = (0.20 \text{ m})(0.25 \text{ m}) = 0.05 \text{ m}^2$$

Then the mass flow rate of air through the duct and the rate of heat loss become

$$\dot{m} = \rho V A_c = (1.046 \text{ kg/m}^3)(5 \text{ m/s})(0.05 \text{ m}^2) = 0.2615 \text{ kg/s}$$

and

$$\begin{aligned} \dot{Q}_{\text{loss}} &= \dot{m} c_p (T_{\text{in}} - T_{\text{out}}) \\ &= (0.2615 \text{ kg/s})(1.007 \text{ kJ/kg} \cdot ^\circ\text{C})(60 - 54)^\circ\text{C} \\ &= \mathbf{1.58 \text{ kJ/s}} \end{aligned}$$

or 5688 kJ/h. The cost of this heat loss to the homeowner is

$$\begin{aligned} \text{Cost of heat loss} &= \frac{(\text{Rate of heat loss})(\text{Unit cost of energy input})}{\text{Furnace efficiency}} \\ &= \frac{(5688 \text{ kJ/h})(\$1.60/\text{therm}) \left(\frac{1 \text{ therm}}{105,500 \text{ kJ}} \right)}{0.80} \\ &= \mathbf{\$0.108/\text{h}} \end{aligned}$$

Discussion The heat loss from the heating ducts in the basement is costing the homeowner 10.8 cents per hour. Assuming the heater operates 2000 hours during a heating season, the annual cost of this heat loss adds up to \$216. Most of this money can be saved by insulating the heating ducts in the unheated areas.

EXAMPLE 1-4 Electric Heating of a House at High Elevation

Consider a house that has a floor space of 2000 ft² and an average height of 9 ft at 5000 ft elevation where the standard atmospheric pressure is 12.2 psia (Fig. 1–22). Initially the house is at a uniform temperature of 50°F. Now the electric heater is turned on, and the heater runs until the air temperature in the house rises to an average value of 70°F. Determine the amount of energy transferred to the air assuming (a) the house is airtight and thus no air escapes during the heating process and (b) some air escapes through the cracks as the heated air in the house expands at constant pressure. Also determine the cost of this heat for each case if the cost of electricity in that area is \$0.075/kWh.

SOLUTION The air in the house is heated by an electric heater. The amount and cost of the energy transferred to the air are to be determined for constant-volume and constant-pressure cases.

Assumptions 1 Air can be treated as an ideal gas with constant properties. 2 Heat loss from the house during heating is negligible. 3 The volume occupied by the furniture and other things is negligible.

Properties The specific heats of air at the average temperature of $(50 + 70)/2 = 60^\circ\text{F}$ are $c_p = 0.240 \text{ Btu/lbm} \cdot \text{R}$ and $c_v = R = 0.171 \text{ Btu/lbm} \cdot \text{R}$ (Tables A–1E and A–15E).

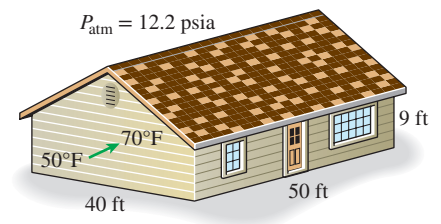


FIGURE 1-22
Schematic for Example 1–4.

Analysis The volume and the mass of the air in the house are

$$V = (\text{Floor area})(\text{Height}) = (2000 \text{ ft}^2)(9 \text{ ft}) = 18,000 \text{ ft}^3$$

$$m = \frac{PV}{RT} = \frac{(12.2 \text{ psia})(18,000 \text{ ft}^3)}{(0.3704 \text{ psia} \cdot \text{ft}^3/\text{lbm} \cdot \text{R})(50 + 460)\text{R}} = 1162 \text{ lbm}$$

(a) The amount of energy transferred to air at constant volume is simply the change in its internal energy and is determined from

$$\begin{aligned} E_{\text{in}} - E_{\text{out}} &= \Delta E_{\text{system}} \\ E_{\text{in, constant volume}} &= \Delta U_{\text{air}} = mc_v \Delta T \\ &= (1162 \text{ lbm})(0.171 \text{ Btu/lbm} \cdot ^\circ\text{F})(70 - 50)^\circ\text{F} \\ &= \mathbf{3974 \text{ Btu}} \end{aligned}$$

At a unit cost of \$0.075/kWh, the total cost of this energy is

$$\begin{aligned} \text{Cost of energy} &= (\text{Amount of energy})(\text{Unit cost of energy}) \\ &= (3974 \text{ Btu})(\$0.075/\text{kWh}) \left(\frac{1 \text{ kWh}}{3412 \text{ Btu}} \right) \\ &= \mathbf{\$0.087} \end{aligned}$$

(b) The amount of energy transferred to air at constant pressure is the change in its enthalpy and is determined from

$$\begin{aligned} E_{\text{in, constant pressure}} &= \Delta H_{\text{air}} = mc_p \Delta T \\ &= (1162 \text{ lbm})(0.240 \text{ Btu/lbm} \cdot ^\circ\text{F})(70 - 50)^\circ\text{F} \\ &= \mathbf{5578 \text{ Btu}} \end{aligned}$$

At a unit cost of \$0.075/kWh, the total cost of this energy is

$$\begin{aligned} \text{Cost of energy} &= (\text{Amount of energy})(\text{Unit cost of energy}) \\ &= (5578 \text{ Btu})(\$0.075/\text{kWh}) \left(\frac{1 \text{ kWh}}{3412 \text{ Btu}} \right) \\ &= \mathbf{\$0.123} \end{aligned}$$

Discussion It costs about 9 cents in the first case and 12 cents in the second case to raise the temperature of the air in this house from 50°F to 70°F. The second answer is more realistic since every house has cracks, especially around the doors and windows, and the pressure in the house remains essentially constant during a heating process. Therefore, the second approach is used in practice. This conservative approach somewhat overpredicts the amount of energy used, however, since some of the air escapes through the cracks before it is heated to 70°F.

1-5 ■ HEAT TRANSFER MECHANISMS

In Sec. 1-1, we defined *heat* as the form of energy that can be transferred from one system to another as a result of temperature difference. A thermodynamic analysis is concerned with the *amount* of heat transfer as a system undergoes a process from one equilibrium state to another. The science that deals with

the determination of the *rates* of such energy transfers is the science of *heat transfer*. The transfer of energy as heat is always from the higher-temperature medium to the lower-temperature one, and heat transfer stops when the two media reach the same temperature.

Heat can be transferred in three different modes: *conduction*, *convection*, and *radiation*. All modes of heat transfer require the existence of a temperature difference, and all modes are from the high-temperature medium to a lower-temperature one. Next we give a brief description of each mode. A detailed study of these modes is given in later chapters of this text.

1-6 ■ CONDUCTION

Conduction is the transfer of energy from the more energetic particles of a substance to the adjacent less energetic ones as a result of interactions between the particles. Conduction can take place in solids, liquids, or gases. In gases and liquids, conduction is due to the *collisions* and *diffusion* of the molecules during their random motion. In solids, it is due to the combination of *vibrations* of the molecules in a lattice and the energy transport by *free electrons*. A cold canned drink in a warm room, for example, eventually warms up to the room temperature as a result of heat transfer from the room to the drink through the aluminum can by conduction.

The *rate* of heat conduction through a medium depends on the *geometry* of the medium, its *thickness*, and the *material* of the medium, as well as the *temperature difference* across the medium. We know that wrapping a hot water tank with glass wool (an insulating material) reduces the rate of heat loss from the tank. The thicker the insulation, the smaller the heat loss. We also know that a hot water tank loses heat at a higher rate when the temperature of the room housing the tank is lowered. Further, the larger the tank, the larger the surface area and thus the rate of heat loss.

Consider steady heat conduction through a large plane wall of thickness $\Delta x = L$ and area A , as shown in Fig. 1-23. The temperature difference across the wall is $\Delta T = T_2 - T_1$. Experiments have shown that the rate of heat transfer $\dot{Q}$ through the wall is *doubled* when the temperature difference ΔT across the wall or the area A normal to the direction of heat transfer is doubled, but is *halved* when the wall thickness L is doubled. Thus we conclude that *the rate of heat conduction through a plane layer is proportional to the temperature difference across the layer and the heat transfer area but is inversely proportional to the thickness of the layer*. That is,

$$\text{Rate of heat conduction} \propto \frac{(\text{Area})(\text{Temperature difference})}{\text{Thickness}}$$

or,

$$\dot{Q}_{\text{cond}} = kA \frac{T_1 - T_2}{\Delta x} = -kA \frac{\Delta T}{\Delta x} \quad (\text{W}) \quad (1-21)$$

where the constant of proportionality k is the **thermal conductivity** of the material, which is a *measure of the ability of a material to conduct heat* (Fig. 1-24). In the limiting case of $\Delta x \rightarrow 0$, Eq. 1-21 reduces to the differential form

$$\dot{Q}_{\text{cond}} = -kA \frac{dT}{dx} \quad (\text{W}) \quad (1-22)$$

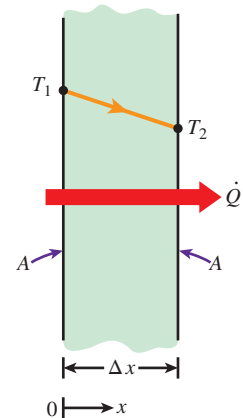


FIGURE 1-23

Heat conduction through a large plane wall of thickness Δx and area A .

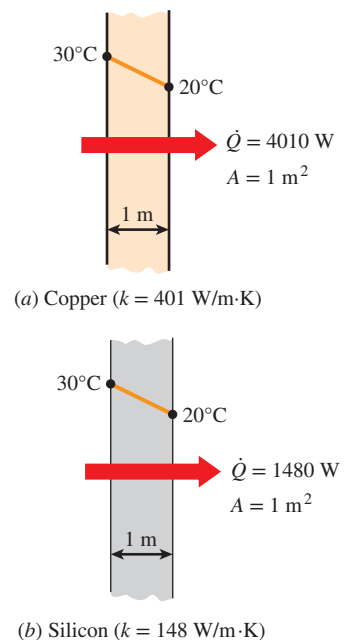


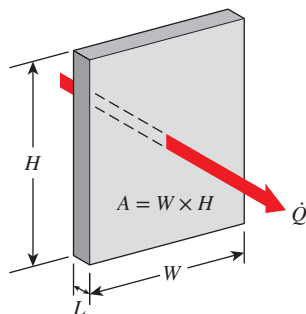
FIGURE 1-24

The rate of heat conduction through a solid is directly proportional to its thermal conductivity.

**FIGURE 1–25****Jean Baptiste Joseph Fourier**

(1768–1830) was a French mathematician and physicist born in Auxerre, France. He is best known for his work on the infinite series of trigonometric functions that bear his name and for his development of the mathematical theory of heat conduction. Fourier established the partial differential equation governing heat diffusion, and he solved it by using the Fourier series. The Fourier transform, Fourier number, and Fourier's law of heat conduction are named in his honor. Fourier is also credited with the discovery of the greenhouse effect in 1824.

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**FIGURE 1–26**

In heat conduction analysis, A represents the area *normal* to the direction of heat transfer.

which is called **Fourier's law of heat conduction** after J. Fourier (Fig. 1–25), who expressed it first in his heat transfer text in 1822. Here dT/dx is the **temperature gradient**, which is the slope of the temperature curve on a T - x diagram (the rate of change of T with x), at location x . Equation 1–22 indicates that the rate of heat conduction in a given direction is proportional to the temperature gradient in that direction. Heat is conducted in the direction of decreasing temperature, and the temperature gradient becomes negative when temperature decreases with increasing x . The *negative sign* in Eq. 1–22 ensures that heat transfer in the positive x direction is a positive quantity.

The heat transfer area A is always *normal* to the direction of heat transfer. For heat loss through a 5-m-long, 3-m-high, and 25-cm-thick wall, for example, the heat transfer area is $A = 15 \text{ m}^2$. Note that the thickness of the wall has no effect on A (Fig. 1–26).

EXAMPLE 1–5 The Cost of Heat Loss Through a Roof

The roof of an electrically heated home is 6 m long, 8 m wide, and 0.25 m thick, and it is made of a flat layer of concrete whose thermal conductivity is $k = 0.8 \text{ W/m}\cdot\text{K}$ (Fig. 1–27). The temperatures of the inner and the outer surfaces of the roof one night are measured to be 15°C and 4°C , respectively, for a period of 10 h. Determine (a) the rate of heat loss through the roof that night and (b) the cost of that heat loss to the homeowner if the cost of electricity is $\$0.08/\text{kWh}$.

SOLUTION The inner and outer surfaces of the flat concrete roof of an electrically heated home are maintained at specified temperatures during a night. The heat loss through the roof and its cost that night are to be determined.

Assumptions 1 Steady operating conditions exist during the entire night since the surface temperatures of the roof remain constant at the specified values. 2 Constant properties can be used for the roof.

Properties The thermal conductivity of the roof is given to be $k = 0.8 \text{ W/m}\cdot\text{K}$.

Analysis (a) Noting that heat transfer through the roof is by conduction and the area of the roof is $A = 6 \text{ m} \times 8 \text{ m} = 48 \text{ m}^2$, the steady rate of heat transfer through the roof is

$$\dot{Q} = kA \frac{T_1 - T_2}{L} = (0.8 \text{ W/m}\cdot\text{K})(48 \text{ m}^2) \frac{(15 - 4)^\circ\text{C}}{0.25 \text{ m}} = \mathbf{1690 \text{ W} = 1.69 \text{ kW}}$$

(b) The amount of heat lost through the roof during a 10-h period and its cost are

$$\begin{aligned} Q &= \dot{Q} \Delta t = (1.69 \text{ kW})(10 \text{ h}) = 16.9 \text{ kWh} \\ \text{Cost} &= (\text{Amount of energy})(\text{Unit cost of energy}) \\ &= (16.9 \text{ kWh})(\$0.08/\text{kWh}) = \mathbf{\$1.35} \end{aligned}$$

Discussion The cost to the homeowner of the heat loss through the roof that night was $\$1.35$. The total heating bill of the house will be much larger since the heat losses through the walls are not considered in these calculations.

Thermal Conductivity

We have seen that different materials store heat differently, and we have defined the property specific heat c_p as a measure of a material's ability to store thermal energy. For example, $c_p = 4.18 \text{ kJ/kg}\cdot^\circ\text{C}$ for water and $c_p = 0.45 \text{ kJ/kg}\cdot^\circ\text{C}$

for iron at room temperature, which indicates that water can store almost 10 times the energy that iron can per unit mass. Likewise, the thermal conductivity k is a measure of a material's ability to conduct heat. For example, $k = 0.607 \text{ W/m}\cdot\text{K}$ for water and $k = 80.2 \text{ W/m}\cdot\text{K}$ for iron at room temperature, which indicates that iron conducts heat more than 100 times faster than water can. Thus we say that water is a poor heat conductor relative to iron, although water is an excellent medium for storing thermal energy.

Equation 1–21 for the rate of conduction heat transfer under steady conditions can also be viewed as the defining equation for thermal conductivity. Thus the **thermal conductivity** of a material can be defined as *the rate of heat transfer through a unit thickness of the material per unit area per unit temperature difference*. The thermal conductivity of a material is a measure of the ability of the material to conduct heat. A high value for thermal conductivity indicates that the material is a good heat conductor, and a low value indicates that the material is a poor heat conductor or *insulator*. The thermal conductivities of some common materials at room temperature are given in Table 1–1. The thermal conductivity of pure copper at room temperature is $k = 401 \text{ W/m}\cdot\text{K}$, which indicates that a 1-m-thick copper wall will conduct heat at a rate of 401 W per m^2 area per K temperature difference across the wall. Note that materials such as copper and silver that are good electric conductors are also good heat conductors and have high values of thermal conductivity. Materials such as rubber, wood, and Styrofoam are poor conductors of heat and have low conductivity values.

A layer of material of known thickness and area can be heated from one side by an electric resistance heater of known output. If the outer surfaces of the heater are well insulated, all the heat generated by the resistance heater will be transferred through the material whose conductivity is to be determined. Then measuring the two surface temperatures of the material when steady heat transfer is reached and substituting them into Eq. 1–21 together with other known quantities give the thermal conductivity (Fig. 1–28).

The thermal conductivities of materials vary over a wide range, as shown in Fig. 1–29. The thermal conductivities of gases such as air vary by a factor of 10^4 from those of pure metals such as copper. Note that pure crystals and metals have the highest thermal conductivities, and gases and insulating materials have the lowest. Aside from the typical insulators listed in Fig. 1–29, one of the best solid insulators with the lowest thermal conductivity compared to the conventional insulators (Table 1–1) is **aerogel**. See Sec. 1–10 for more details about aerogels.

Temperature is a measure of the kinetic energies of the particles such as the molecules or atoms of a substance. In a liquid or gas, the kinetic energy of the molecules is due to their random translational motion as well as their vibrational and rotational motions. When two molecules possessing different kinetic energies collide, part of the kinetic energy of the more energetic (higher-temperature) molecule is transferred to the less energetic (lower-temperature) molecule, much the same as when two elastic balls of the same mass at different velocities collide and part of the kinetic energy of the faster ball is transferred to the slower one. The higher the temperature, the faster the molecules move, and the higher the number of such collisions, the better the heat transfer.

The *kinetic theory* of gases predicts, and the experiments confirm, that the thermal conductivity of gases is proportional to the *square root of the*

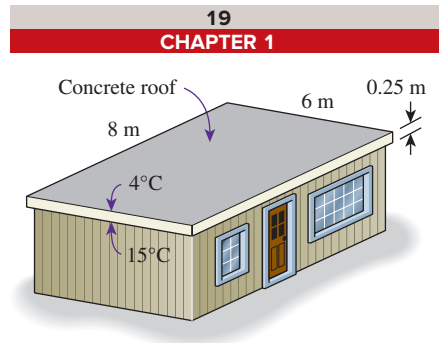


FIGURE 1–27

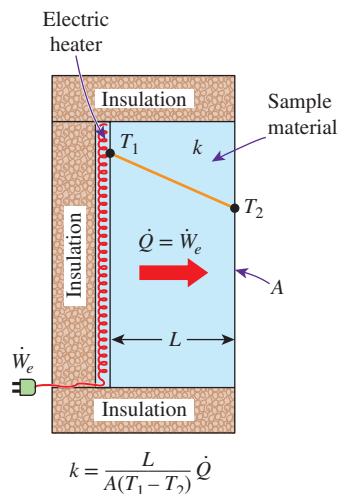
Schematic for Example 1–5.

TABLE 1–1

The thermal conductivities of some materials at room temperature

Material	$k, \text{W/m}\cdot\text{K}^*$
Diamond	2300
Silver	429
Copper	401
Gold	317
Aluminum	237
Iron	80.2
Mercury (l)	8.54
Glass	0.78
Brick	0.72
Water (l)	0.607
Human skin	0.37
Wood (oak)	0.17
Helium (g)	0.152
Soft rubber	0.13
Glass fiber	0.043
Air (g)	0.026
Urethane, rigid foam	0.026

*Multiply by 0.5778 to convert to $\text{Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$.

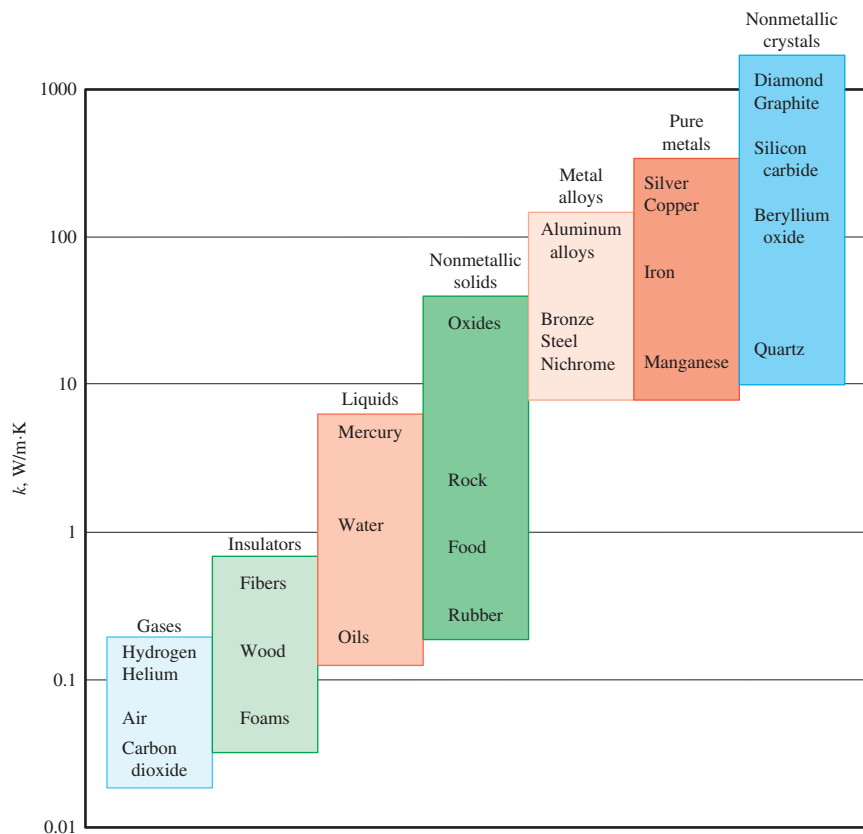
**FIGURE 1-28**

A simple experimental setup to determine the thermal conductivity of a material.

thermodynamic temperature T and inversely proportional to the square root of the molar mass M . Therefore, for a particular gas (fixed M), the thermal conductivity increases with increasing temperature, and at a fixed temperature the thermal conductivity decreases with increasing M . For example, at a fixed temperature of 1000 K, the thermal conductivity of helium ($M = 4$) is 0.343 W/m·K, and that of air ($M = 29$) is 0.0667 W/m·K, which is much lower than that of helium.

The thermal conductivities of *gases* at 1 atm pressure are listed in Table A-16. However, they can also be used at pressures other than 1 atm since the thermal conductivity of gases is *independent of pressure* in a wide range of pressures encountered in practice.

The mechanism of heat conduction in a *liquid* is complicated by the fact that the molecules are more closely spaced, and they exert a stronger intermolecular force field. The thermal conductivities of liquids usually lie between those of solids and gases. The thermal conductivity of a substance is normally highest in the solid phase and lowest in the gas phase. The thermal conductivity of liquids is generally insensitive to pressure except near the thermodynamic critical point. Unlike gases, the thermal conductivities of most liquids decrease with increasing temperature, with water being a notable exception. Like gases, the conductivity of liquids decreases with increasing molar mass. Liquid metals such as mercury and sodium have high thermal conductivities and are very suitable for use in applications where a high heat transfer rate to a liquid is desired, as in nuclear power plants.

**FIGURE 1-29**

The range of thermal conductivity of various materials at room temperature.

In *solids*, heat conduction is due to two effects: (1) the *lattice vibrational waves* induced by the vibrational motions of the molecules positioned at relatively fixed positions in a periodic manner called a lattice and (2) the energy transported via the *free flow of electrons* in the solid (Fig. 1–30). The thermal conductivity of a solid is obtained by adding the lattice and electronic components. The relatively high thermal conductivities of pure metals are primarily due to the electronic component. The lattice component of thermal conductivity strongly depends on the way the molecules are arranged. For example, diamond, which is a highly ordered crystalline solid, has the highest known thermal conductivity at room temperature.

Unlike metals, which are good electrical and heat conductors, *crystalline solids* such as diamond and semiconductors such as silicon are good heat conductors but poor electrical conductors. As a result, such materials find widespread use in the electronics industry. Despite their higher price, diamond heat sinks are used in the cooling of sensitive electronic components because of the excellent thermal conductivity of diamond. Silicon oils and gaskets are commonly used in the packaging of electronic components because they provide both good thermal contact and good electrical insulation.

Pure metals have high thermal conductivities, and one would think that *metal alloys* should also have high conductivities. One would expect an alloy made of two metals of thermal conductivities k_1 and k_2 to have a conductivity k between k_1 and k_2 . But this turns out not to be the case. The thermal conductivity of an alloy of two metals is usually much lower than that of either metal, as shown in Table 1–2. Even small amounts in a pure metal of “foreign” molecules that are good conductors themselves seriously disrupt the transfer of heat in that metal. For example, the thermal conductivity of steel containing just 1 percent of chrome is 62 W/m·K, while the thermal conductivities of iron and chromium are 83 and 95 W/m·K, respectively.

The thermal conductivities of materials vary with temperature (Table 1–3). The variation of thermal conductivity over certain temperature ranges is negligible for some materials but significant for others, as shown in Fig. 1–31. The thermal conductivities of certain solids exhibit dramatic increases at temperatures near absolute zero, when these solids become *superconductors*. For example, the conductivity of copper reaches a maximum value of about 20,000 W/m·K at 20 K, which is about 50 times the conductivity at room temperature. The thermal conductivities and other thermal properties of various materials are given in Tables A–3 to A–17.

As shown in Table 1–1 and Fig. 1–31, most liquids have relatively low thermal conductivities when compared to the thermal conductivity of solids. The thermal conductivity of commonly used heat transfer fluids (such as water or oil) can be considerably increased by adding small solid particles to the fluid. The name **nanofluid** refers to the suspension of nanometer-sized particles (with typical length scales of 1 to 100 nm), called nanoparticles, in conventional fluids. The nanoparticles used in nanofluids are typically made of metals, oxides, carbides, or carbon nanotubes. Common base fluids include water, ethylene glycol, and oil. Nanofluids have been shown to enhance the thermal conductivity and convective heat transfer performance of the working (base) liquids.

Values of thermal conductivity given in Tables A–3 to A–17 are appropriate when the physical dimensions of the material under consideration are

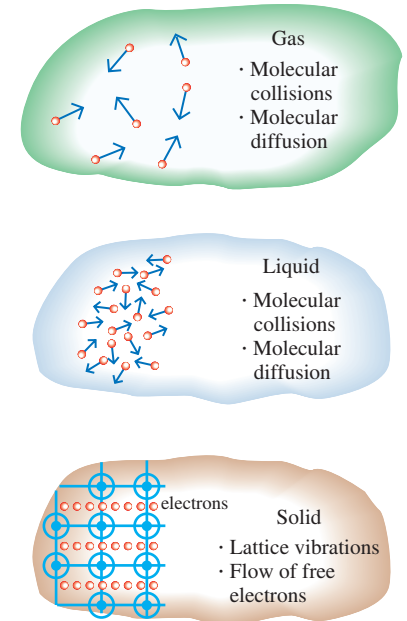


FIGURE 1–30

The mechanisms of heat conduction in different phases of a substance.

TABLE 1–2

The thermal conductivity of an alloy is usually much lower than the thermal conductivity of either metal of which it is composed

Pure Metal or Alloy	k , W/m·K, at 300 K
Copper	401
Nickel	91
Constantan (55% Cu, 45% Ni)	23
Copper	401
Aluminum	237
Commercial bronze (90% Cu, 10% Al)	52

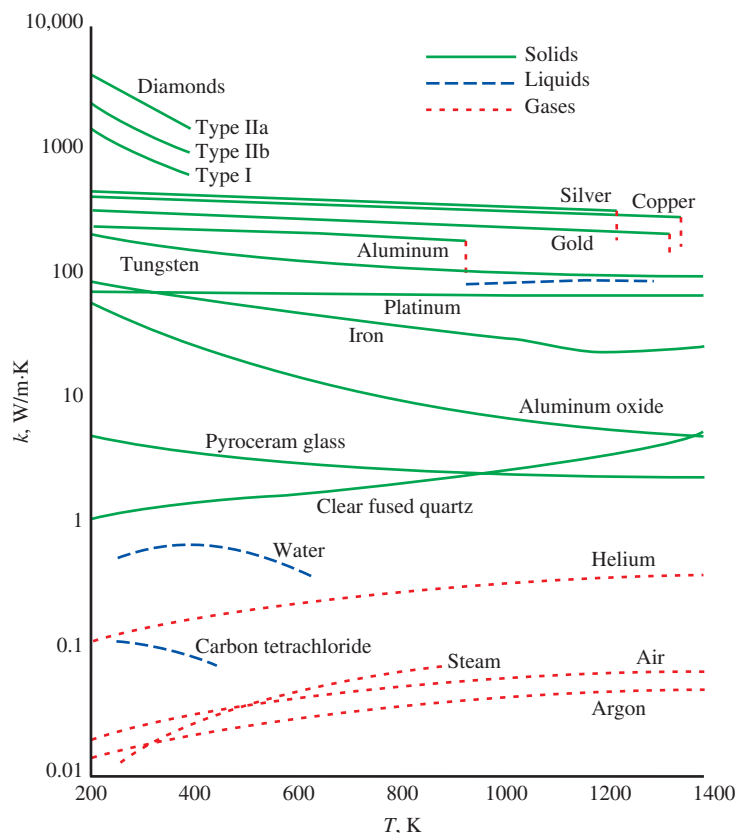
TABLE 1-3

Thermal conductivities of materials vary with temperature

T, K	$k, W/m \cdot K$	
	Copper	Aluminum
100	482	302
200	413	237
300	401	237
400	393	240
600	379	231
800	366	218

FIGURE 1-31

The variation of the thermal conductivity of various solids, liquids, and gases with temperature.



relatively large. In some of the emerging areas of technology, such as micro-electronics, the physical dimensions are typically in the order of micro or nanometers. For these applications the small physical dimensions most likely will influence the value of thermal conductivity in the solid and liquid states. In these situations, as the physical dimensions decrease, the average net distance traveled by the energy carriers typically decreases, and this reduces the value of thermal conductivity.

The temperature dependence of thermal conductivity causes considerable complexity in conduction analysis. Therefore, it is common practice to evaluate the thermal conductivity k at the *average temperature* and treat it as a *constant* in calculations.

In heat transfer analysis, a material is normally assumed to be *isotropic*; that is, to have uniform properties in all directions. This assumption is realistic for most materials, except those that exhibit different structural characteristics in different directions, such as laminated composite materials and wood. The thermal conductivity of wood across the grain, for example, is different than that parallel to the grain.

Thermal Diffusivity

The product ρc_p , which is often encountered in heat transfer analysis, is called the **heat capacity** of a material. Both the specific heat c_p and the heat capacity ρc_p represent the heat storage capability of a material. But c_p expresses it *per*

unit mass, whereas ρc_p expresses it *per unit volume*, as can be noticed from their units J/kg·K and J/m³·K, respectively.

Another material property that appears in the transient heat conduction analysis is the **thermal diffusivity**, which represents how fast heat diffuses through a material and is defined as

$$\alpha = \frac{\text{Heat conduction}}{\text{Heat storage}} = \frac{k}{\rho c_p} \quad (\text{m}^2/\text{s}) \quad (1-23)$$

Note that the thermal conductivity k represents how well a material conducts heat, and the heat capacity ρc_p represents how much energy a material stores per unit volume. Therefore, the thermal diffusivity of a material can be viewed as the ratio of the *heat conducted* through the material to the *heat stored* per unit volume. A material that has a high thermal conductivity or a low heat capacity will obviously have a large thermal diffusivity. The larger the thermal diffusivity, the faster the propagation of heat into the medium. A small value of thermal diffusivity means that heat is mostly absorbed by the material and a small amount of heat is conducted further.

The thermal diffusivities of some common materials at 20°C are given in Table 1–4. Note that the thermal diffusivity ranges from $\alpha = 0.14 \times 10^{-6} \text{ m}^2/\text{s}$ for water to $149 \times 10^{-6} \text{ m}^2/\text{s}$ for silver, which is a difference of more than a thousand times. Also note that the thermal diffusivities of beef and water are the same. This is not surprising, since meat as well as fresh vegetables and fruits are mostly water, and thus they possess the thermal properties of water.

TABLE 1–4

The thermal diffusivities of some materials at room temperature

Material	$\alpha, \text{m}^2/\text{s}^*$
Silver	149×10^{-6}
Gold	127×10^{-6}
Copper	113×10^{-6}
Aluminum	97.5×10^{-6}
Iron	22.8×10^{-6}
Mercury (l)	4.7×10^{-6}
Marble	1.2×10^{-6}
Ice	1.2×10^{-6}
Concrete	0.75×10^{-6}
Brick	0.52×10^{-6}
Heavy soil (dry)	0.52×10^{-6}
Glass	0.34×10^{-6}
Glass wool	0.23×10^{-6}
Water (l)	0.14×10^{-6}
Beef	0.14×10^{-6}
Wood (oak)	0.13×10^{-6}

*Multiply by 10.76 to convert to ft²/s.

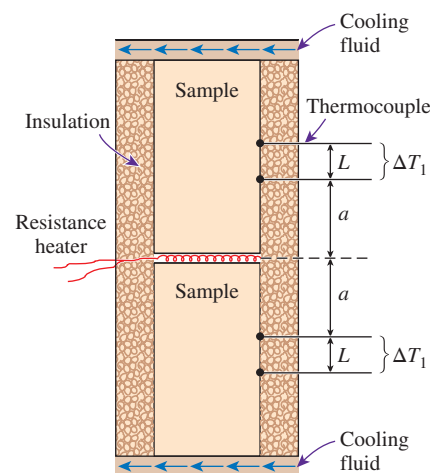
EXAMPLE 1–6 Measuring the Thermal Conductivity of a Material

A common way of measuring the thermal conductivity of a material is to sandwich an electric thermofoil heater between two identical samples of the material, as shown in Fig. 1–32. The thickness of the resistance heater, including its cover, which is made of thin silicon rubber, is usually less than 0.5 mm. A circulating fluid such as tap water keeps the exposed ends of the samples at constant temperature. The lateral surfaces of the samples are well insulated to ensure that heat transfer through the samples is one-dimensional. Two thermocouples are embedded into each sample some distance L apart, and a differential thermometer reads the temperature drop ΔT across this distance along each sample. When steady operating conditions are reached, the total rate of heat transfer through both samples becomes equal to the electric power drawn by the heater.

In a certain experiment, cylindrical samples of diameter 5 cm and length 10 cm are used. The two thermocouples in each sample are placed 3 cm apart. After initial transients, the electric heater is observed to draw 0.4 A at 110 V, and both differential thermometers read a temperature difference of 15°C. Determine the thermal conductivity of the sample.

SOLUTION The thermal conductivity of a material is to be determined by ensuring one-dimensional heat conduction and by measuring temperatures when steady operating conditions are reached.

Assumptions 1 Steady operating conditions exist since the temperature readings do not change with time. 2 Heat losses through the lateral surfaces of the apparatus are negligible since those surfaces are well insulated, and thus all heat generated by the heater is conducted through the samples. 3 The apparatus possesses thermal symmetry.

**FIGURE 1–32**

Apparatus to measure the thermal conductivity of a material using two identical samples and a thin resistance heater (Example 1–6).

Analysis The electrical power consumed by the resistance heater and converted to heat is

$$\dot{W}_e = VI = (110 \text{ V})(0.4 \text{ A}) = 44 \text{ W}$$

The rate of heat flow through each sample is

$$\dot{Q} = \frac{1}{2} \dot{W}_e = \frac{1}{2} \times (44 \text{ W}) = 22 \text{ W}$$

since only half of the heat generated flows through each sample because of symmetry. Reading the same temperature difference across the same distance in each sample also confirms that the apparatus possesses thermal symmetry. The heat transfer area is the area normal to the direction of heat transfer, which is the cross-sectional area of the cylinder in this case:

$$A = \frac{1}{4} \pi D^2 = \frac{1}{4} \pi (0.05 \text{ m})^2 = 0.001963 \text{ m}^2$$

Noting that the temperature drops by 15°C within 3 cm in the direction of heat flow, the thermal conductivity of the sample is determined to be

$$\dot{Q} = kA \frac{\Delta T}{L} \rightarrow k = \frac{\dot{Q}L}{A\Delta T} = \frac{(22 \text{ W})(0.03 \text{ m})}{(0.001963 \text{ m}^2)(15^\circ\text{C})} = \mathbf{22.4 \text{ W/m}\cdot\text{K}}$$

Discussion Perhaps you are wondering if we really need to use two samples in the apparatus, since the measurements on the second sample do not give any additional information. It seems like we can replace the second sample with insulation. Indeed, we do not need the second sample; however, it enables us to verify the temperature measurements on the first sample and provides thermal symmetry, which reduces experimental error.

EXAMPLE 1-7 Conversion Between SI and English Units

An engineer who is working on the heat transfer analysis of a brick building in English units needs the thermal conductivity of brick. But the only value he can find from his handbooks is 0.72 W/m·°C, which is in SI units. To make matters worse, the engineer does not have a direct conversion factor between the two unit systems for thermal conductivity. Can you help him out?

SOLUTION The situation this engineer is facing is not unique, and most engineers often find themselves in a similar position. When converting units, one must be careful to avoid some common pitfalls that lead to costly mistakes. Although unit conversion is a simple process, it requires care and careful reasoning.

The conversion factors for W and m are straightforward and are given in conversion tables to be

$$\begin{aligned} 1 \text{ W} &= 3.41214 \text{ Btu/h} \\ 1 \text{ m} &= 3.2808 \text{ ft} \end{aligned}$$

But the conversion of °C into °F is not so simple, and it can be a source of error if one is not careful. Perhaps the first thought that comes to mind is to replace °C by (°F – 32)/1.8 since $T(^{\circ}\text{C}) = [T(^{\circ}\text{F}) - 32]/1.8$. But this will be wrong since the °C in the unit W/m·°C represents per °C change in temperature. Noting that 1°C change in temperature corresponds to 1.8°F, so the proper conversion factor is

$$1^{\circ}\text{C} = 1.8^{\circ}\text{F}$$

Substituting, we get

$$1 \text{ W/m} \cdot ^\circ\text{C} = \frac{3.41214 \text{ Btu/h}}{(3.2808 \text{ ft})(1.8^\circ\text{F})} = 0.5778 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}$$

which is the desired conversion factor. Therefore, the thermal conductivity of the brick in English units is

$$\begin{aligned} k_{\text{brick}} &= 0.72 \text{ W/m} \cdot ^\circ\text{C} \\ &= 0.72 \times (0.5778 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}) \\ &= \mathbf{0.42 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}} \end{aligned}$$

Discussion Note that the thermal conductivity value of a material in English units is about half that in SI units (Fig. 1–33). Also note that we rounded the result to two significant digits (the same number in the original value) since expressing the result in more significant digits (such as 0.4160 instead of 0.42) would falsely imply a more accurate value than the original one.

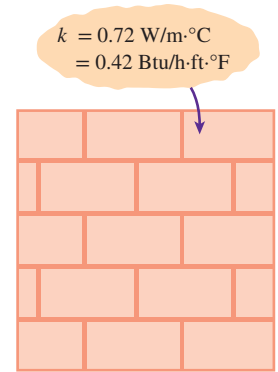


FIGURE 1–33

The thermal conductivity value in English units is obtained by multiplying the value in SI units by 0.5778.

1–7 ■ CONVECTION

Convection is the mode of energy transfer between a solid surface and the adjacent liquid or gas that is in motion, and it involves the combined effects of *conduction* and *fluid motion*. The faster the fluid motion, the greater the convection heat transfer. In the absence of any bulk fluid motion, heat transfer between a solid surface and the adjacent fluid is by pure conduction. The presence of bulk motion of the fluid enhances the heat transfer between the solid surface and the fluid, but it also complicates the determination of heat transfer rates.

Consider the cooling of a hot block by blowing cool air over its top surface (Fig. 1–34). Heat is first transferred to the air layer adjacent to the block by conduction. This heat is then carried away from the surface by convection, that is, by the combined effects of conduction within the air that is due to the random motion of air molecules and the bulk or macroscopic motion of the air that removes the heated air near the surface and replaces it with the cooler air.

Convection is called **forced convection** if the fluid is forced to flow over the surface by external means such as a fan, pump, or the wind. In contrast, convection is called **natural (or free) convection** if the fluid motion is caused by buoyancy forces that are induced by density differences due to the variation of temperature in the fluid (Fig. 1–35). For example, in the absence of a fan, heat transfer from the surface of the hot block in Fig. 1–34 is by natural convection since any motion in the air in this case is due to the rise of the warmer (and thus lighter) air near the surface and the fall of the cooler (and thus heavier) air to fill its place. Heat transfer between the block and the surrounding air is by conduction if the temperature difference between the air and the block is not large enough to overcome the resistance of air to movement and thus to initiate natural convection currents.

Heat transfer processes that involve *change of phase* of a fluid are also considered to be convection because of the fluid motion induced during the process, such as the rise of the vapor bubbles during boiling or the fall of the liquid droplets during condensation.

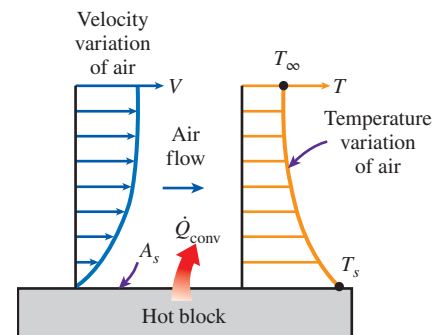


FIGURE 1–34

Heat transfer from a hot surface to air by convection.

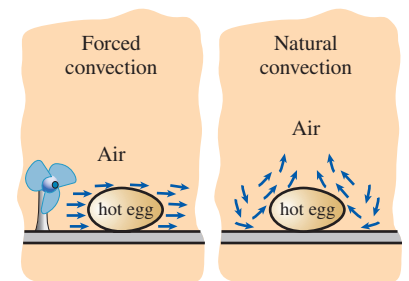


FIGURE 1–35

The cooling of a boiled egg by forced and natural convection.

**FIGURE 1–36**

Sir Isaac Newton (1642–1727) was an English mathematician, physicist, and astronomer, born in Lincolnshire, England. Newton is regarded as one of the greatest scientists and mathematicians in history. His contributions to mathematics include the development of the binomial theorem, the differential calculus, and the integral calculus. He is said to have conceived the idea of the law of gravity upon observing the fall of an apple in 1665. With the three fundamental laws that bear his name and are described in *Philosophiae Naturalis Principia Mathematica*, Newton is known as the father of classical mechanics. Newton showed that each of Kepler's three laws on the motion of planets and stars could be derived from the single law of gravity. Newton is also credited for the discovery of the composite nature of white light and the separation of different colors by a prism. The law of cooling that governs the rate of heat transfer from a hot surface to a cooler surrounding fluid is also attributed to Newton.

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Despite the complexity of convection, the rate of *convection heat transfer* is observed to be proportional to the temperature difference, and it is conveniently expressed by **Newton's law of cooling** as (Fig. 1–36)

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) \quad (\text{W}) \quad (1-24)$$

where h is the *convection heat transfer coefficient* in $\text{W/m}^2\cdot\text{K}$ or $\text{Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$, A_s is the surface area through which convection heat transfer takes place, T_s is the surface temperature, and T_∞ is the temperature of the fluid sufficiently far from the surface. Note that at the surface, the fluid temperature equals the surface temperature of the solid.

The convection heat transfer coefficient h is not a property of the fluid. It is an experimentally determined parameter whose value depends on all the variables influencing convection such as the surface geometry, the nature of fluid motion, the properties of the fluid, and the bulk fluid velocity. Typical values of h are given in Table 1–5.

Some people do not consider convection to be a fundamental mechanism of heat transfer since it is essentially heat conduction in the presence of fluid motion. But we still need to give this combined phenomenon a name, unless we are willing to keep referring to it as “conduction with fluid motion.” Thus, it is practical to recognize convection as a separate heat transfer mechanism despite the valid arguments to the contrary.

EXAMPLE 1–8 Measuring Convection Heat Transfer Coefficient

A 2-m-long, 0.3-cm-diameter electrical wire extends across a room at 15°C , as shown in Fig. 1–37. Heat is generated in the wire as a result of resistance heating, and the surface temperature of the wire is measured to be 152°C in steady operation. Also, the voltage drop and electric current through the wire are measured to be 60 V and 1.5 A, respectively. Disregarding any heat transfer by radiation, determine the convection heat transfer coefficient for heat transfer between the outer surface of the wire and the air in the room.

SOLUTION The convection heat transfer coefficient for heat transfer from an electrically heated wire to air is to be determined by measuring temperatures when steady operating conditions are reached and the electric power consumed.

Assumptions 1 Steady operating conditions exist since the temperature readings do not change with time. 2 Radiation heat transfer is negligible.

Analysis When steady operating conditions are reached, the rate of heat loss from the wire equals the rate of heat generation in the wire as a result of resistance heating. That is,

$$\dot{Q} = \dot{E}_{\text{generated}} = VI = (60 \text{ V})(1.5 \text{ A}) = 90 \text{ W}$$

The surface area of the wire is

$$A_s = \pi DL = \pi(0.003 \text{ m})(2 \text{ m}) = 0.01885 \text{ m}^2$$

Newton's law of cooling for convection heat transfer is expressed as

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty)$$

Disregarding any heat transfer by radiation and thus assuming all the heat loss from the wire occurs by convection, the convection heat transfer coefficient is determined to be

$$h = \frac{\dot{Q}_{\text{conv}}}{A_s(T_s - T_\infty)} = \frac{90 \text{ W}}{(0.01885 \text{ m}^2)(152 - 15)^\circ\text{C}} = 34.9 \text{ W/m}^2\cdot\text{K}$$

Discussion Note that the simple setup described above can be used to determine the average heat transfer coefficients from a variety of surfaces in air. Also, heat transfer by radiation can be eliminated by keeping the surrounding surfaces at the temperature of the wire.

1-8 ■ RADIATION

Radiation is the energy emitted by matter in the form of *electromagnetic waves* (or *photons*) as a result of the changes in the electronic configurations of the atoms or molecules. Unlike conduction and convection, the transfer of heat by radiation does not require the presence of an *intervening medium*. In fact, heat transfer by radiation is fastest (it occurs at the speed of light), and it suffers no attenuation in a vacuum. This is how the energy of the sun reaches the earth.

In heat transfer studies we are interested in *thermal radiation*, which is the form of radiation emitted by bodies because of their temperature. It differs from other forms of electromagnetic radiation such as X-rays, gamma rays, microwaves, radio waves, and television waves that are not related to temperature. All bodies at a temperature above absolute zero emit thermal radiation.

Radiation is a *volumetric phenomenon*, and all solids, liquids, and gases emit, absorb, or transmit radiation to varying degrees. However, radiation is usually considered to be a *surface phenomenon* for solids that are opaque to thermal radiation such as metals, wood, and rocks since the radiation emitted by the interior regions of such material can never reach the surface, and the radiation incident on such bodies is usually absorbed within a few microns from the surface.

The maximum rate of radiation that can be emitted from a surface at a thermodynamic temperature T_s (in K or R) is given by the **Stefan–Boltzmann law** as

$$\dot{Q}_{\text{emit, max}} = \sigma A_s T_s^4 \quad (\text{W}) \quad (1-25)$$

where $\sigma = 5.670 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4$ or $0.1714 \times 10^{-8} \text{ Btu/h}\cdot\text{ft}^2\cdot\text{R}^4$ is the *Stefan–Boltzmann constant*. The idealized surface that emits radiation at this maximum rate is called a **blackbody**, and the radiation emitted by a blackbody is called **blackbody radiation** (Fig. 1–38). The radiation emitted by all real surfaces is less than the radiation emitted by a blackbody at the same temperature and is expressed as

$$\dot{Q}_{\text{emit}} = \epsilon \sigma A_s T_s^4 \quad (\text{W}) \quad (1-26)$$

where ϵ is the **emissivity** of the surface. The property emissivity, whose value is in the range $0 \leq \epsilon \leq 1$, is a measure of how closely a surface approximates a blackbody for which $\epsilon = 1$. The emissivities of some surfaces are given in Table 1–6.

TABLE 1–5

Typical values of convection heat transfer coefficient

Type of Convection	h , W/m ² ·K*
Free convection of gases	2–25
Free convection of liquids	10–1000
Forced convection of gases	25–250
Forced convection of liquids	50–20,000
Boiling and condensation	2500–100,000

*Multiply by 0.176 to convert to Btu/h·ft²·°F.

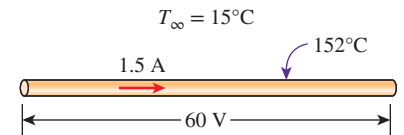


FIGURE 1–37
Schematic for Example 1–8.

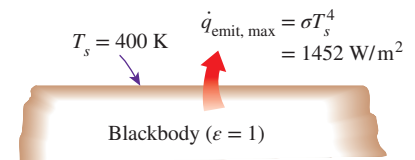


FIGURE 1–38
Blackbody radiation represents the *maximum amount of radiation that can be emitted from a surface at a specified temperature.*

TABLE 1–6

Emissivities of some materials at 300 K

Material	Emissivity
Aluminum foil	0.07
Anodized aluminum	0.82
Polished copper	0.03
Polished gold	0.03
Polished silver	0.02
Polished stainless steel	0.17
Black paint	0.98
White paint	0.90
White paper	0.92–0.97
Asphalt pavement	0.85–0.93
Red brick	0.93–0.96
Human skin	0.95
Wood	0.82–0.92
Soil	0.93–0.96
Water	0.96
Vegetation	0.92–0.96

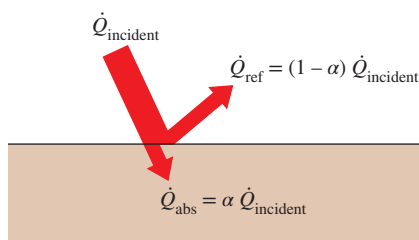


FIGURE 1–39

The absorption of radiation incident on an opaque surface of absorptivity α .

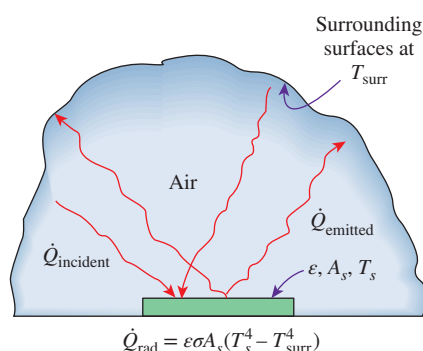


FIGURE 1–40

Radiation heat transfer between a surface and the surfaces surrounding it.

Another important radiation property of a surface is its **absorptivity** α , which is the fraction of the radiation energy incident on a surface that is absorbed by the surface. Like emissivity, its value is in the range $0 \leq \alpha \leq 1$. A blackbody absorbs all radiation incident on it. That is, a blackbody is a perfect absorber ($\alpha = 1$) as well as a perfect emitter.

In general, both ϵ and α of a surface depend on the temperature and the wavelength of the radiation. **Kirchhoff's law** of radiation states that the emissivity and the absorptivity of a surface at a given temperature and wavelength are equal. In many practical applications, the surface temperature and the temperature of the source of incident radiation are of the same order of magnitude, and the average absorptivity of a surface is taken to be equal to its average emissivity. The rate at which a surface absorbs radiation is determined from (Fig. 1–39)

$$\dot{Q}_{\text{absorbed}} = \alpha \dot{Q}_{\text{incident}} \quad (\text{W}) \quad (1-27)$$

where $\dot{Q}_{\text{incident}}$ is the rate at which radiation is incident on the surface and α is the absorptivity of the surface. For opaque (nontransparent) surfaces, the portion of incident radiation not absorbed by the surface is reflected back.

The difference between the rates of radiation emitted by the surface and the radiation absorbed is the *net* radiation heat transfer. If the rate of radiation absorption is greater than the rate of radiation emission, the surface is said to be *gaining* energy by radiation. Otherwise, the surface is said to be *losing* energy by radiation. In general, the determination of the net rate of heat transfer by radiation between two surfaces is a complicated matter since it depends on the properties of the surfaces, their orientation relative to each other, and the interaction of the medium between the surfaces with radiation.

When a surface of emissivity ϵ and surface area A_s at a *thermodynamic temperature* T_s is *completely enclosed* by a much larger (or black) surface at thermodynamic temperature T_{surr} separated by a gas (such as air) that does not intervene with radiation, the net rate of radiation heat transfer between these two surfaces is given by (Fig. 1–40)

$$\dot{Q}_{\text{rad}} = \epsilon \sigma A_s (T_s^4 - T_{\text{surr}}^4) \quad (\text{W}) \quad (1-28)$$

In this special case, the emissivity and the surface area of the surrounding surface do not have any effect on the net radiation heat transfer.

Radiation heat transfer to or from a surface surrounded by a gas such as air occurs *parallel* to conduction (or convection, if there is bulk gas motion) between the surface and the gas. Thus the total heat transfer is determined by *adding* the contributions of both heat transfer mechanisms. For simplicity and convenience, this is often done by defining a **combined heat transfer coefficient** h_{combined} that includes the effects of both convection and radiation. Then the *total* heat transfer rate to or from a surface by convection and radiation is expressed as

$$\begin{aligned} \dot{Q}_{\text{total}} &= \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = h_{\text{conv}} A_s (T_s - T_{\text{surr}}) + \epsilon \sigma A_s (T_s^4 - T_{\text{surr}}^4) \\ \dot{Q}_{\text{total}} &= h_{\text{combined}} A_s (T_s - T_{\text{surr}}) \quad (\text{W}) \\ h_{\text{combined}} &= h_{\text{conv}} + h_{\text{rad}} = h_{\text{conv}} + \epsilon \sigma (T_s + T_{\text{surr}})(T_s^2 + T_{\text{surr}}^2) \end{aligned} \quad (1-29)$$

Note that the combined heat transfer coefficient is essentially a convection heat transfer coefficient modified to include the effects of radiation.

Radiation is usually significant relative to conduction or natural convection, but negligible relative to forced convection. Thus radiation in forced convection applications is usually disregarded, especially when the surfaces involved have low emissivities and low to moderate temperatures.

EXAMPLE 1–9 Radiation Effect on Thermal Comfort

It is a common experience to feel chilly in winter and warm in summer in our homes even when the thermostat setting is kept the same. This is due to the so-called radiation effect resulting from radiation heat exchange between our bodies and the surrounding surfaces of the walls and the ceiling.

Consider a person standing in a room maintained at 22°C at all times. The inner surfaces of the walls, floors, and the ceiling of the house are at an average temperature of 10°C in winter and 25°C in summer. Determine the rate of radiation heat transfer between this person and the surrounding surfaces if the exposed surface area and the average outer surface temperature of the person are 1.4 m² and 30°C, respectively (Fig. 1–41).

SOLUTION The rates of radiation heat transfer between a person and the surrounding surfaces at specified temperatures are to be determined in summer and winter.

Assumptions 1 Steady operating conditions exist. 2 Heat transfer by convection is not considered. 3 The person is completely surrounded by the interior surfaces of the room. 4 The surrounding surfaces are at a uniform temperature.

Properties The emissivity of a person is $\varepsilon = 0.95$ (Table 1–6).

Analysis The net rates of radiation heat transfer from the body to the surrounding walls, ceiling, and floor in winter and summer are

$$\begin{aligned}\dot{Q}_{\text{rad, winter}} &= \varepsilon \sigma A_s (T_s^4 - T_{\text{surr, winter}}^4) \\ &= (0.95)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1.4 \text{ m}^2) \\ &\quad \times [(30 + 273)^4 - (10 + 273)^4] \text{ K}^4 \\ &= \mathbf{152 \text{ W}}\end{aligned}$$

and

$$\begin{aligned}\dot{Q}_{\text{rad, summer}} &= \varepsilon \sigma A_s (T_s^4 - T_{\text{surr, summer}}^4) \\ &= (0.95)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1.4 \text{ m}^2) \\ &\quad \times [(30 + 273)^4 - (25 + 273)^4] \text{ K}^4 \\ &= \mathbf{40.9 \text{ W}}\end{aligned}$$

Discussion Note that we must use *thermodynamic* (i.e., *absolute*) temperatures in radiation calculations. Also note that the rate of heat loss from the person by radiation is almost four times as large in winter as it is in summer, which explains the chill we feel in winter even if the thermostat setting is kept the same.

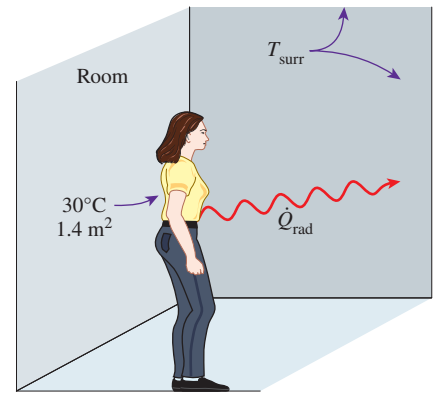
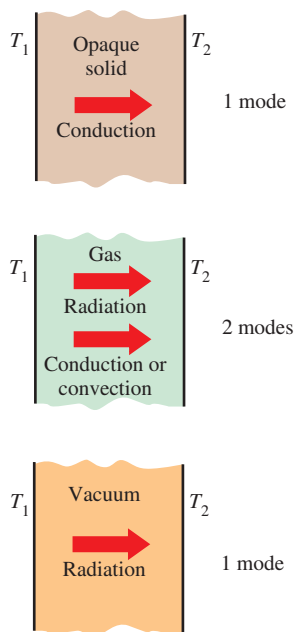


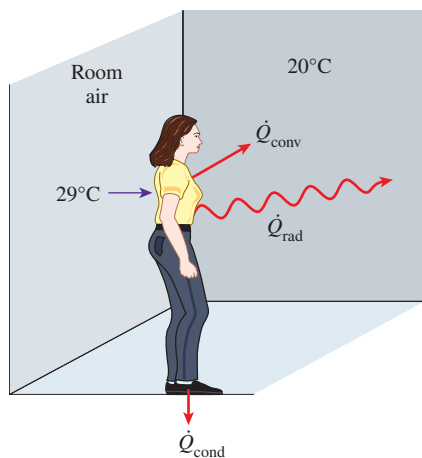
FIGURE 1–41
Schematic for Example 1–9.

1–9 ■ SIMULTANEOUS HEAT TRANSFER MECHANISMS

We mentioned that there are three mechanisms of heat transfer, but not all three can exist simultaneously in a medium. For example, heat transfer is only by conduction in *opaque solids*, but by conduction and radiation in *semitransparent solids*. Thus, a solid may involve conduction and radiation but not

**FIGURE 1-42**

Although there are three mechanisms of heat transfer, a medium may involve only two of them simultaneously.

**FIGURE 1-43**

Heat transfer from the person described in Example 1-10.

convection. However, a solid may involve heat transfer by convection and/or radiation on its surfaces exposed to a fluid or other surfaces. For example, the outer surfaces of a cold piece of rock will warm up in a warmer environment as a result of heat gain by convection (from the air) and radiation (from the sun or the warmer surrounding surfaces). But the inner parts of the rock will warm up as this heat is transferred to the inner region of the rock by conduction.

Heat transfer is by conduction and possibly by radiation in a *still fluid* (no bulk fluid motion) and by convection and radiation in a *flowing fluid*. In the absence of radiation, heat transfer through a fluid is either by conduction or convection, depending on the presence of any bulk fluid motion. Convection can be viewed as combined conduction and fluid motion, and conduction in a fluid can be viewed as a special case of convection in the absence of any fluid motion (Fig. 1-42).

Thus, when we deal with heat transfer through a *fluid*, we have either *conduction* or *convection*, but not both. Also, gases are practically transparent to radiation, except that some gases are known to absorb radiation strongly at certain wavelengths. Ozone, for example, strongly absorbs ultraviolet radiation. But in most cases, a gas between two solid surfaces does not interfere with radiation, and it acts effectively as a vacuum. Liquids, on the other hand, are usually strong absorbers of radiation.

Finally, heat transfer through a *vacuum* is by radiation only since conduction or convection requires the presence of a material medium.

EXAMPLE 1-10 Heat Loss from a Person

Consider a person standing in a breezy room at 20°C. Determine the total rate of heat transfer from this person if the exposed surface area and the average outer surface temperature of the person are 1.6 m² and 29°C, respectively, and the convection heat transfer coefficient is 6 W/m²·K (Fig. 1-43).

SOLUTION The total rate of heat transfer from a person by both convection and radiation to the surrounding air and surfaces at specified temperatures is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The person is completely surrounded by the interior surfaces of the room. 3 The surrounding surfaces are at the same temperature as the air in the room. 4 Heat conduction to the floor through the feet is negligible.

Properties The emissivity of a person is $\epsilon = 0.95$ (Table 1-6).

Analysis The heat transfer between the person and the air in the room is by convection (instead of conduction) since it is conceivable that the air in the vicinity of the skin or clothing warms up and rises as a result of heat transfer from the body, initiating natural convection currents. It appears that the experimentally determined value for the rate of convection heat transfer in this case is 6 W per unit surface area (m²) per unit temperature difference (in K or °C) between the person and the air away from the person. Thus, the rate of convection heat transfer from the person to the air in the room is

$$\begin{aligned}\dot{Q}_{\text{conv}} &= hA_s(T_s - T_\infty) \\ &= (6 \text{ W/m}^2 \cdot \text{K})(1.6 \text{ m}^2)(29 - 20)^\circ\text{C} \\ &= 86.4 \text{ W}\end{aligned}$$

The person also loses heat by radiation to the surrounding wall surfaces. We take the temperature of the surfaces of the walls, ceiling, and floor to be equal to the air temperature in this case for simplicity, but we recognize that this does not need to be the case. These surfaces may be at a higher or lower temperature than the average temperature of the room air, depending on the outdoor conditions and the structure of the walls. Considering that air does not intervene with radiation and the person is completely enclosed by the surrounding surfaces, the net rate of radiation heat transfer from the person to the surrounding walls, ceiling, and floor is

$$\begin{aligned}\dot{Q}_{\text{rad}} &= \varepsilon \sigma A_s (T_s^4 - T_{\text{surr}}^4) \\ &= (0.95)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1.6 \text{ m}^2) \\ &\quad \times [(29 + 273)^4 - (20 + 273)^4] \text{ K}^4 \\ &= 81.7 \text{ W}\end{aligned}$$

Note that we must use *thermodynamic* temperatures in radiation calculations. Also note that we used the emissivity value for the skin and clothing at room temperature since the emissivity is not expected to change significantly at a slightly higher temperature.

Then the rate of total heat transfer from the body is determined by adding these two quantities:

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = (86.4 + 81.7) \text{ W} \cong \mathbf{168 \text{ W}}$$

Discussion The heat transfer would be much higher if the person were not dressed since the exposed surface temperature would be higher. Thus, an important function of the clothes is to serve as a barrier against heat transfer.

In these calculations, heat transfer through the feet to the floor by conduction, which is usually very small, is neglected. Heat transfer from the skin by perspiration, which is the dominant mode of heat transfer in hot environments, is not considered here.

Also, the units $\text{W/m}^2 \cdot ^\circ\text{C}$ and $\text{W/m}^2 \cdot \text{K}$ for heat transfer coefficient are equivalent and can be interchanged.

EXAMPLE 1–11 Heat Transfer Between Two Isothermal Plates

Consider steady heat transfer between two large parallel plates at constant temperatures of $T_1 = 300 \text{ K}$ and $T_2 = 200 \text{ K}$ that are $L = 1 \text{ cm}$ apart, as shown in Fig. 1–44. Assuming the surfaces to be black (emissivity $\varepsilon = 1$), determine the rate of heat transfer between the plates per unit surface area, assuming the gap between the plates is (a) filled with atmospheric air, (b) evacuated, (c) filled with urethane insulation, and (d) filled with superinsulation that has an apparent thermal conductivity of $0.00002 \text{ W/m} \cdot \text{K}$.

SOLUTION The total rate of heat transfer between two large parallel plates at specified temperatures is to be determined for four different cases.

Assumptions 1 Steady operating conditions exist. 2 There are no natural convection currents in the air between the plates. 3 The surfaces are black, and thus $\varepsilon = 1$.

Properties The thermal conductivity at the average temperature of 250 K is $k = 0.0219 \text{ W/m} \cdot \text{K}$ for air (Table A–15), $0.026 \text{ W/m} \cdot \text{K}$ for urethane insulation (Table A–6), and $0.00002 \text{ W/m} \cdot \text{K}$ for the superinsulation.

Analysis (a) The rates of conduction and radiation heat transfer between the plates through the air layer are

$$\dot{Q}_{\text{cond}} = kA \frac{T_1 - T_2}{L} = (0.0219 \text{ W/m} \cdot \text{K})(1 \text{ m}^2) \frac{(300 - 200) \text{ K}}{0.01 \text{ m}} = 219 \text{ W}$$

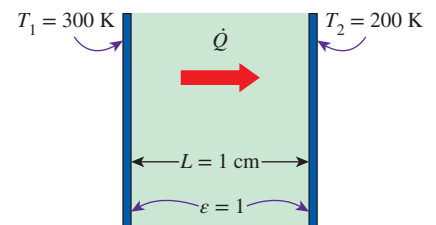


FIGURE 1–44
Schematic for Example 1–11.

and

$$\begin{aligned}\dot{Q}_{\text{rad}} &= \varepsilon \sigma A (T_1^4 - T_2^4) \\ &= (1)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(1 \text{ m}^2)[(300 \text{ K})^4 - (200 \text{ K})^4] = 369 \text{ W}\end{aligned}$$

Therefore,

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{cond}} + \dot{Q}_{\text{rad}} = 219 + 369 = \mathbf{588 \text{ W}}$$

The heat transfer rate in reality will be higher because of the natural convection currents that are likely to occur in the airspace between the plates.

(b) When the airspace between the plates is evacuated, there will be no conduction or convection, and the only heat transfer between the plates will be by radiation. Therefore,

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{rad}} = \mathbf{369 \text{ W}}$$

(c) An opaque solid material placed between two plates blocks direct radiation heat transfer between the plates. Also, the thermal conductivity of an insulating material accounts for the radiation heat transfer that may be occurring through the voids in the insulating material. The rate of heat transfer through the urethane insulation is

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{cond}} = kA \frac{T_1 - T_2}{L} = (0.026 \text{ W/m} \cdot \text{K})(1 \text{ m}^2) \frac{(300 - 200) \text{ K}}{0.01 \text{ m}} = \mathbf{260 \text{ W}}$$

Note that heat transfer through the urethane material is less than the heat transfer through the air determined in (a), although the thermal conductivity of the insulation is higher than that of air. This is because the insulation blocks the radiation, whereas air transmits it.

(d) The layers of the superinsulation prevent any direct radiation heat transfer between the plates. However, radiation heat transfer between the sheets of superinsulation does occur, and the apparent thermal conductivity of the superinsulation accounts for this effect. Therefore,

$$\dot{Q}_{\text{total}} = kA \frac{T_1 - T_2}{L} = (0.00002 \text{ W/m} \cdot \text{K})(1 \text{ m}^2) \frac{(300 - 200) \text{ K}}{0.01 \text{ m}} = \mathbf{0.2 \text{ W}}$$

which is $\frac{1}{1845}$ of the heat transfer through the vacuum. The results of this example are summarized in Fig. 1–45 to put them into perspective.

Discussion This example demonstrates the effectiveness of superinsulations and explains why they are the insulation of choice in critical applications despite their high cost.

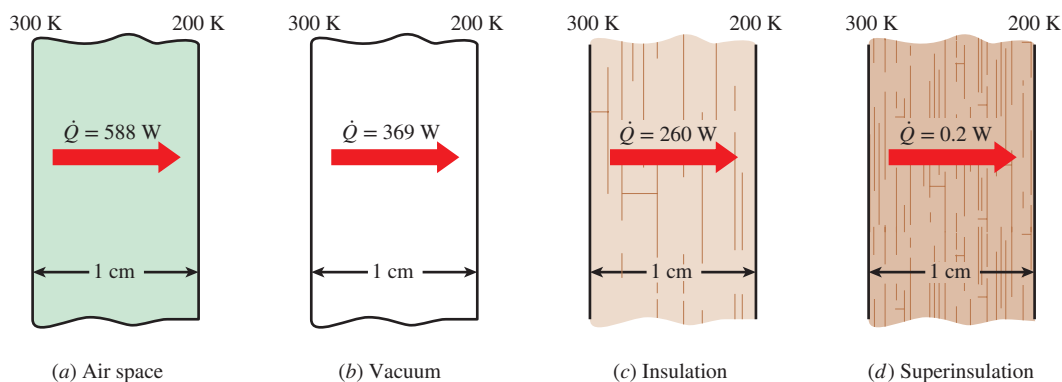


FIGURE 1–45

Different ways of reducing heat transfer between two isothermal plates, and their effectiveness.

EXAMPLE 1-12 Heat Transfer in Conventional and Microwave Ovens

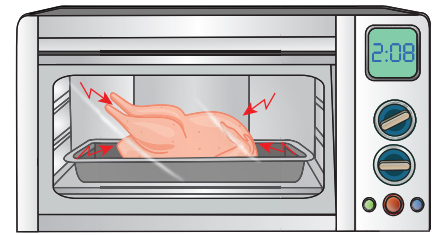
The fast and efficient cooking of microwave ovens has made them one of the essential appliances in modern kitchens (Fig. 1–46). Discuss the heat transfer mechanisms associated with the cooking of a chicken in microwave and conventional ovens, and explain why cooking in a microwave oven is more efficient.

SOLUTION Food is cooked in a microwave oven by absorbing the electromagnetic radiation energy generated by the microwave tube, called the magnetron. The radiation emitted by the magnetron is not thermal radiation since its emission is not due to the temperature of the magnetron; rather, it is due to the conversion of electrical energy into electromagnetic radiation at a specified wavelength. The wavelength of the microwave radiation is such that it is *reflected* by metal surfaces; *transmitted* by the cookware made of glass, ceramic, or plastic; and *absorbed* and converted to internal energy by food (especially the water, sugar, and fat) molecules.

In a microwave oven, the *radiation* that strikes the chicken is absorbed by the skin of the chicken and the outer parts. As a result, the temperature of the chicken at and near the skin rises. Heat is then *conducted* toward the inner parts of the chicken from its outer parts. Of course, some of the heat absorbed by the outer surface of the chicken is lost to the air in the oven by *convection*.

In a conventional oven, the air in the oven is first heated to the desired temperature by the electric or gas heating element. This preheating may take several minutes. The heat is then transferred from the air to the skin of the chicken by *natural convection* in older ovens or by *forced convection* in the newer convection ovens that utilize a fan. The air motion in convection ovens increases the convection heat transfer coefficient and thus decreases the cooking time. Heat is then *conducted* toward the inner parts of the chicken from its outer parts as in microwave ovens.

Microwave ovens replace the slow convection heat transfer process in conventional ovens by the instantaneous radiation heat transfer. As a result, microwave ovens transfer energy to the food at full capacity the moment they are turned on, and thus they cook faster while consuming less energy.

**FIGURE 1-46**

A chicken being cooked in a microwave oven (Example 1–12).

EXAMPLE 1-13 Heating of a Plate by Solar Energy

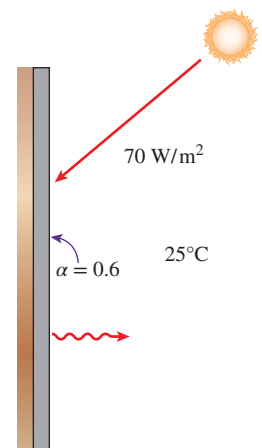
A thin metal plate is insulated on the back and exposed to solar radiation at the front surface (Fig. 1–47). The exposed surface of the plate has an absorptivity of 0.6 for solar radiation. If solar radiation is incident on the plate at a rate of 700 W/m^2 and the surrounding air temperature is 25°C , determine the surface temperature of the plate when the heat loss by convection and radiation equals the solar energy absorbed by the plate. Assume the combined convection and radiation heat transfer coefficient to be $50 \text{ W/m}^2\cdot\text{K}$.

SOLUTION The back side of the thin metal plate is insulated, and the front side is exposed to solar radiation. The surface temperature of the plate is to be determined when it stabilizes.

Assumptions 1 Steady operating conditions exist. 2 Heat transfer through the insulated side of the plate is negligible. 3 The heat transfer coefficient remains constant.

Properties The solar absorptivity of the plate is given to be $\alpha = 0.6$.

Analysis The absorptivity of the plate is 0.6, and thus 60 percent of the solar radiation incident on the plate is absorbed continuously. As a result, the temperature of the plate

**FIGURE 1-47**

Schematic for Example 1–13.

risks, and the temperature difference between the plate and the surroundings increases. This increasing temperature difference causes the rate of heat loss from the plate to the surroundings to increase. At some point, the rate of heat loss from the plate equals the rate of solar energy absorbed, and the temperature of the plate no longer changes. The temperature of the plate when steady operation is established is determined from

$$\dot{E}_{\text{gained}} = \dot{E}_{\text{lost}} \quad \text{or} \quad \alpha A_s \dot{q}_{\text{incident, solar}} = h_{\text{combined}} A_s (T_s - T_{\infty})$$

Solving for T_s and substituting, the plate surface temperature is determined to be

$$T_s = T_{\infty} + \alpha \frac{\dot{q}_{\text{incident, solar}}}{h_{\text{combined}}} = 25^{\circ}\text{C} + \frac{0.6 \times (700 \text{ W/m}^2)}{50 \text{ W/m}^2 \cdot \text{K}} = 33.4^{\circ}\text{C}$$

Discussion Note that the heat losses prevent the plate temperature from rising above 33.4°C . Also, the combined heat transfer coefficient accounts for the effects of both convection and radiation, and thus it is very convenient to use in heat transfer calculations when its value is known with reasonable accuracy.

1-10 ■ AEROGEL—A REMARKABLE SUPERINSULATING MATERIAL*

Aerogels—often called “frozen smoke”—are lightweight and have an ethereal blue haze. An aerogel in your hand feels like a piece of air—but it is a solid. What makes them so interesting is their unique nanoporous structure. This structure is formed through the sol-gel process, whereby a sol (a particulate-containing solution) undergoes a polymerization reaction to form a nanoporous wet gel. The key to making an aerogel is to carefully dry the wet gel. This can be difficult because, unless care is taken, capillary forces due to surface tension will cause the nanoporous structure to collapse as it dries. Supercritical extraction is commonly used to make aerogels. If the wet gel is heated and pressurized to a state above the critical point of the solvent that is in the wet gel, surface tension can be eliminated and the wet gel can be dried to form an aerogel. Aerogels can be made from many different precursor materials. The most common type of aerogel is the silica aerogel. Figure 1-48, a famous aerogel picture from the NASA Stardust site, demonstrates the insulating properties of silica aerogel.

When an aerogel is made by the sol-gel method, it forms a nanoporous structure comprised of clusters of silica spheres on the order of 5–10 nm that connect and form long, chain-like connecting structures with pores that range from 5 to 100 nm in size, as shown in Fig. 1-49. Because of this structure, silica aerogels are very porous (90–99 percent air) and have low density ($0.05\text{--}0.4 \text{ g/cm}^3$) and large surface areas ($500\text{--}1400 \text{ m}^2/\text{g}$). They are also highly translucent and have low thermal conductivity ($0.01\text{--}0.3 \text{ W/m}\cdot\text{K}$, which is lower than the thermal conductivity of most of the insulators shown in Fig. 1-29).

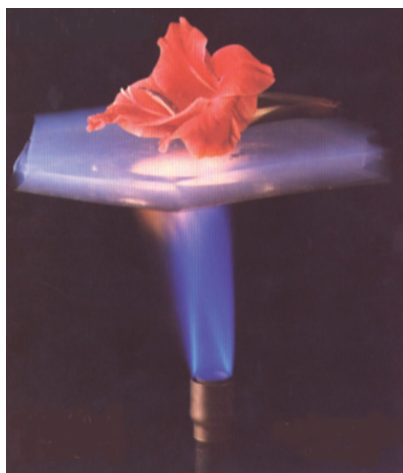
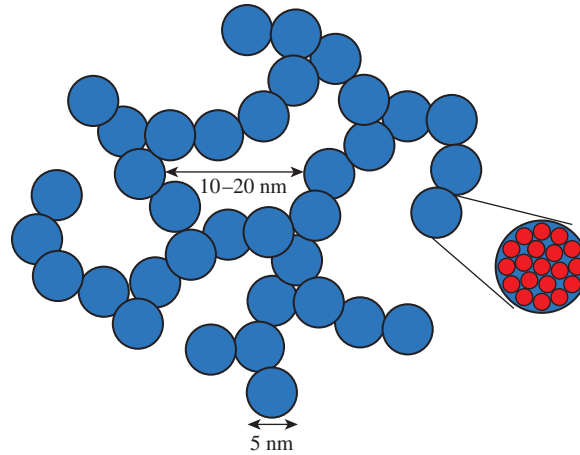


FIGURE 1-48

A silica aerogel insulates a flower from the heat of a Bunsen burner.

Source: JPL/NASA

*This section is contributed by Professor Ann M. Anderson, Mechanical Engineering Department, Union College, Schenectady, NY.

**FIGURE 1-49**

Schematic representation of an aerogel nanostructure.

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In general, heat is transferred by conduction, convection, and radiation. Convection through an aerogel is hindered by the small pore size, and at low temperatures radiation is not significant. Therefore, the primary method for heat transfer through an aerogel is by conduction in both the solid and gaseous phases. Solid phase conduction is related to the thermal conductivity of the solid and the amount of that solid. Since aerogels are so porous, there is very little solid material, and what is there follows a tortuous path with many dead ends (formed by the original sol-gel process). For these reasons, the thermal conductivity of an aerogel is very low.

Aerogels are commercially available in an impregnated blanket form, as granules, or as monolithic pieces. They are used in a variety of applications, including comet dust collectors, oil spill clean-up material, and chemical sensors. The most common application for aerogels is as insulation where, for example, they are used in daylighting products (insulating windows), high-performance clothing, and pipe insulation.

A monolithic aerogel under slight vacuum can have a thermal conductivity as low as 0.01 W/m·K. Because they are also translucent, aerogels can be used in daylighting and window applications. For example, consider a 6-mm-thick single-pane glass window ($k = 1.0$ W/m·K) that is 25 cm $\times$ 16 cm with an inside temperature of 30°C and an outside temperature of 10°C. The heat loss through the window can be calculated as:

$$\dot{Q} = kA \frac{\Delta T}{\Delta x} = \left(1.0 \frac{\text{W}}{\text{m} \cdot \text{K}} \right) (0.25 \text{ m} \times 0.16 \text{ m}) \frac{(30 - 10)^\circ\text{C}}{0.006 \text{ m}} = 133.3 \text{ W}$$

If we could make the window pane out of a monolithic aerogel under slight vacuum, the heat loss would be:

$$\dot{Q} = kA \frac{\Delta T}{\Delta x} = \left(0.01 \frac{\text{W}}{\text{m} \cdot \text{K}} \right) (0.25 \text{ m} \times 0.16 \text{ m}) \frac{(30 - 10)^\circ\text{C}}{0.006 \text{ m}} = 1.3 \text{ W}$$

The use of aerogel would reduce the heat loss by a factor of 100!

So why don't we have more aerogel windows? There are two major limiting factors to the more widespread use of aerogels: cost and fragility. Supercritical

extraction involves expensive equipment, which drives up the cost of aerogels. The unique nanoporous structure of the aerogel that gives it the remarkable insulating properties is also very fragile. Much of the current research in the aerogel community is focused on developing methods that reduce the cost of aerogels and make them stronger.

1-11 ■ PREVENTION THROUGH DESIGN*

The emphasis on safety is not unusual in the fields of engineering. It is not surprising that the first of the fundamental canons of ethics for engineers is to “hold paramount the safety, health, and welfare of the public” when fulfilling their professional duties (*Code of Ethics for Engineers*, 2007). In 2007, the National Institute for Occupational Safety and Health (NIOSH) launched the National Prevention through Design (PtD) initiative, with the mission to prevent or reduce work-related injuries, illnesses, and fatalities by including prevention considerations in all circumstances that affect people in their workplaces (*Prevention Through Design: Plan for the National Initiative*, 2010). PtD involves applying the means of reducing risks and preventing hazards in the design of equipment, tools, processes, and work facilities. We present example problems and end-of-chapter problems on PtD throughout this text not only to provide discussions of interesting real-world applications, but also to introduce the concepts of PtD to the minds of tomorrow’s engineers so that they may influence a change in culture toward more emphasis on safe designs.

The National PtD Initiative is largely being discussed in the context of preventing hazards in working environments. Nevertheless, the concepts of PtD can also be rationally applied to preventing failures and damages to devices, products, and systems. Failures and damages often have negative impacts on the environment, profitability, and the society at large. Within the context of heat and mass transfer, the PtD concepts can be presented along with the physical mechanisms involved and their practical applications. Issues such as prevention of thermal burn, fire hazard, and thermal failure in systems are topics that can relate the concepts of PtD with the basic science of heat and mass transfer. The process of solving heat and mass transfer problems, along with the application of PtD concepts, involves incorporating prescribed PtD criteria, be it the prevention of burn injury, fire hazard, or system failure, to the solutions (Fig. 1–50). To successfully arrive at a solution that satisfies prescribed PtD criteria requires understanding how the physical mechanisms of heat and mass transfer interrelate with the concepts of PtD.

Thermal burn occurs when skin tissues are exposed to temperatures that are higher than their physiological temperature. When this happens, the protein and cellular structures of skin tissues are altered as the higher temperatures cause thermal shocks and disrupt the skin cells. The degree of injury as a result of thermal burn is influenced by the temperature to which the

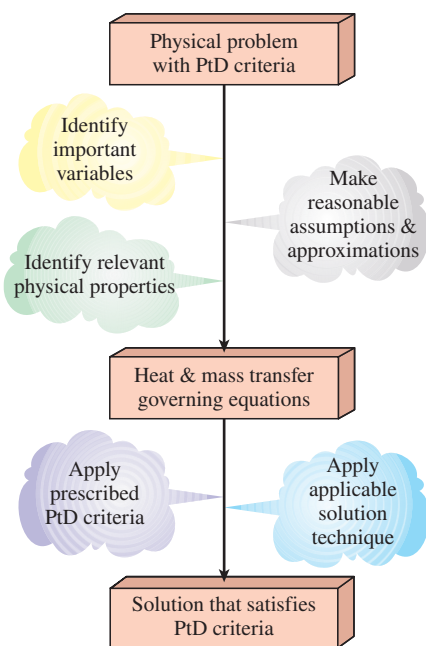


FIGURE 1–50

Process of solving problems with application of PtD concepts.

*This section is contributed by Professor Clement C. Tang, Mechanical Engineering Department, University of North Dakota, Grand Forks, ND.

skin is exposed and the duration of exposure. The skin reaction to thermal burn is dependent upon the heat transfer rate from the hot medium to the skin. When exposed to temperatures above approximately 43°C, the human skin can begin to feel discomfort and pain; damage can occur if the exposure is sufficiently long. Exposure to a temperature of about 48°C can cause human skin to receive a first-degree burn injury, and temperatures above 70°C can cause instantaneous damage to human skin (ASTM Standard C1055-03, 2009).

Injury from thermal burn can result from all three modes of heat transfer: conduction, convection, and radiation. Thermal burn by conduction heat transfer occurs when the skin is in contact with a hot solid surface. The thermal properties, roughness, and temperature of the hot solid surface, along with the contact duration and the nature of the skin, all influence the degree of this thermal burn injury. Scalding of skin from hot fluid is a thermal burn caused by convection heat transfer to the skin. Scalding on skin tissue can occur through spilling of hot liquid or accidental immersion into hot liquid. Thermal burn injuries from convection heat transfer of hot gas are not only external but also can be internal, affecting the upper respiratory tract and even the lungs. Thermal burn by radiation can be caused from laser or other sources such as nuclear explosion. Although human skin can reflect much of the incident energy from radiation, exposure to high-energy laser beams can cause thermal burn on skin. The severity of thermal burn from laser is influenced by the range of wavelength that allows the laser beam to penetrate into the skin tissue and the duration of exposure.

In the prevention of fire hazard, the understanding of flash point is important. Flash point is the lowest temperature at which a liquid can vaporize and form a mixture with air to become ignitable. When a liquid reaches its flash point, a source of ignition (e.g., open flame, spark, static electricity, and hot object) will cause the liquid to ignite. Therefore, designs for fire hazard prevention must be in place when storing, handling, or transporting a liquid above its flash point so that it is not exposed to an ignition source. The lower the flash point of a liquid, the more susceptible it is to a fire hazard. Flash point is also used for characterizing liquids into fire hazard categories, such as combustible and flammable liquids. The National Fire Protection Association (NFPA) has classified a liquid with a flash point of 37.8°C (100°F) or higher as a combustible liquid. A liquid with a flash point below 37.8°C is classified as flammable liquid. Another concept important to fire hazard prevention is the autoignition temperature (AIT), which is the minimum temperature at which a substance will spontaneously ignite without an ignition source. A substance is autoignited when it is heated above a temperature at which the rate of heat release from the exothermic oxidation reaction becomes significantly greater than the rate of heat lost to the surroundings. Factors influencing the autoignition temperature include atmospheric pressure, humidity, and oxygen concentration.

The science of heat and mass transfer can be coupled with the concepts of PtD to mitigate the risks of thermal failure in systems. Thermal stress can compromise the integrity of parts and components in a system. Extreme temperature can alter the physical properties of a material, which can cause a component to lose its functionality. Cold temperature on the morning of January 28, 1986,

affected the elasticity of the O-ring on a solid rocket booster of the space shuttle *Challenger*. The loss of the O-ring's elasticity and ability to seal allowed hot combustion gas to leak through a solid rocket booster, which led to the tragic disaster.

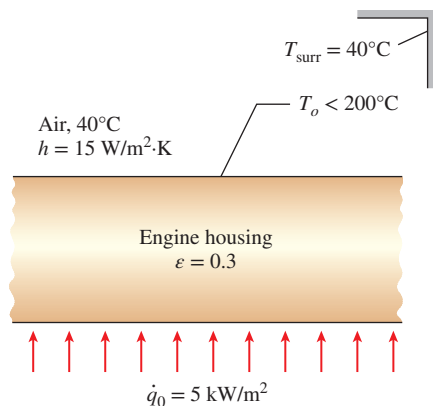


FIGURE 1–51

Schematic for Example 1–14.

PtD

EXAMPLE 1–14

Fire Hazard Prevention of Oil Leakage on Hot Engine Surface

Oil leakage and spillage on hot engine surfaces can create fire hazards. Some engine oils have an autoignition temperature above 250°C . When oil leakage comes in contact with a hot engine surface that has a higher temperature than its autoignition temperature, the oil can ignite spontaneously. Consider the outer surface of an engine situated in a place where there is a possibility of being in contact with oil leakage. The engine surface has an emissivity of 0.3, and when it is in operation, its inner surface is subjected to 5 kW/m^2 of heat flux (Fig. 1–51). The engine is in an environment where the ambient air and surrounding temperature is 40°C , while the convection heat transfer coefficient is $15 \text{ W/m}^2\cdot\text{K}$. To prevent a fire hazard, the temperature of the engine surface should be kept below 200°C . Determine whether oil on the engine surface is at a risk of autoignition. If there is a risk of autoignition, discuss a possible prevention measure that can be implemented.

SOLUTION In this example, the concepts of PtD are applied with the basic understanding of simultaneous heat transfer mechanisms via convection and radiation. The inner surface of an engine is subject to a heat flux of 5 kW/m^2 . We must determine whether the engine surface temperature is below 200°C to prevent spontaneous ignition in the event of oil leakage onto the engine surface.

Assumptions 1 Steady operating conditions exist. 2 The surrounding surfaces are at the same temperature as the ambient air. 3 Heat conduction through the engine housing is one-dimensional. 4 The engine inner surface is subjected to uniform heat flux.

Properties Emissivity of the engine surface is given as $\varepsilon = 0.3$.

Analysis When in operation, the inner surface of the engine is subjected to a uniform heat flux, which is equal to the sum of heat fluxes transferred by convection and radiation on the outer surface. Therefore,

$$\begin{aligned}\dot{q}_0 &= h(T_o - T_\infty) + \varepsilon\sigma(T_o^4 - T_{\text{surr}}^4) \\ 5000 \text{ W/m}^2 &= (15 \text{ W/m}^2\cdot\text{K}) [T_o - (40 + 273)]\text{K} \\ &\quad + (0.3)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[T_o^4 - (40 + 273)^4]\text{K}^4\end{aligned}$$

Using appropriate software and solving implicitly for the engine outer surface temperature (T_o), the result is $T_o = 552 \text{ K} = 279^\circ\text{C}$.

Discussion The solution reveals that the engine outer surface temperature is greater than 200°C , which presents the risk of autoignition in the event of oil leakage onto the engine outer surface. To mitigate the risk of fire hazards, the outer surface of the engine should be insulated. In practice, the engine surface temperature is not uniform; instead, high local surface temperatures result in hot spots on the engine surface. Engine housings generally come in irregular shapes, thus making the prediction of hot spots on the engine surface difficult. However, using handheld infrared thermometers, engine operators can quickly identify the areas that are prone to hot spots and take proper prevention measures.

1-12 ■ ENGINEERING CODES AND STANDARDS*

Engineering codes and standards are ubiquitous. When we consider any given device that we use daily, the path, from the design to the prototype and finally to the production of the final product is guided by standards and regulated by codes. A product that has been manufactured to meet certain standards tells the user about the product's quality and that it has met certain levels of scrutiny set forth by those standards. In essence, engineering standards communicate to the user that the product is expected to function safely and meets quality objectives. At the industrial level, engineering codes and standards provide efficiency in engineering processes (from design to manufacturing), maintain safety and uniformity of products that allows interchangeability, and improve commerce. Engineering codes and standards are evolutionary—they are continually revised to reflect technical advances, such as new technologies, new designs, new procedures, and new materials.

The development of engineering codes and standards is rooted in the engineers' code of ethics to “hold paramount the safety, health, and welfare of the public” in their professions (*Code of Ethics for Engineers*, 2007). The founding of the American Society of Mechanical Engineers (ASME) in 1880 provided a forum for engineers, designers, and machine shop managers to discuss and collaborate to solve issues brought on by the industrial age. Within four years, the first ASME standard, “Code for the Conduct of Trials of Steam Boilers,” was issued. This was the first ASME Performance Test Code (PTC). Today, there are close to 50 ASME PTCs, and they cover components, systems, and facilities. Performance test codes are used by manufacturers to ensure the reliability of their products, and users can use the PTCs to protect themselves from products that function poorly.

Some of the very first engineering codes and standards were needed because of engineering failures that caused the loss of life and property. In 1905, a boiler exploded in a shoe factory in Brockton, Massachusetts. In the next year, a similar incident happened in another shoe factory in Lynn, Massachusetts. Within a decade, the first edition of the Boiler and Pressure Vessel Code (BPVC) was issued by the ASME in 1914 and published in 1915. Today, the ASME BPVC has been adopted by the United States and Canada, and it is being used in over 100 countries around the world (*ASME Boiler & Pressure Vessel Code*, 2015).

A recent study done by ASME Vision 2030 Project reveals that almost 50 percent of engineers in their early career are unfamiliar with engineering codes and standards (ASME Vision 2030 Project, n.d.). As companies and manufacturers expand operations globally, there is a greater need to harmonize codes and standards across jurisdictions and disciplines. The need for engineers to know about codes and standards is growing, and having this knowledge allows engineers to innovate and manufacture competitive products.

In the subject of heat and mass transfer, there are several engineering codes and standards that are relevant. These codes and standards have been issued and published by professional associations, such as the ASME and

*This section is contributed by Professor Clement C. Tang, Mechanical Engineering Department, University of North Dakota, Grand Forks, ND.

the ASHRAE, and standards organizations, such as ASTM International, the ANSI, and the ISO. For instance, the 2015 ASME Boiler & Pressure Vessel Code, Section IV (ASME BPVC.IV-2015) provided the permitted O-ring materials and their temperature limits for water tube attachments in hot water boilers (*ASME Boiler & Pressure Vessel Code*, 2015). Another example is the standard specification issued by ASTM International for polyethylene insulation used on electrical cable (ASTM D1351-14, 2014), which is important as overheating can rapidly deteriorate the insulation. Extreme temperatures and thermal cycles can cause materials to deteriorate and thereby affect the functionality of components. The knowledge of heat and mass transfer, along with adherence to the relevant codes and standards, allow engineers to analyze, design, and build components and systems to function within the design conditions. Throughout the text, as appropriate, example problems and end-of-chapter problems related to codes and standards will be presented to introduce this concept to tomorrow's engineers.

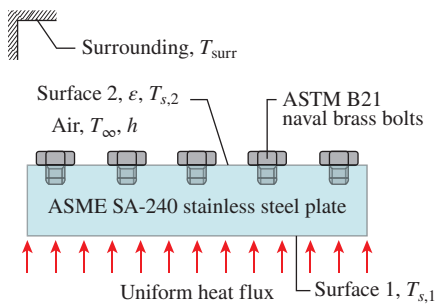


FIGURE 1-52

Schematic for Example 1-15.

C&S **EXAMPLE 1-15**

Compliance of ASME Codes for a Component Experiencing Simultaneous Heat Transfer Mechanisms

A component consists of a series of ASTM B21 naval brass bolts bolted on the upper surface of an ASME SA-240 stainless steel plate ($k = 15 \text{ W/m}\cdot\text{K}$) with a thickness of 10 cm (Fig. 1-52). One side of the stainless steel plate is subjected to a uniform heat flux of 2.2 kW/m^2 , while another side experiences convection and thermal radiation with ambient air at 30°C and surroundings at 40°C . The plate surface has an emissivity of 0.3, and the convection heat transfer coefficient is $5 \text{ W/m}^2\cdot\text{K}$. The maximum use temperature for the B21 bolts permitted by the ASME Code for Process Piping (ASME B31.3-2014) is 149°C . Also, the ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HF-300) limits the maximum use temperature for the SA-240 plate to 260°C . An engineer is tasked to ensure that both the B21 bolts and the SA-240 plate comply with the ASME codes. What would be the engineer's conclusion?

SOLUTION In this example, the engineering codes developed by the ASME are incorporated in a problem that involves simultaneous heat transfer mechanisms. To ensure compliance with the ASME Boiler and Pressure Vessel Code and the ASME Code for Process Piping, it becomes necessary to first determine the surface temperature of the plate on the side that is subjected to convection and thermal radiation. By inspection, one can conclude that the minimum temperature of the plate occurs at surface 2, while the maximum temperature occurs at surface 1.

Assumptions 1 Heat transfer is steady. 2 One-dimensional heat conduction through the metal plate. 3 Uniform heat flux at bottom surface. 4 Uniform surface temperature at the upper plate surface. 5 The temperature of the bolts is equal to the upper surface temperature of the plate.

Properties Emissivity of the engine surface is given as $\epsilon = 0.3$.

Analysis The uniform heat flux subjected on the bottom plate surface is equal to the heat flux transferred by convection and radiation on the upper surface.

$$\dot{q}_o = \dot{q}_{\text{cond}} = \dot{q}_{\text{conv}} + \dot{q}_{\text{rad}} = 2200 \text{ W/m}^2$$

For the convection and radiation heat transfer, we have

$$\dot{q}_{\text{conv}} + \dot{q}_{\text{rad}} = h(T_{s,2} - T_{\infty}) + \varepsilon \sigma (T_{s,2}^4 - T_{\text{surr}}^4) = 2200 \text{ W/m}^2$$

Assuming the temperature of the bolts is equal to the upper surface temperature of the plate, with $T_{\infty} = 303 \text{ K}$ and $T_{\text{surr}} = 313 \text{ K}$, we determine

$$T_{\text{bolt}} = T_{s,2} = \mathbf{522 \text{ K} = 249^{\circ}\text{C}} > 149^{\circ}\text{C}$$

To ensure that the entire plate is below 260°C , the temperature at the plate surface subjected to the uniform heat flux needs to be determined. From Fourier's law of heat conduction, we have

$$\dot{q}_o = k \frac{T_{s,1} - T_{s,2}}{L} = 2200 \text{ W/m}^2 \rightarrow T_{s,1} = \dot{q}_o \frac{L}{k} + T_{s,2}$$

Thus

$$\begin{aligned} T_{s,1} &= \dot{q}_o \frac{L}{k} + T_{s,2} = (2200 \text{ W/m}^2) \frac{0.1 \text{ m}}{15 \text{ W/m}\cdot\text{K}} + 249^{\circ}\text{C} \\ &= \mathbf{264^{\circ}\text{C}} > 260^{\circ}\text{C} \end{aligned}$$

Discussion The temperature of the B21 bolts exceeds the maximum allowable use temperature by 100°C . Although surface 2 is below the permitted temperature for the SA-240 plate, surface 1 exceeds the maximum allowable use temperature by 4°C . Thus, under the given conditions, both the ASTM B21 bolts and the ASME SA-240 plate are not suitable for this component.

1-13 ■ PROBLEM-SOLVING TECHNIQUE

The first step in learning any science is to grasp the fundamentals. The next step is to master the fundamentals by testing this knowledge. This is done by solving significant real-world problems. Solving such problems, especially complicated ones, requires a systematic approach. By using a step-by-step approach, an engineer can reduce the solution of a complicated problem into the solution of a series of simple problems (Fig. 1-53). When you are solving a problem, we recommend that you use the following steps zealously as applicable. This will help you to avoid some of the common pitfalls associated with problem solving.

Step 1: Problem Statement

In your own words, briefly state the problem, the key information given, and the quantities to be found. This is to make sure that you understand the problem and the objectives before you attempt to solve the problem.

Step 2: Schematic

Draw a realistic sketch of the physical system involved, and list the relevant information on the figure. The sketch does not have to be something elaborate,

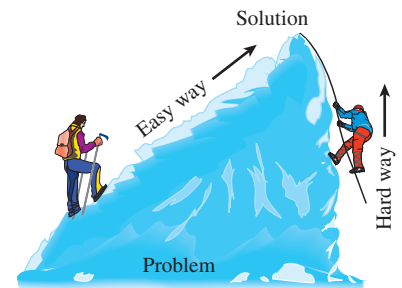


FIGURE 1-53

A step-by-step approach can greatly simplify problem solving.

Given: Air temperature in Denver

To be found: Density of air

Missing information: Atmospheric pressure

Assumption #1: Take $P = 1$ atm
(Inappropriate. Ignores effect of altitude. Will cause more than 15% error.)

Assumption #2: Take $P = 0.83$ atm
(Appropriate. Ignores only minor effects such as weather.)

FIGURE 1–54

The assumptions made while solving an engineering problem must be reasonable and justifiable.

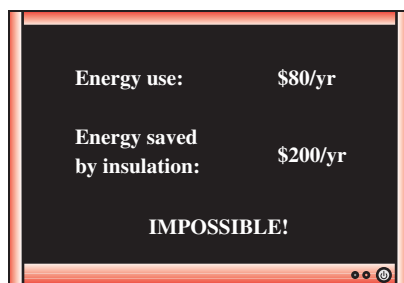


FIGURE 1–55

The results obtained from an engineering analysis must be checked for reasonableness.

but it should resemble the actual system and show the key features. Indicate any energy and mass interactions with the surroundings. Listing the given information on the sketch helps one to see the entire problem at once.

Step 3: Assumptions and Approximations

State any appropriate assumptions and approximations made to simplify the problem to make it possible to obtain a solution. Justify the questionable assumptions. Assume reasonable values for missing quantities that are necessary. For example, in the absence of specific data for atmospheric pressure, it can be taken to be 1 atm. However, it should be noted in the analysis that the atmospheric pressure decreases with increasing elevation. For example, it drops to 0.83 atm in Denver (elevation 1610 m) (Fig. 1–54).

Step 4: Physical Laws

Apply all the relevant basic physical laws and principles (such as the conservation of energy), and reduce them to their simplest form by utilizing the assumptions made. However, the region to which a physical law is applied must be clearly identified first.

Step 5: Properties

Determine the unknown properties necessary to solve the problem from property relations or tables. List the properties separately, and indicate their source, if applicable.

Step 6: Calculations

Substitute the known quantities into the simplified relations and perform the calculations to determine the unknowns. Pay particular attention to the units and unit cancellations, and remember that a dimensional quantity without a unit is meaningless. Also, don't give a false implication of high precision by copying all the digits from the calculator—round the results to an appropriate number of significant digits (see Sec. 1–15).

Step 7: Reasoning, Verification, and Discussion

Check to make sure that the results obtained are reasonable and intuitive, and verify the validity of the questionable assumptions. Repeat the calculations that resulted in unreasonable values. For example, insulating a water heater that uses \$80 worth of natural gas a year cannot result in savings of \$200 a year (Fig. 1–55).

Also, point out the significance of the results, and discuss their implications. State the conclusions that can be drawn from the results and any recommendations that can be made from them. Emphasize the limitations under which the results are applicable, and caution against any possible misunderstandings and using the results in situations where the underlying assumptions do not apply. For example, if you determined that wrapping a water heater with a \$20 insulation jacket will reduce the energy cost by \$30 a year, indicate that the insulation will pay for itself from the energy it saves in less than a year. However, also indicate that the analysis does not consider labor costs and that this will be the case if you install the insulation yourself.

Keep in mind that the solutions you present to your instructors, and any engineering analysis presented to others, are a form of communication. Therefore neatness, organization, completeness, and visual appearance are of utmost importance for maximum effectiveness (Fig. 1–56). Besides, neatness also serves as a great checking tool since it is very easy to spot errors and inconsistencies in neat work. Carelessness and skipping steps to save time often end up costing more time and unnecessary anxiety.

The approach described here is used in the solved example problems without explicitly stating each step, and it is also used in the Solutions Manual for this text. For some problems, some of the steps may not be applicable or necessary. However, we cannot overemphasize the importance of a logical and orderly approach to problem solving. Most difficulties encountered while solving a problem are not due to a lack of knowledge; rather, they are due to a lack of organization. You are strongly encouraged to follow these steps in problem solving until you develop your own approach that works best for you.



FIGURE 1–56

Neatness and organization are highly valued by employers.

1–14 ■ ENGINEERING SOFTWARE PACKAGES

You may be wondering why we are about to undertake an in-depth study of the fundamentals of another engineering science. After all, almost all problems we are likely to encounter in practice can be solved using one of several sophisticated software packages readily available today. These software packages not only give the desired numerical results, but they also supply the outputs in graphical form for impressive presentations. It is unthinkable to practice engineering today without using some of these packages. This tremendous computing power available to us at the touch of a button is both a blessing and a curse. It certainly enables engineers to solve problems easily and quickly, but it also opens the door for abuses and misinformation. In the hands of poorly educated people, these software packages are as dangerous as sophisticated, powerful weapons in the hands of poorly trained soldiers.

Thinking that a person who can use engineering software without proper training on fundamentals can practice engineering is like thinking that a person who can use a wrench can work as a mechanic. If it were true that engineering students no longer need fundamentals courses because of computers, then it would also be true that employers would no longer need high-salaried engineers since any person who knows how to use a word-processing program can also learn how to use the software. However, the statistics show that the need for engineers is on the rise, not on the decline, despite the availability of software.

We should always remember that the computing power and the engineering software available today are just tools, and tools have meaning only in the hands of masters. Having the best word-processing program does not make a person a good writer, but it certainly makes the job of a good writer much easier and makes the writer more productive (Fig. 1–57). Hand calculators did not eliminate the need to teach our children how to add or subtract, and sophisticated medical software did not take the place of medical school training. Neither will engineering software replace the traditional engineering education. It will simply cause a shift in emphasis in the courses from mathematics to physics. That is, more time will be spent in the classroom discussing the physical aspects of the problems in greater detail, and less time on the mechanics of solution procedures.



FIGURE 1–57

An excellent word-processing program does not make a person a good writer; it simply makes a good writer a better and more efficient writer.

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All these marvelous and powerful tools available today put an extra burden on today's engineers. They must still have a thorough understanding of the fundamentals, develop a “feel” for the physical phenomena, be able to put the data into proper perspective, and make sound engineering judgments, just like their predecessors. However, they must do it much better, and much faster, using more realistic models because of the powerful tools available today. The engineers in the past had to rely on hand calculations, slide rules, and later hand calculators and computers. Today they rely on software packages. The easy access to such power and the possibility of a simple misunderstanding or misinterpretation causing great damage make it more important today than ever to have solid training in the fundamentals of engineering. In this text we make an extra effort to put the emphasis on developing an intuitive and physical understanding of natural phenomena instead of on the mathematical details of solution procedures.

Equation Solvers

You are probably familiar with the equation solving capabilities of spreadsheets such as Microsoft Excel. Despite its simplicity, Excel is commonly used in solving systems of equations in engineering as well as finance. It enables the user to conduct parametric studies, plot the results, and ask “what if” questions. It can also solve simultaneous equations if properly set up. There are also numerous sophisticated equation solvers commonly used in engineering practice, such as the Engineering Equation Solver (EES), which is a program that easily solves systems of linear or nonlinear algebraic or differential equations numerically. It has a large library of built-in thermodynamic property functions as well as mathematical functions, and it allows the user to supply additional property data.

Unlike some software packages, equation solvers do not solve engineering problems; they only solve the equations supplied by the user. Therefore, the user must understand the problem and formulate it by applying any relevant physical laws and relations. Equation solvers save the user considerable time and effort by simply solving the resulting mathematical equations. This makes it possible to attempt significant engineering problems not suitable for hand calculations and to conduct parametric studies quickly and conveniently.

EXAMPLE 1–16 Solving a System of Equations Numerically

The difference of two numbers is 4, and the sum of the squares of these two numbers is equal to the sum of the numbers plus 20. Determine these two numbers.

SOLUTION Relations are given for the difference and the sum of the squares of two numbers. The two numbers are to be determined.

Analysis We first solve the problem using EES. We start the EES program by double-clicking on its icon, open a new file, and type the following on the blank screen that appears:

$$\begin{aligned}x - y &= 4 \\x^2 + y^2 &= x + y + 20\end{aligned}$$

which is an exact mathematical expression of the problem statement with x and y denoting the unknown numbers. The solution to this system of equations (one linear and one nonlinear) with two unknowns is obtained by a single click on the “calculator” icon on the taskbar. It gives (Fig. 1–58)

$$x = 5 \text{ and } y = 1$$

We now solve the same problem using Excel. Start Excel. File/Options/Add-Ins/Solver Add-In/OK, where the underline means to click on that option and the slash separates each sequential option. Choose a cell for x and a cell for y and enter initial guesses there (we chose cells C25 and D25 and guessed 0.5 and 0.5). We must rewrite the two equations so that no variables are on the right-hand side (RHS): $x - y = 4$ and $x^2 + y^2 - x - y = 20$. Choose a cell for the RHS of each equation and enter the formula there (we chose cells D20 and D21; see the equations in Fig. 1–59a). Data/Solver. Set the cell for the RHS of the first equation (D20) as the “Objective” with a value of 4, set the cells for x and y (C25:D25) as those subject to constraints, and set the constraint such that the cell for the RHS of the second equation (D21) must equal 20. Solve/OK. The solution iterates to the correct final values of $x = 5$ and $y = 1$, respectively (Fig. 1–59b). *Note:* For better convergence, the precision, number of allowed iterations, etc., can be changed in Data/Solver/Options.

Discussion Note that all we did was formulate the problem as we would on paper; EES or Excel took care of all the mathematical details of the solution. Also note that equations can be linear or nonlinear, and they can be entered in any order with unknowns on either side. Friendly equation solvers such as EES allow the user to concentrate on the physics of the problem without worrying about the mathematical complexities associated with the solution of the resulting system of equations.

1–15 ■ ACCURACY, PRECISION, AND SIGNIFICANT DIGITS

In engineering calculations, the supplied information is not known to more than a certain number of significant digits, usually three digits. Consequently, the results obtained cannot possibly be precise to more significant digits. Reporting results in more significant digits falsely implies greater precision than exists, and it should be avoided.

Regardless of the system of units employed, engineers must be aware of three principles that govern the proper use of numbers: accuracy, precision, and significant digits. For engineering measurements, they are defined as follows:

- **Accuracy error** (*inaccuracy*) is the value of one reading minus the true value. In general, the accuracy of a set of measurements refers to the closeness of the average reading to the true value. Accuracy is generally associated with repeatable, fixed errors.
- **Precision error** is the value of one reading minus the average of readings. In general, the precision of a set of measurements refers to the fineness of the resolution and the repeatability of the instrument. Precision is generally associated with unrepeatable, random errors.
- **Significant digits** are digits that are relevant and meaningful.

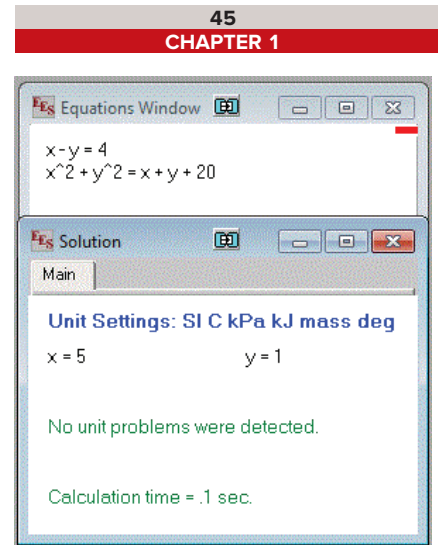


FIGURE 1–58

EES screen images for Example 1–16.

Source: Engineering Equation Solver

	A	B	C	D
19		Equation		RHS of equation
20		$x - y = 4$		=C25-D25
21		$x^2 + y^2 - x - y = 20$		=C25^2 + D25^2 - C25 - D25
23		Variable values:		
24			x	y
25			0.5	0.5

(a)

	A	B	C	D
19		Equation		RHS of equation
20		$x - y = 4$		4
21		$x^2 + y^2 - x - y = 20$		20
23		Variable values:		
24			x	y
25			5	1

(b)

FIGURE 1–59

Excel screen images for Example 1–16. (a) Equations, with initial guesses highlighted. (b) Final results after using Excel’s Solver, with converged values highlighted.

Source: Microsoft Excel

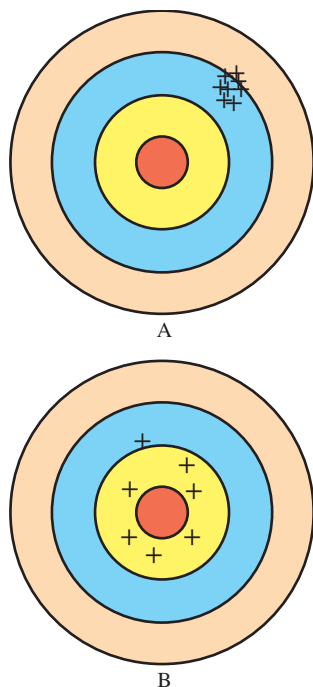
**FIGURE 1-60**

Illustration of accuracy versus precision. Shooter A is more precise, but less accurate, while shooter B is more accurate, but less precise.

A measurement or calculation can be very precise without being very accurate, and vice versa. For example, suppose the true value of the temperature is 25.00°C . Two thermocouples A and B take five temperature readings each:

Thermocouple A: 25.50, 25.69, 25.52, 25.58, and 25.61°C . Average of all readings = 25.58°C .

Thermocouple B: 26.3, 24.5, 23.9, 26.8, and 23.6°C . Average of all readings = 25.02°C .

Clearly, thermocouple A is more precise, since none of the readings differs by more than 0.11°C from the average. However, the average is 25.58°C , 0.58°C greater than the true temperature; this indicates significant **bias error**, also called **constant error** or **systematic error**. On the other hand, thermocouple B is not very precise, since its readings swing wildly from the average; but its overall average is much closer to the true value. Hence, thermocouple B is more accurate than thermocouple A, at least for this set of readings, even though it is less precise. The difference between accuracy and precision can be illustrated effectively by analogy to shooting arrows at a target, as sketched in Fig. 1–60. Shooter A is very precise, but not very accurate, while shooter B has better overall accuracy, but less precision.

Many engineers do not pay proper attention to the number of significant digits in their calculations. The least significant numeral in a number implies the precision of the measurement or calculation. For example, a result written as 1.23 (three significant digits) *implies* that the result is precise to within one digit in the second decimal place; i.e., the number is somewhere between 1.22 and 1.24. Expressing this number with any more digits would be misleading. The number of significant digits is most easily evaluated when the number is written in exponential notation; the number of significant digits can then simply be counted, including zeroes. Alternatively, the least significant digit can be underlined to indicate the author's intent. Some examples are shown in Table 1–7.

When performing calculations or manipulations of several parameters, the final result is generally only as precise as the least precise parameter in the problem. For example, suppose A and B are multiplied to obtain C . If $A = 2.3601$ (five significant digits), and $B = 0.34$ (two significant digits), then $C = 0.80$ (only two digits are significant in the final result). Note that most students are tempted to write $C = 0.802434$, with six significant digits, since that is what is displayed on a calculator after multiplying these two numbers.

Let's analyze this simple example carefully. Suppose the exact value of B is 0.33501, which is read by the instrument as 0.34. Also suppose A is exactly 2.3601, as measured by a more accurate and precise instrument. In this case, $C = A \times B = 0.79066$ to five significant digits. Note that our first answer, $C = 0.80$, is off by one digit in the second decimal place. Likewise, if B is 0.34499 and is read by the instrument as 0.34, the product of A and B would be 0.81421 to five significant digits. Our original answer of 0.80 is again off by one digit in the second decimal place. The main point here is that 0.80 (to two significant digits) is the best one can expect from this multiplication since, to begin with, one of the values had only two significant digits.

TABLE 1-7

Significant digits

Number	Exponential Notation	Number of Significant Digits
12.3	1.23×10^1	3
123,000	1.23×10^5	3
0.00123	1.23×10^{-3}	3
40,300	4.03×10^4	3
40,300	4.0300×10^4	5
0.005600	5.600×10^{-3}	4
0.0056	5.6×10^{-3}	2
0.006	$6. \times 10^{-3}$	1

Finally, when the number of significant digits is unknown, the accepted engineering standard is three significant digits. Therefore, if the length of a pipe is given to be 40 m, we will assume it to be 40.0 m in order to justify using three significant digits in the final results.

[illegible]

A result with more significant digits than that of given data falsely implies more accuracy.

TOPIC OF SPECIAL INTEREST*

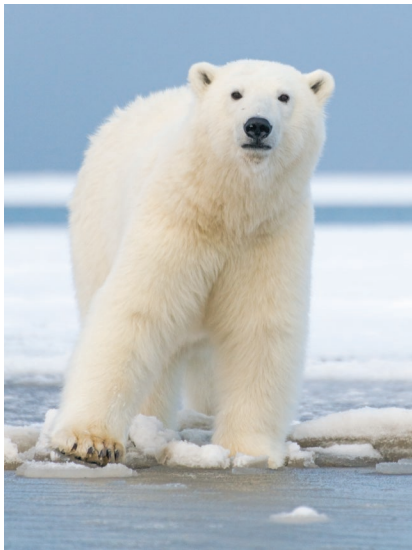


FIGURE 1–62

Most animals come into this world with fur, but human beings come with a delicate skin that provides little thermal insulation.

©Dawn Wilson Photography/Getty Images

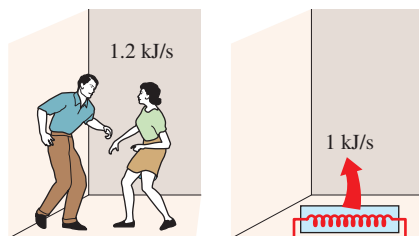


FIGURE 1–63

Two fast-dancing people supply more heat to a room than a 1-kW resistance heater.

Thermal Comfort

Unlike animals such as a fox or a bear that are born with fur, human beings come into this world with little protection against harsh environmental conditions (Fig. 1–62). Therefore, we can claim that the search for thermal comfort dates back to the beginning of human history. It is believed that early human beings lived in caves that provided shelter as well as protection from extreme thermal conditions. Probably the first form of heating system used was *open fire*, followed by fire in dwellings through the use of a *chimney* to vent out the combustion gases. The concept of *central heating* dates back to the times of the Romans, who heated homes by utilizing double-floor construction techniques and passing the fire's fumes through the opening between the two floor layers. The Romans were also the first to use *transparent windows* made of mica or glass to keep the wind and rain out while letting the light in. Wood and coal were the primary energy sources for heating, and oil and candles were used for lighting. The ruins of south-facing houses indicate that the value of *solar heating* was recognized early in our history.

The term **air-conditioning** is usually used in a restricted sense to imply cooling, but in its broad sense it means *to condition* the air to the desired level by heating, cooling, humidifying, dehumidifying, cleaning, and/or deodorizing. The purpose of the air-conditioning system of a building is to provide *complete thermal comfort* for its occupants. Therefore, we need to understand the thermal aspects of the *human body* in order to design an effective air-conditioning system.

The building blocks of living organisms are *cells*, which resemble miniature factories performing various functions necessary for the survival of organisms. The human body contains about 100 trillion cells with an average diameter of 0.01 mm. In a typical cell, thousands of chemical reactions occur every second during which some molecules are broken down (catabolic pathways). Energy is released in the form of heat, and some energy is stored in usable forms and some new molecules are formed (anabolic pathways) using the stored energy. The high level of chemical activity in all cells helps to maintain the human body temperature of 37.0°C (98.6°F) while performing the necessary bodily functions to sustain life. The combination of all of these processes is called the **metabolism**. In simple terms, metabolism refers to the burning of foods such as carbohydrates, fat, and protein. The metabolizable energy content of foods is usually expressed by nutritionists in terms of the capitalized Calorie. One Calorie is equivalent to 1 Cal = 1 kcal = 4.1868 kJ. Excess consumed Calories are stored in the body for potential use at a later time. Interestingly, these excess stored Calories alter the thermal properties and thus thermal comfort of the human body.

*This section can be skipped without loss in continuity.

The rate of metabolism at the resting state is called the *basal metabolic rate*, which is the rate of metabolism required to keep a body performing the necessary bodily functions such as breathing and blood circulation at zero external activity level. The metabolic rate can also be interpreted as the energy consumption rate for a body. For an *average man* (30 years old, 70 kg, 1.73 m high, 1.8 m² surface area), the basal metabolic rate is 84 W. That is, the body converts chemical energy of the food (or of the body fat if the person had not eaten) into heat at a rate of 84 J/s, which is then dissipated to the surroundings. The metabolic rate increases with the *level of activity*, and it may exceed 10 times the basal metabolic rate when someone is doing strenuous exercise. That is, two people doing heavy exercising in a room may be supplying more energy to the room than a 1-kW resistance heater (Fig. 1–63). An average man generates heat at a rate of 108 W while reading, writing, typing, or listening to a lecture in a classroom in a seated position. The maximum metabolic rate of an average man is 1250 W at age 20 and 730 at age 70. The corresponding rates for women are about 30 percent lower. Maximum metabolic rates of trained athletes can exceed 2000 W.

Metabolic rates during various activities are given in Table 1–8 per unit body surface area. The **surface area** of a nude body was given by D. DuBois in 1916 as

$$A_s = 0.202 m^{0.425} h^{0.725} \quad (\text{m}^2) \quad (1-30)$$

where m is the mass of the body in kg and h is the height in m. *Clothing* increases the surface area of a person by up to about 50 percent and may provide additional resistance to dissipation of heat. The metabolic rates given in the table are sufficiently accurate for most purposes, but there is considerable uncertainty at high activity levels, and these values are dependent on the tissue composition of the body (e.g., fat vs. lean muscle). More accurate values can be determined by measuring the rate of respiratory *oxygen consumption*, which ranges from about 0.25 L/min for an average resting man to more than 2 L/min during extremely heavy work. All energy released during metabolism can be assumed to be released as *heat* (in sensible or latent forms) since the external mechanical work done by the muscles is very small. Besides, the work done during most activities such as walking or riding an exercise bicycle is eventually converted to heat through friction.

The comfort of the human body depends primarily on three environmental factors: the temperature, relative humidity, and air motion. The temperature of the environment is the single most important index of comfort. Extensive research is done on human subjects to determine the “**thermal comfort zone**” and to identify the conditions under which the body feels comfortable in an environment. It has been observed that most normally clothed people resting or doing light work feel comfortable in the *operative temperature* (roughly, the average temperature of air and surrounding surfaces) range of 23°C to 27°C or 73°F to 80°F (Fig. 1–64). For unclothed people, this range is 29°C to 31°C. Interestingly, the operative temperature

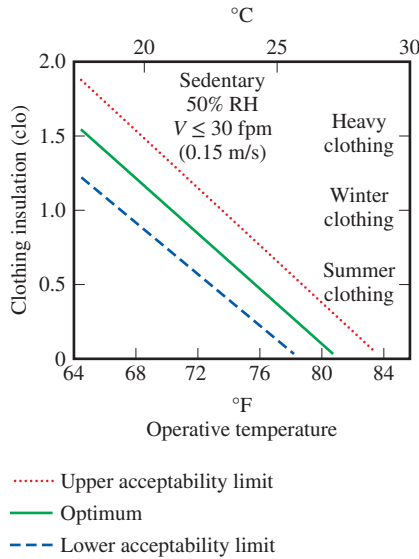
TABLE 1–8

Metabolic rates during various activities

Activity	Metabolic Rate*
	W/m ²
Resting:	
Sleeping	40
Reclining	45
Seated, quiet	60
Standing, relaxed	70
Walking (on the level):	
2 mph (0.89 m/s)	115
3 mph (1.34 m/s)	150
4 mph (1.79 m/s)	220
Office Activities:	
Reading, seated	55
Writing	60
Typing	65
Filing, seated	70
Filing, standing	80
Walking about	100
Lifting/packing	120
Driving/Flying:	
Car	60–115
Aircraft, routine	70
Heavy vehicle	185
Miscellaneous Occupational Activities:	
Cooking	95–115
Cleaning house	115–140
Machine work:	
Light	115–140
Heavy	235
Handling 50-kg bags	235
Pick and shovel work	235–280
Miscellaneous Leisure Activities:	
Dancing, social	140–255
Calisthenics/exercise	175–235
Tennis, singles	210–270
Basketball	290–440
Wrestling, competitive	410–505

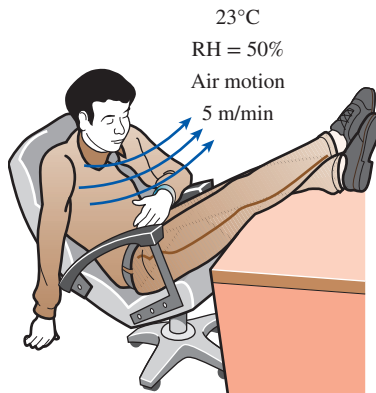
*Multiply by 1.8 m² to obtain metabolic rates for an average man. Multiply by 0.3171 to convert to Btu/h·ft².

From American Society of Heating, Refrigeration, and Air-Conditioning Engineers, Handbook of Fundamentals, 2017, Chap. 9, Table 4.

**FIGURE 1-64**

The effect of clothing on the environmental temperature that feels comfortable ($1 \text{ clo} = 0.155 \text{ m}^2 \cdot ^\circ\text{C}/\text{W} = 0.880 \text{ ft}^2 \cdot ^\circ\text{F} \cdot \text{h}/\text{Btu}$).

From ASHRAE Standard 55-1981

**FIGURE 1-65**

A thermally comfortable environment.

is higher than the average temperature of many highly populated areas in the world, which tend to have an average yearly temperature in the range of 15°C to 20°C or 59°F to 68°F (Table 1-9). Relative humidity also has a considerable effect on comfort since it is a measure of air's ability to absorb moisture, and thus it affects the amount of heat a body can dissipate by evaporation. High relative humidity slows down heat rejection by evaporation, especially at high temperatures, and low relative humidity speeds it up. The desirable level of *relative humidity* is the broad range of 30 to 70 percent, with 50 percent being the most desirable level. Most people at these conditions feel neither hot nor cold, and the body does not need to activate any of the defense mechanisms to maintain the normal body temperature (Fig. 1-65).

Another factor that has a major effect on thermal comfort is **excessive air motion** or **draft**, which causes undesired local cooling of the human body. Draft is identified by many as a most annoying factor in work places, automobiles, and airplanes. Experiencing discomfort by draft is most common among people wearing indoor clothing and doing light sedentary work and least common among people with high activity levels. The air velocity should be kept below 9 m/min (30 ft/min) in winter and 15 m/min (50 ft/min) in summer to minimize discomfort by draft, especially when the air is cool. A low level of air motion is desirable as it removes the warm, moist air that builds around the body and replaces it with fresh air. Therefore, air motion should be strong enough to remove heat and moisture from the vicinity of the body but gentle enough to be unnoticed. High-speed air motion causes discomfort outdoors as well. For example, an environment at 10°C (50°F) with 48 km/h winds feels as cold as an environment at -7°C (20°F) with 3 km/h winds because of the chilling effect of the air motion (the wind-chill factor).

A comfort system should provide *uniform conditions* throughout the living space to avoid discomfort caused by nonuniformities such as *drafts*, *asymmetric thermal radiation*, *hot or cold floors*, and *vertical temperature stratification*. **Asymmetric thermal radiation** is caused by the *cold surfaces* of large windows, uninsulated walls, or cold products and the *warm surfaces* of gas or electric radiant heating panels on the walls or ceiling, solar-heated masonry walls or ceilings, and warm machinery. Asymmetric radiation causes discomfort by exposing different sides of the body to surfaces at different temperatures and thus to different heat loss or gain by radiation. A person whose left side is exposed to a cold window, for example, will feel like heat is being drained from that side of his or her body (Fig. 1-66). For thermal comfort, the radiant temperature asymmetry should not exceed 5°C in the vertical direction and 10°C in the horizontal direction. The unpleasant effect of radiation asymmetry can be minimized by properly sizing and installing heating panels, using double-pane windows, and providing generous insulation at the walls and the roof.

Direct contact with **cold** or **hot floor surfaces** also causes localized discomfort in the feet. The temperature of the floor depends on the way it is *constructed* (being directly on the ground or on top of a heated room,

being made of wood or concrete, the use of insulation, etc.) as well as the *floor covering used* such as pads, carpets, rugs, and linoleum. A floor temperature of 23 to 25°C is comfortable to most people. The floor asymmetry loses its significance for people with footwear. An effective and economical way of raising the floor temperature is to use radiant heating panels instead of turning the thermostat up. Another nonuniform condition that causes discomfort is **temperature stratification** in a room that exposes the head and the feet to different temperatures. For thermal comfort, the temperature difference between the head and foot levels should not exceed 5°C. This effect can be minimized by using destratification fans.

It should be noted that no thermal environment will please everyone. No matter what we do, some people will express some discomfort. The thermal comfort zone is based on a 90 percent acceptance rate. That is, an environment is deemed comfortable if only 10 percent of the people are dissatisfied with it. Metabolism decreases somewhat with *age*, but it has no effect on the comfort zone. Research indicates that there is no appreciable difference between the environments preferred by old and young people. Experiments also show that *men* and *women* prefer almost the same environment. The metabolism rate of women is somewhat lower, but this is compensated by their slightly lower skin temperature, lower surface area, and lower evaporative loss as compared with men. Also, there is no significant variation in the comfort zone from one part of the world to another and from winter to summer even though large environmental temperature differences may exist in different parts of the world and even within the same region during different seasons (Table 1–9). Therefore, the same thermal comfort conditions can be used *throughout the world* in any season. Also, people cannot *acclimatize* themselves to prefer different comfort conditions since these preferences are based on factors that are difficult to control, such as basal metabolic rate.

In a **cold environment**, the rate of heat loss from the body may exceed the rate of metabolic heat generation. Average specific heat of the human body is 3.49 kJ/kg·°C, and thus each 1°C drop in body temperature corresponds to a deficit of 244 kJ in body heat content for an average 70-kg man. A drop of 0.5°C in mean body temperature causes noticeable but acceptable discomfort. A drop of 2.6°C causes extreme discomfort. A sleeping person wakes up when his or her mean body temperature drops by 1.3°C (which normally shows up as a 0.5°C drop in the core body and 3°C in the skin area). The drop of core body temperature below 35°C may damage the body temperature regulation mechanism, while a core below 28°C may be fatal and a drop to 23°C will be fatal (Table 1–9). Sedentary people report feeling *comfortable* at a *mean skin temperature* of 33.3°C, *uncomfortably cold* at 31°C, *shivering cold* at 30°C, and *extremely cold* at 29°C. People doing heavy work report feeling comfortable at much lower temperatures, which shows that the activity level and thus metabolism rate affects human performance and comfort. The extremities of the body such as hands and feet are most easily affected by cold weather, and their temperature is a better indication of comfort and performance. The extremities

TABLE 1–9

Comparison of biologically relevant and environmental temperatures

Location or Activity	Temperature
	°C
Record high world temperature, Death Valley, California, USA	57
Death (due to proteins denaturing)	44
Normal core body temperature	37
Average yearly temperature in Singapore	28
Average yearly temperature in Corumba, Brazil	25
Death (due to hypothermia)	23
Nominal interior temperature	22
Average yearly temperature in Los Angeles, California, USA	19
Average yearly temperature in Shanghai, China	16
Average yearly temperature in New York, New York, USA	13
Average yearly temperature in Anchorage, Alaska, USA	2
Record low world temperature, Vostok Station, Antarctica	–89

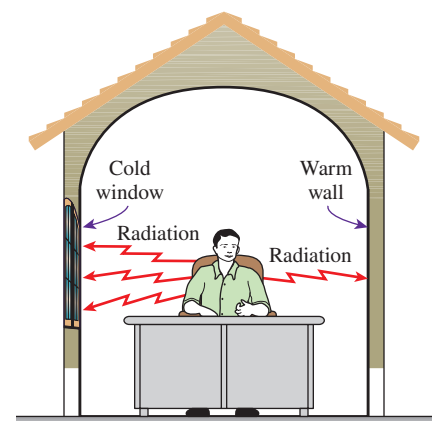


FIGURE 1–66

Cold surfaces cause excessive heat loss from the body by radiation, and thus discomfort on the affected side of the body.

**FIGURE 1–67**

The rate of metabolic heat generation may go up by six times the resting level during total body shivering in cold weather.

**FIGURE 1–68**

In hot environments, a body can dissipate a large amount of metabolic heat by sweating since the sweat absorbs the body heat and evaporates.

can and normally do experience more drastic changes in temperature as compared with the core temperature or skin temperature closer to the core. A hand-skin temperature of 20°C is perceived to be uncomfortably cold, 15°C to be extremely cold, and 5°C to be painfully cold. Useful work can be performed by hands without difficulty as long as the skin temperature of fingers remains above 16°C (American Society of Heating, Refrigeration, and Air-Conditioning Engineers, 2017, Chapter 9).

The first line of defense of the body against excessive heat loss in a cold environment is to *reduce the skin temperature* and thus the rate of heat loss from the skin by constricting the arteries, thus decreasing the blood flow to the skin. This measure decreases the temperature of the tissues subjacent to the skin but maintains the core body temperature. The next preventive measure is increasing the rate of *metabolic heat generation* in the body by *shivering*, unless the person does it voluntarily by increasing his or her level of activity or puts on additional clothing. Shivering begins slowly in small muscle groups and may double the rate of metabolic heat production of the body at its initial stages. In the extreme case of total body shivering, the rate of heat production may reach six times the resting levels (Fig. 1–67). Shivering converts stored chemical energy into mechanical work and dissipated heat. If this measure also proves inadequate, the core body temperature starts *falling*. Body parts farthest away from the core such as the hands and feet are at greatest danger for tissue damage.

In **hot environments**, the rate of heat loss from the body may drop below the metabolic heat generation rate. This time the body activates the opposite mechanisms. First the body increases the *blood flow* and thus heat transport to the skin, causing the temperature of the skin and the subjacent tissues to rise and approach the deep body temperature. Under extreme heat conditions, the *heart rate* may reach 150 beats per minute in order to maintain adequate blood supply to the brain and the skin. At higher heart rates, the *volumetric efficiency* of the heart drops because the very short time between heartbeats does not allow for complete filling of the heart chambers, and thus the blood supply to the skin, and more importantly to the brain, drops. This causes the person to faint as a result of *heat exhaustion*. Dehydration makes the problem worse, since physiological mechanisms will activate to reduce blood flow to the extremities and skin. A similar thing happens when a person working very hard for a long time stops suddenly and becomes sedentary. The blood that has flooded the skin and other tissues that were in use (e.g., muscles) has difficulty returning to the heart in this case since the relaxed muscles no longer force the blood back to the heart, and thus there is less blood available for pumping to the brain. Therefore, a “cooling down” period after exercising is just as important as a “warming up” period.

The next line of defense is releasing water from sweat glands and resorting to *evaporative cooling*, unless the person removes some clothing and reduces the activity level (Fig. 1–68). The body can maintain its core temperature at 37°C in this evaporative cooling mode indefinitely, even

in environments at higher temperatures (as high as 200°C during military endurance tests), if the person drinks plenty of liquids to replenish his or her water reserves and the ambient air is sufficiently dry to allow the sweat to evaporate instead of rolling down the skin. If this measure proves inadequate, the body will have to start absorbing the metabolic heat and the deep body temperature will rise. A person can tolerate a temperature rise of 1.4°C without major discomfort but may *collapse* when the core body temperature rise reaches 2.8°C. People feel sluggish and their efficiency drops considerably when the core body temperature rises above 39°C. A core temperature above 41°C may damage most innate proteins, resulting in cessation of sweating, increased heat production by shivering, and a *heat stroke* with irreversible and life-threatening damage. Death can occur above 43°C (Table 1–9).

A surface temperature of 46°C causes pain on the skin. Therefore, direct contact with a metal block at this temperature or above is painful. However, a person can stay in a room at 100°C for up to 30 min without any damage or pain on the skin because of the convective resistance at the skin surface and evaporative cooling. We can even put our hands into an oven at 200°C for a short time without getting burned as long as the skin does not touch a hot surface. This is also because the thermal conductivity of air to skin is much lower than metal to skin.

The processes that have been described in the previous few paragraphs fall into a category of physiology termed *thermoregulation*. The body has mechanisms to act as a thermostat, when the core body temperature deviates from the normal resting value. With an increase in temperature, skin blood vessels dilate and sweat is produced, while a decrease in temperature induces skin blood vessel constriction and shivering. It is also important to point out that not all living beings use these same mechanisms to regulate the temperature of their bodies. A dog, which is an endotherm similar to humans, loses heat by panting because sweat evaporation would be minimal since the fluid particles would be trapped by the fur. Ectotherms must derive heat from the environment—for example, a lizard warms up by standing on a sun-heated rock. It is also important to note that the metabolisms of endotherms are not dependent on the surrounding environment, whereas ectotherm metabolism is highly dependent on the environment. Thermoregulation in the human body is achieved by keeping a tight balance between heat gain and heat loss. The bioheat transfer equation introduced in Chap. 3 is used to calculate the heat transfer between a human body and its surroundings.

Thermoregulation can be adjusted by both behavioral changes and physiological changes. We have discussed many behavioral changes, such as relocating to a more desirable environment within the structure or putting on more clothing. Physiological changes include blood vessel diameter changes and the production of sweat. However, under normal conditions, few of these changes are needed because of the efficient organization of arteries and veins; they are arranged as a countercurrent heat exchanger (for more information, refer to Chap. 11 Topic of Special

Interest, “The Human Cardiovascular System as a Countercurrent Heat Exchanger”). The primary function of this countercurrent heat exchanger in the human body is to minimize the loss of heat to the external environment. In nearly all systems, an artery, which delivers “warm” blood from the body core, is located directly adjacent to a vein, which returns “cool” blood to the core of the body. Recall that the extremities may be at a very different temperature than the core, and this would be manifested directly in the blood returning from the limb. However, since the vein runs adjacent to an artery, there is a continual conduction of heat along the entire blood vessel length from the artery to the vein. This acts to prevent cool blood from entering the core of the body and reducing the core temperature.

Another factor that affects thermal comfort, health, and productivity is **ventilation**. Fresh outdoor air can be provided to a building *naturally* by doing nothing, or *forcefully* by a mechanical ventilation system. In the first case, which is the norm in residential buildings, the necessary ventilation is provided by *infiltration through cracks and leaks* in the living space and by the opening of the windows and doors. The additional ventilation needed in the bathrooms and kitchens is provided by *air vents with dampers* or *exhaust fans* due to the high humidity/smoke generated in these rooms. With this kind of uncontrolled ventilation, however, the fresh air supply will be either too high, wasting energy, or too low, causing poor indoor air quality. But the current practice is not likely to change for residential buildings since there is not a public outcry about energy waste or air quality, and thus it is difficult to justify the cost and complexity of mechanical ventilation systems.

Mechanical ventilation systems are part of any heating and air conditioning system in *commercial buildings*, providing the necessary amount of fresh outdoor air and distributing it uniformly throughout the building. This is not surprising since many rooms in large commercial buildings have no windows and thus rely on mechanical ventilation. Even the rooms with windows are in the same situation since the windows are tightly sealed and cannot be opened in most buildings. It is not a good idea to oversize the ventilation system just to be on the “safe side” since exhausting the heated or cooled indoor air wastes energy. On the other hand, reducing the ventilation rates below the required minimum to conserve energy should also be avoided so that the indoor air quality can be maintained at the required levels. The minimum fresh air ventilation requirements are listed in Table 1–10. The values are based on controlling CO₂ and other contaminants with an adequate margin of safety, which requires each person be supplied with at least 7.5 L/s (15 ft³/min) of fresh air.

Another function of the mechanical ventilation system is to **clean** the air by filtering it as it enters the building. Various types of filters are available for this purpose, depending on the cleanliness requirements and the allowable pressure drop.

TABLE 1–10

Minimum fresh air requirements in buildings

Application	Requirement (per person)	
	L/s	ft ³ /min
Classrooms, libraries, supermarkets	8	15
Dining rooms, conference rooms, offices	10	20
Hospital rooms	13	25
Hotel rooms	15 (per room)	30 (per room)
Smoking lounges	30	60
Retail stores	1.0–1.5 (per m ²)	0.2–0.3 (per ft ²)
Residential buildings	0.35 air change per hour, but not less than 7.5 L/s (or 15 ft ³ /min) per person	

From ASHRAE Standard 62–1989

SUMMARY

In this chapter, the basics of heat transfer are introduced and discussed. The science of *thermodynamics* deals with the amount of heat transfer as a system undergoes a process from one equilibrium state to another, whereas the science of *heat transfer* deals with the rate of heat transfer, which is the main focus of interest in the design and evaluation of heat transfer equipment. The sum of all forms of energy of a system is called *total energy*, and it includes the internal, kinetic, and potential energies. The *internal energy* represents the molecular energy of a system, and it consists of sensible, latent, chemical, and nuclear forms. The sensible and latent forms of internal energy can be transferred from one medium to another as a result of a temperature difference, and they are referred to as *heat* or *thermal energy*. Thus, *heat transfer* is the exchange of the sensible and latent forms of internal energy between two media as a result of a temperature difference. The amount of heat transferred per unit time is called the *heat transfer rate* and is denoted by $\dot{Q}$. The rate of heat transfer per unit area is called *heat flux*, $\dot{q}$.

A system of fixed mass is called a *closed system*, and a system that involves mass transfer across its boundaries is called an *open system* or *control volume*. The *first law of thermodynamics* or the *energy balance* for any system undergoing any process can be expressed as

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

When a stationary closed system involves heat transfer only and no work interactions across its boundary, the energy balance relation reduces to

$$Q = mc_p \Delta T$$

where Q is the amount of net heat transfer to or from the system. When heat is transferred at a constant rate of $\dot{Q}$, the amount of heat transfer during a time interval Δt can be determined from $Q = \dot{Q} \Delta t$.

Under steady conditions and in the absence of any work interactions, the conservation of energy relation for a control volume with one inlet and one exit with negligible changes in kinetic and potential energies can be expressed as

$$\dot{Q} = \dot{m} c_p \Delta T$$

where $\dot{m} = \rho V A_c$ is the mass flow rate and $\dot{Q}$ is the rate of net heat transfer into or out of the control volume.

Heat can be transferred in three different modes: conduction, convection, and radiation. *Conduction* is the transfer of heat from the more energetic particles of a substance to the adjacent less energetic ones as a result of interactions between

the particles and is expressed by *Fourier's law of heat conduction* as

$$\dot{Q}_{\text{cond}} = -kA \frac{dT}{dx}$$

where k is the *thermal conductivity* of the material in W/m·K or Btu/h·ft·R, A is the *area* normal to the direction of heat transfer, and dT/dx is the *temperature gradient*. The magnitude of the rate of heat conduction across a plane layer of thickness L is given by

$$\dot{Q}_{\text{cond}} = kA \frac{\Delta T}{L}$$

where ΔT is the temperature difference across the layer.

Convection is the mode of heat transfer between a solid surface and the adjacent liquid or gas that is in motion, and it involves the combined effects of conduction and fluid motion. The rate of convection heat transfer is expressed by *Newton's law of cooling* as

$$\dot{Q}_{\text{convection}} = hA_s(T_s - T_{\infty})$$

where h is the *convection heat transfer coefficient* in W/m²·K or Btu/h·ft²·R, A_s is the *surface area* through which convection heat transfer takes place, T_s is the *surface temperature*, and T_{∞} is the *temperature of the fluid* sufficiently far from the surface.

Radiation is the energy emitted by matter in the form of electromagnetic waves (or photons) as a result of the changes in the electronic configurations of the atoms or molecules. The maximum rate of radiation that can be emitted from a surface at a thermodynamic temperature T_s is given by the *Stefan-Boltzmann law* as $\dot{Q}_{\text{emit, max}} = \sigma A_s T_s^4$, where $\sigma = 5.67 \times 10^{-8}$ W/m²·K⁴ or 0.1714×10^{-8} Btu/h·ft²·R⁴ is the *Stefan-Boltzmann constant*.

When a surface of emissivity ϵ and surface area A_s at a temperature T_s is completely enclosed by a much larger (or black) surface at a temperature T_{surr} separated by a gas (such as air) that does not block radiation, the net rate of radiation heat transfer between these two surfaces is given by

$$\dot{Q}_{\text{rad}} = \epsilon \sigma A_s (T_s^4 - T_{\text{surr}}^4)$$

In this case, the emissivity (ϵ) and the surface area of the surrounding surface do not have any effect on the net radiation heat transfer.

The rate at which a surface absorbs radiation is determined from $\dot{Q}_{\text{absorbed}} = \alpha \dot{Q}_{\text{incident}}$ where $\dot{Q}_{\text{incident}}$ is the rate at which radiation is incident on the surface and α is the absorptivity of the surface.

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PROBLEMS*

Thermodynamics and Heat Transfer

1-1C How does the science of heat transfer differ from the science of thermodynamics?

1-2C What is the driving force for (a) heat transfer, (b) electric current flow, and (c) fluid flow?

1-3C How do rating problems in heat transfer differ from the sizing problems?

1-4C What is the difference between the analytical and experimental approaches to heat transfer? Discuss the advantages and disadvantages of each approach.

1-5C What is the importance of modeling in engineering? How are the mathematical models for engineering processes prepared?

1-6C When modeling an engineering process, how is the right choice made between a simple but crude and a complex but accurate model? Is the complex model necessarily a better choice since it is more accurate?

*Problems designated by a "C" are concept questions, and students are encouraged to answer them all. Problems designated by an "E" are in English units, and the SI users can ignore them. Problems with the  icon are comprehensive in nature and are intended to be solved with appropriate software. Problems with the  icon are Prevention through Design problems. Problems with the  icon are Codes and Standards problems.

1-7C On a hot summer day, a student turns his fan on when he leaves his room in the morning. When he returns in the evening, will his room be warmer or cooler than the neighboring rooms? Why? Assume all the doors and windows are kept closed.

1-8C Consider two identical rooms, one with a refrigerator in it and the other without one. If all the doors and windows are closed, will the room that contains the refrigerator be cooler or warmer than the other room? Why?

1-9C It is often claimed that you should leave the heating system in your house on all the time rather than turning it off when you are not home because it takes more energy to heat the house back to its normal temperature. What do you think?

1-10C Someone claims that turning the thermostat of the central air conditioning of a warm house to the lowest level will cool the house a lot faster. Is there any truth to this claim?

1-11C Consider a perfectly insulated (no heat loss from the skin) 80-L electric water heater set at 60°C. Observing that one tank of hot water is sufficient to meet the hot water needs of a family for an entire day, someone suggests using a timer to heat the water once a day to the set temperature and then turning it off to save energy and money. Do you think the family should take this advice and invest in a timer? Assume there is a single tariff for electricity for all hours of the day.

1-12C An ideal gas is heated from 50°C to 80°C (*a*) at constant volume and (*b*) at constant pressure. For which case do you think the energy required will be greater? Why?

1-13C What is heat flux? How is it related to the heat transfer rate?

1-14C What are the mechanisms of energy transfer to a closed system? How is heat transfer distinguished from the other forms of energy transfer?

1-15E A logic chip used in a computer dissipates 3 W of power in an environment at 120°F and has a heat transfer surface area of 0.08 in^2 . Assuming the heat transfer from the surface to be uniform, determine (*a*) the amount of heat this chip dissipates during an eight-hour workday in kWh and (*b*) the heat flux on the surface of the chip in W/in^2 .

1-16 Consider a 150-W incandescent lamp. The filament of the lamp is 5 cm long and has a diameter of 0.5 mm. The diameter of the glass bulb of the lamp is 8 cm. Determine the heat flux in W/m^2 (*a*) on the surface of the filament and (*b*) on the surface of the glass bulb, and (*c*) calculate how much it will cost per year to keep that lamp on for eight hours a day every day if the unit cost of electricity is $\$0.08/\text{kWh}$.

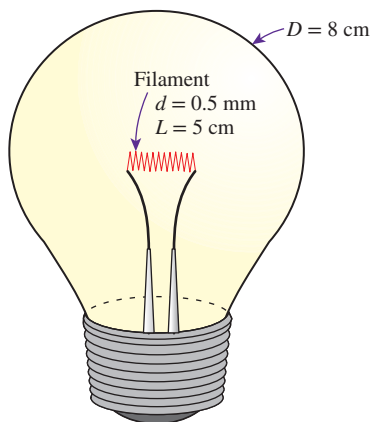


FIGURE P1-16

1-17 A 15-cm-diameter aluminum ball is to be heated from 80°C to an average temperature of 200°C . Taking the average density and specific heat of aluminum in this temperature range to be $\rho = 2700\text{ kg/m}^3$ and $c_p = 0.90\text{ kJ/kg}\cdot\text{K}$, respectively, determine the amount of energy that needs to be transferred to the aluminum ball. *Answer: 515 kJ*

1-18E A 60-gallon water heater is initially filled with water at 50°F . Determine how much energy (in Btu) needs to be transferred to the water to raise its temperature to 120°F . Evaluate the water properties at an average water temperature of 85°F .

1-19 Infiltration of cold air into a warm house during winter through the cracks around doors, windows, and other openings is a major source of energy loss since the cold air that enters

needs to be heated to the room temperature. The infiltration is often expressed in terms of ACH (air changes per hour). An ACH of 2 indicates that all air in the house is replaced twice every hour by the cold air outside.

Consider an electrically heated house that has a floor space of 200 m^2 and an average height of 3 m at 1000 m elevation, where the standard atmospheric pressure is 89.6 kPa. The house is maintained at a temperature of 22°C , and the infiltration losses are estimated to amount to 0.7 ACH. Assuming the pressure and the temperature in the house remain constant, determine the amount of energy loss from the house due to infiltration for a day during which the average outdoor temperature is 5°C . Also, determine the cost of this energy loss for that day if the unit cost of electricity in that area is $\$0.082/\text{kWh}$. *Answers: 53.8 kWh/day, \\$4.41/day*

1-20 **PtD** Liquid ethanol is a flammable fluid and can release vapors that form explosive mixtures at temperatures above its flash point at 16.6°C . In a chemical plant, liquid ethanol ($c_p = 2.44\text{ kJ/kg}\cdot\text{K}$, $\rho = 789\text{ kg/m}^3$) is being transported in a pipe with an inside diameter of 5 cm. The pipe is located in a hot area with the presence of an ignition source, where an estimated 20 kW of heat is added to the ethanol. Your task as an engineer is to design a pumping system to transport the ethanol safely and to prevent fire hazard. If the inlet temperature of the ethanol is 10°C , determine the volume flow rate that is necessary to keep the temperature of the ethanol in the pipe below its flash point.

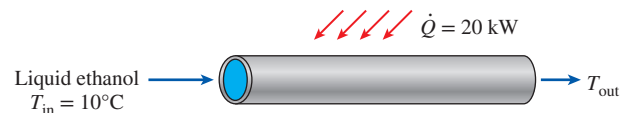


FIGURE P1-20

1-21 **PtD** In many manufacturing plants, individuals are often working around high-temperature surfaces. Exposed hot surfaces that can cause thermal burns on human skin are considered hazards in the workplace. Metallic surfaces above 70°C can instantly damage skin that touches them. Consider an AISI 1010 carbon steel strip ($\rho = 7832\text{ kg/m}^3$) 2 mm thick and 3 cm wide that is conveyed into a chamber to be cooled at a constant speed of 1 m/s. The steel strip enters the cooling chamber at 597°C . Determine the rate at which heat needs to be removed so that the steel strip exits the chamber at 47°C . Discuss how the conveyance speed can affect the required rate of heat removal.

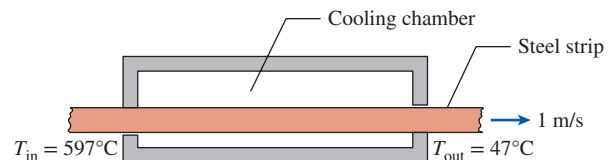


FIGURE P1-21

1–22 1.2 kg of liquid water initially at 15°C is to be heated to 95°C in a teapot equipped with a 1200-W electric heating element inside. The teapot is 0.5 kg and has an average specific heat of $0.7\text{ kJ/kg}\cdot\text{K}$. Taking the specific heat of water to be $4.18\text{ kJ/kg}\cdot\text{K}$ and disregarding any heat loss from the teapot, determine how long it will take for the water to be heated.

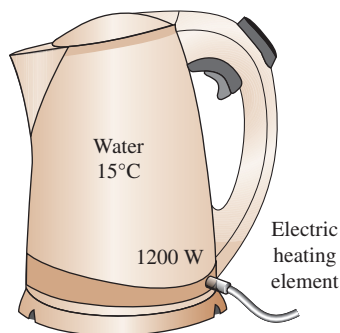


FIGURE P1–22

1–23 **C&S** A water heater uses 100 kW to heat 60 gallons of liquid water from an initial temperature of 20°C . The average density and specific heat of the water are 970 kg/m^3 and $4.18\text{ kJ/kg}\cdot\text{K}$, respectively. According to the service restrictions of the ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HG-101), water heaters should not be operating at temperatures exceeding 120°C at or near the heater outlet. Determine the heating duration such that the final water temperature that would exit the water heater would be in compliance with the ASME Boiler and Pressure Vessel Code.

1–24 **C&S** A boiler with a mass of 10 kg uses 50 kW to heat 20 gallons of liquid water initially at 15°C . The average specific heats for the boiler material and the water are $0.48\text{ kJ/kg}\cdot\text{K}$ and $4.18\text{ kJ/kg}\cdot\text{K}$, respectively. The average density of the water is 850 kg/m^3 . According to the service restrictions of the ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HG-101), hot water boilers should not be operating at temperatures exceeding 120°C at or near the boiler outlet. Determine whether the final water temperature that would exit the boiler after 30 min of heating at 50 kW would be in compliance with the ASME Boiler and Pressure Vessel Code.

1–25 Water is heated in an insulated, constant diameter tube by a 7-kW electric resistance heater. If the water enters the heater steadily at 15°C and leaves at 70°C , determine the mass flow rate of the water.

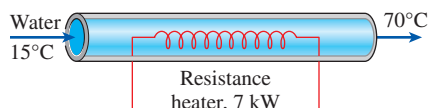


FIGURE P1–25

1–26 A room is heated by a baseboard resistance heater. When the heat losses from the room on a winter day amount to 7000 kJ/h , it is observed that the air temperature in the room remains constant even though the heater operates continuously. Determine the power rating of the heater in kW.

1–27 A $5\text{-m} \times 6\text{-m} \times 8\text{-m}$ room is to be heated by an electrical resistance heater placed in a short duct in the room. Initially, the room is at 15°C , and the local atmospheric pressure is 98 kPa . The room is losing heat steadily to the outside at a rate of 200 kJ/min . A 300-W fan circulates the air steadily through the duct and the electric heater at an average mass flow rate of 50 kg/min . The duct can be assumed to be adiabatic, and there is no air leaking in or out of the room. If it takes 18 min for the room air to reach an average temperature of 25°C , find (a) the power rating of the electric heater and (b) the temperature rise that the air experiences each time it passes through the heater.

1–28 A hair dryer is basically a duct in which a few layers of electric resistors are placed. A small fan pulls the air in and forces it to flow over the resistors, where it is heated. Air enters a 1200-W hair dryer at 100 kPa and 22°C and leaves at 47°C . The cross-sectional area of the hair dryer at the exit is 60 cm^2 . Neglecting the power consumed by the fan and the heat losses through the walls of the hair dryer, determine (a) the volume flow rate of air at the inlet and (b) the velocity of the air at the exit.

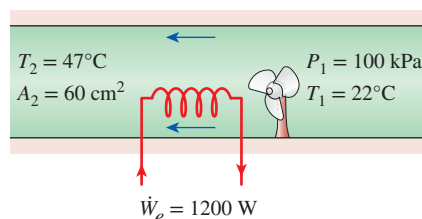


FIGURE P1–28

1–29E Air enters the duct of an air-conditioning system at 15 psia and 50°F at a volume flow rate of $450\text{ ft}^3/\text{min}$. The diameter of the duct is 10 in, and heat is transferred to the air in the duct from the surroundings at a rate of 2 Btu/s . Determine (a) the velocity of the air at the duct inlet and (b) the temperature of the air at the exit. *Answers: (a) 825 ft/min , (b) 64°F*

1–30 **C&S** Liquid water with a specific heat of $4.18\text{ kJ/kg}\cdot\text{K}$ is heated in a circular tube by an electrical heater at 10 kW. The water enters the tube at 10°C with a mass flow rate of 10 g/s . The inner surface of the tube is lined with polyvinylidene chloride (PVDC) lining. According to the ASME Code for Process Piping (ASME B31.3-2014, A323), the recommended maximum temperature for PVDC lining is 79°C .

Determine whether the water exiting the tube would be in compliance with the ASME Code for Process Piping. What is the minimum mass flow rate of the water needed to keep the exit temperature from exceeding the recommended maximum temperature for PVDC lining?

Heat Transfer Mechanisms

1-31C Define thermal conductivity, and explain its significance in heat transfer.

1-32C Which is a better heat conductor, diamond or silver?

1-33C How do the thermal conductivity of gases and liquids vary with temperature?

1-34C Why is the thermal conductivity of superinsulation orders of magnitude lower than the thermal conductivity of ordinary insulation?

1-35C Why do we characterize the heat conduction ability of insulators in terms of their apparent thermal conductivity instead of their ordinary thermal conductivity?

1-36C What are the mechanisms of heat transfer? How are they distinguished from each other?

1-37C Write down the expressions for the physical laws that govern each mode of heat transfer, and identify the variables involved in each relation.

1-38C How does heat conduction differ from convection?

1-39C Does any of the energy of the sun reach the earth by conduction or convection?

1-40C How does forced convection differ from natural convection?

1-41C What is the physical mechanism of heat conduction in a solid, a liquid, and a gas?

1-42C Consider heat transfer through a windowless wall of a house on a winter day. Discuss the parameters that affect the rate of heat conduction through the wall.

1-43C Consider heat loss through two walls of a house on a winter night. The walls are identical except that one of them has a tightly fit glass window. Through which wall will the house lose more heat? Explain.

1-44C Consider two houses that are identical except that the walls are built using bricks in one house and wood in the other. If the walls of the brick house are twice as thick, which house do you think will be more energy efficient?

1-45C Consider two walls of a house that are identical except that one is made of 10-cm-thick wood while the other is made of 25-cm-thick brick. Through which wall will the house lose more heat in winter?

1-46C Define emissivity and absorptivity. What is Kirchhoff's law of radiation?

1-47C What is a blackbody? How do real bodies differ from blackbodies?

1-48 A wood slab with a thickness of 0.05 m is subjected to a heat flux of 40 W/m^2 . The left and right surface temperatures of the wood slab are kept at constant temperatures of 40°C and 20°C , respectively. What is the thermal conductivity of the wood slab?

1-49 The inner and outer surfaces of a $4\text{-m} \times 7\text{-m}$ brick wall of thickness 30 cm and thermal conductivity $0.69 \text{ W/m}\cdot\text{K}$ are maintained at temperatures of 20°C and 5°C , respectively. Determine the rate of heat transfer through the wall in W.

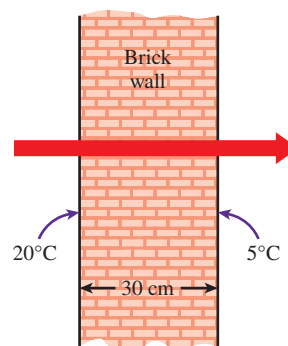


FIGURE P1-49

1-50E The inner and outer glasses of a $4\text{-ft} \times 4\text{-ft}$ double-pane window are at 60°F and 48°F , respectively. If the 0.25-in space between the two glasses is filled with still air, determine the rate of heat transfer through the window. *Answer: 131 Btu/h*

1-51 The inner and outer surfaces of a 0.5-cm thick $2\text{-m} \times 2\text{-m}$ window glass in winter are 10°C and 3°C , respectively. If the thermal conductivity of the glass is $0.78 \text{ W/m}\cdot\text{K}$, determine the amount of heat loss through the glass over a period of 5 h. What would your answer be if the glass were 1 cm thick? *Answers: 78.6 MJ, 39.3 MJ*

1-52  Reconsider Prob. 1-51. Using appropriate software, plot the amount of heat loss through the glass as a function of the window glass thickness in the range of 0.1 cm to 1.0 cm. Discuss the results.

1-53 An aluminum pan whose thermal conductivity is $237 \text{ W/m}\cdot\text{K}$ has a flat bottom with diameter 15 cm and thickness 0.4 cm. Heat is transferred steadily to boiling water in the pan through its bottom at a rate of 800 W. If the inner surface of the bottom of the pan is at 105°C , determine the temperature of the outer surface of the bottom of the pan.

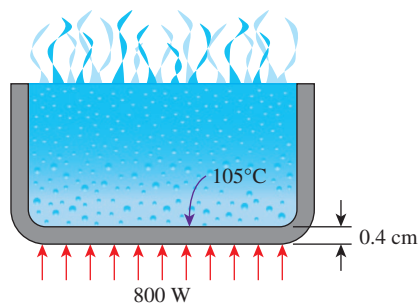


FIGURE P1-53

1-54E The north wall of an electrically heated home is 20 ft long, 10 ft high, and 1 ft thick and is made of brick whose thermal conductivity is $k = 0.42 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$. On a certain winter night, the temperatures of the inner and the outer surfaces of the wall are measured to be at about 62°F and 25°F , respectively, for a period of 8 h. Determine (a) the rate of heat loss through the wall that night and (b) the cost of that heat loss to the homeowner if the cost of electricity is $\$0.07/\text{kWh}$.

1-55 In a certain experiment, cylindrical samples of diameter 4 cm and length 7 cm are used (see Fig. 1-32). The two thermocouples in each sample are placed 3 cm apart. After initial transients, the electric heater is observed to draw 0.6 A at 110 V, and both differential thermometers read a temperature difference of 10°C . Determine the thermal conductivity of the sample. *Answer: $78.8 \text{ W/m}\cdot\text{K}$*

1-56 One way of measuring the thermal conductivity of a material is to sandwich an electric thermofoil heater between two identical rectangular samples of the material and to heavily insulate the four outer edges, as shown in Fig. P1-56. Thermocouples attached to the inner and outer surfaces of the samples record the temperatures.

During an experiment, two 0.5-cm-thick samples $10 \text{ cm} \times 10 \text{ cm}$ in size are used. When steady operation is reached, the heater is observed to draw 25 W of electric power, and the temperature of each sample is observed to drop from 82°C at the inner surface to 74°C at the outer surface. Determine the thermal conductivity of the material at the average temperature.

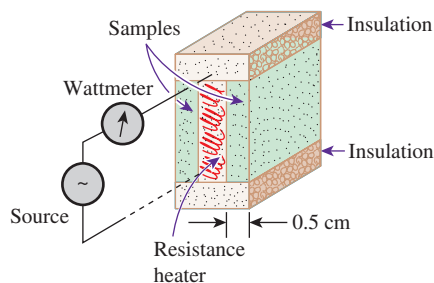


FIGURE P1-56

1-57 **PtD** Silicon wafer is susceptible to warping when the wafer is subjected to a temperature difference across its thickness. Thus, steps needed to be taken to prevent the temperature gradient across the wafer thickness from getting large. As an engineer in a semiconductor company, your task is to determine the maximum allowable heat flux on the bottom surface of the wafer while maintaining the upper surface temperature at 27°C . To prevent the wafer from warping, the temperature difference across its thickness of $500 \mu\text{m}$ can not exceed 1°C .

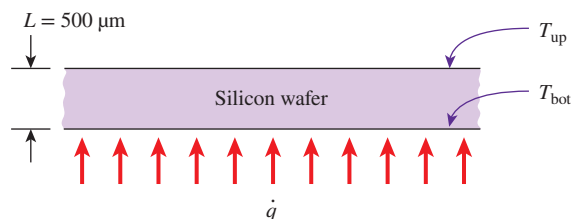


FIGURE P1-57

1-58 A concrete wall with a surface area of 20 m^2 and a thickness of 0.30 m separates conditioned room air from ambient air. The temperature of the inner surface of the wall (T_1) is maintained at 25°C . (a) Determine the heat loss $\dot{Q}(\text{W})$ through the concrete wall for three thermal conductivity values of 0.75, 1, and $1.25 \text{ W/m}\cdot\text{K}$ and outer wall surface temperatures of $T_2 = -15, -10, -5, 0, 5, 10, 15, 20, 25, 30,$ and 38°C (a total of 11 data points for each thermal conductivity value). Tabulate the results for all three cases in one table. Also provide a computer-generated graph [Heat loss, $\dot{Q}(\text{W})$ vs. Outside wall temperature, $T_2(^\circ\text{C})$] for the display of your results. The results for all three cases should be plotted on the same graph. (b) Discuss your results for the three cases.

1-59 A hollow spherical iron container with outer diameter 20 cm and thickness 0.2 cm is filled with iced water at 0°C . If the outer surface temperature is 5°C , determine the approximate rate of heat gain by the iced water in kW and the rate at which ice melts in the container. The heat of fusion of water is 333.7 kJ/kg . Treat the spherical shell as a plain wall, and use the outer area.

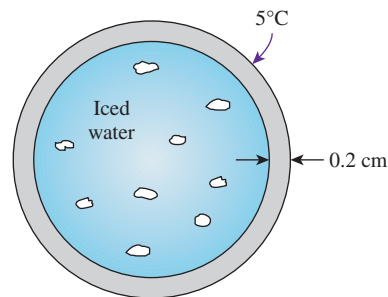


FIGURE P1-59

1-60  Reconsider Prob. 1-59. Using appropriate software, plot the rate at which ice melts as a function of the container thickness in the range of 0.1 cm to 1.0 cm. Discuss the results.

1-61  A heating cable is embedded in a concrete slab to melt snow on a 5-m $\times$ 5-m surface area. The heating cable is heated electrically with joule heating. The concrete has a thermal conductivity of 1.4 W/m $\cdot$ K, and the layer between the slab surface and the heater surface is 50 mm. When the surface is covered with snow, the heat generated from the heating cable can melt snow at a rate of 0.1 kg/s. According to the National Electrical Code (NFPA 70), the power density (heat flux) for embedded snow-melting equipment should not exceed 1300 W/m². Under the given conditions, does the concrete slab with the embedded heating cable operate in compliance with the NFPA 70 code? Also, to minimize thermal stress in the concrete, the temperature difference between the heater surface (T_1) and the slab surface (T_2) should not exceed 21°C (2015 ASHRAE Handbook—HVAC Applications, Chap. 51). Do the current operating conditions allow $T_1 - T_2 \leq 21^\circ\text{C}$?

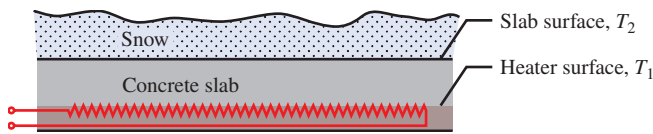


FIGURE P1-61

1-62E An engineer who is working on the heat transfer analysis of a house in English units needs the convection heat transfer coefficient on the outer surface of the house. But the only value he can find from his handbooks is 14 W/m²·K, which is in SI units. The engineer does not have a direct conversion factor between the two unit systems for the convection heat transfer coefficient. Using the conversion factors between W and Btu/h, m and ft, and °C and °F, express the given convection heat transfer coefficient in Btu/h·ft²·°F.
Answer: 2.47 Btu/h·ft²·°F

1-63 Air at 20°C with a convection heat transfer coefficient of 20 W/m²·K blows over a pond. The surface temperature of the pond is at 40°C. Determine the heat flux between the surface of the pond and the air.

1-64  A series of ASME SA-193 carbon steel bolts are bolted to the upper surface of a metal plate. The bottom surface of the plate is subjected to a uniform heat flux of 5 kW/m². The upper surface of the plate is exposed to ambient air with a temperature of 30°C and a convection heat transfer coefficient of 10 W/m²·K. The ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HF-300) limits the maximum allowable use temperature to 260°C for the SA-193

bolts. Determine whether the use of these SA-193 bolts complies with the ASME code under these conditions. If the temperature of the bolts exceeds the maximum allowable use temperature of the ASME code, discuss a possible solution to lower the temperature of the bolts.

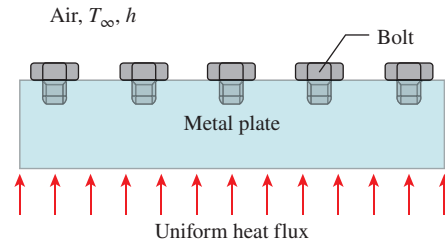


FIGURE P1-64

1-65 In a power plant, pipes transporting superheated vapor are very common. Superheated vapor is flowing at a rate of 0.3 kg/s inside a pipe 5 cm in diameter and 10 m in length. The pipe is located in a power plant at 20°C and has a uniform surface temperature of 100°C. If the temperature drop between the inlet and exit of the pipe is 30°C, and the specific heat of the vapor is 2190 J/kg·K, determine the heat transfer coefficient as a result of convection between the pipe surface and the surroundings.

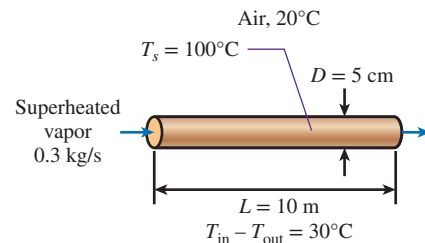


FIGURE P1-65

1-66  A boiler supplies hot water to a commercial dishwasher through a pipe with a surface temperature of 50°C. The hot water exits the boiler at 95°C, and it is transported in a pipe that has an outside diameter of 20 mm. The distance between the boiler and the dishwasher is 20 m. The section of the pipe between the boiler and the dishwasher is exposed to convection with a heat transfer coefficient of 100 W/m²·K at an ambient temperature of 20°C. The hot water flows steadily in the pipe at 60 g/s, and its average specific heat is 4.20 kJ/kg·K. The National Sanitation Foundation standard for commercial warewashing equipment (ANSI/NSF 3)

requires the final rinse water temperature to be at least 82°C . Under these conditions, does the hot water entering the dishwasher meet the ANSI/NSF 3 standard? If not, discuss some possible ways to increase the water temperature entering the dishwasher.

1-67 **C&S** Hot liquid is flowing in a steel pipe with a 22-mm inner diameter and a 27-mm outer diameter. The pipe outer surface is subjected to a uniform flux of 1200 W/m^2 for a length of 1 m. The pipe inner surface is coated with a thin polyvinylidene fluoride (PVDF) lining. The hot liquid flowing inside the pipe has a mean temperature of 120°C and a convection heat transfer coefficient of $50\text{ W/m}^2\cdot\text{K}$. The ASME Code for Process Piping (ASME B31.3-2014, A323) recommends a maximum temperature for PVDF lining to be 135°C . Does the surface temperature of the PVDF lining comply with the recommendation of the code?

1-68 An electric current of 5 A passing through a resistor has a measured voltage of 6 V across the resistor. The resistor is cylindrical with a diameter of 2.5 cm and length of 15 cm. The resistor has a uniform temperature of 90°C , and the room air temperature is 20°C . Assuming that heat transfer by radiation is negligible, determine the heat transfer coefficient by convection.

1-69 **C&S** An electric current of 1 A passing through a cable has a measured voltage of 30 V across the cable. The cable is covered with polyethylene insulation, and convection occurs at the outer surface of the insulation. The ambient temperature is 20°C with a convection heat transfer coefficient of $5\text{ W/m}^2\cdot\text{K}$, and the surface area subjected to the convection is 0.1 m^2 . The ASTM D1351 standard specifies that thermoplastic polyethylene insulation is suitable for use on electrical cable with operation at temperatures up to 75°C . Under these conditions, will the polyethylene insulation for the cable meet the ASTM D1351 standard? If the polyethylene insulation does not meet the ASTM D1351 standard, then discuss possible solutions to meet the standard.

1-70 **PtD** An AISI 316 stainless steel spherical container is used for storing chemicals undergoing exothermic reaction that provides a uniform heat flux of 60 kW/m^2 to the container's inner surface. The container has an inner diameter of 1 m and a wall thickness of 5 cm. To prevent thermal burns on individuals working around the container, it is necessary to keep the container's outer surface temperature below 50°C . If the ambient temperature is 23°C , determine the necessary convection heat transfer coefficient to keep the container's outer surface temperature below 50°C . Is the necessary convection heat transfer coefficient feasible with free convection of air? If not, discuss other options to prevent the container's outer surface temperature from causing thermal burns.

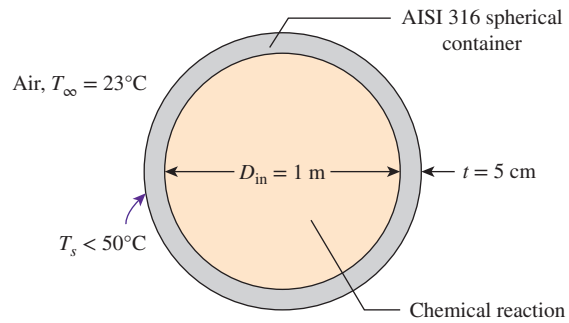


FIGURE P1-70

1-71 A transistor with a height of 0.4 cm and a diameter of 0.6 cm is mounted on a circuit board. The transistor is cooled by air flowing over it with an average heat transfer coefficient of $30\text{ W/m}^2\cdot\text{K}$. If the air temperature is 55°C and the transistor case temperature is not to exceed 70°C , determine the amount of power this transistor can dissipate safely. Disregard any heat transfer from the transistor base.

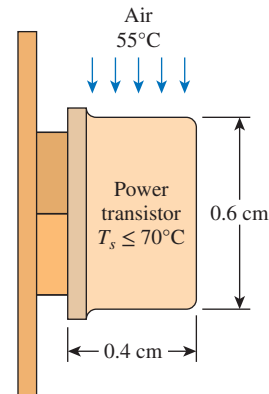


FIGURE P1-71

1-72E A 200-ft-long section of a steam pipe whose outer diameter is 4 in passes through an open space at 50°F . The average temperature of the outer surface of the pipe is measured to be 280°F , and the average heat transfer coefficient on that surface is determined to be $6\text{ Btu/h}\cdot\text{ft}^2\cdot^{\circ}\text{F}$. Determine (a) the rate of heat loss from the steam pipe and (b) the annual cost of this energy loss if steam is generated in a natural gas furnace having an efficiency of 86 percent and the price of natural gas is $\$1.10/\text{therm}$ (1 therm = 100,000 Btu). **Answers:** (a) 289,027 Btu/h, (b) \\$32,384/yr

1-73 The boiling temperature of nitrogen at atmospheric pressure at sea level (1 atm) is -196°C . Therefore, nitrogen is commonly used in low-temperature scientific studies

since the temperature of liquid nitrogen in a tank open to the atmosphere remains constant at -196°C until the liquid nitrogen in the tank is depleted. Any heat transfer to the tank results in the evaporation of some liquid nitrogen, which has a heat of vaporization of 198 kJ/kg and a density of 810 kg/m^3 at 1 atm .

Consider a 4-m-diameter spherical tank initially filled with liquid nitrogen at 1 atm and -196°C . The tank is exposed to 20°C ambient air with a heat transfer coefficient of $25\text{ W/m}^2\cdot\text{K}$. The temperature of the thin-shelled spherical tank is observed to be almost the same as the temperature of the nitrogen inside. Disregarding any radiation heat exchange, determine the rate of evaporation of the liquid nitrogen in the tank as a result of the heat transfer from the ambient air.

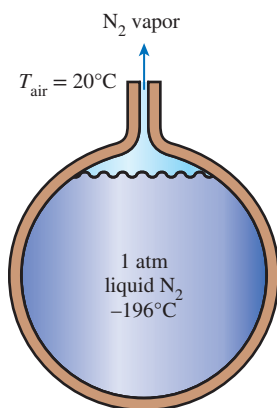


FIGURE P1-73

1-74 A series of experiments were conducted by passing 40°C air over a long 25-mm-diameter cylinder with an embedded electrical heater. The objective of these experiments was to determine the power per unit length required ($\dot{W}/L$) to maintain the surface temperature of the cylinder at 300°C for different air velocities (V). The results of these experiments are given in the following table:

$V\text{ (m/s)}$	1	2	4	8	12
$\dot{W}/L\text{ (W/m)}$	450	658	983	1507	1963

(a) Assuming a uniform temperature over the cylinder, negligible radiation between the cylinder surface and surroundings, and steady-state conditions, determine the convection heat transfer coefficient (h) for each velocity (V). Plot the results in terms of $h\text{ (W/m}^2\cdot\text{K)}$ vs. $V\text{ (m/s)}$. Provide a computer-generated graph for the display of your results, and tabulate the data used for the graph.

(b) Assume that the heat transfer coefficient and velocity can be expressed in the form $h = CV^n$. Determine the values of the constants C and n from the results of part (a) by plotting h vs. V on log-log coordinates and choosing a C value that assures a match at $V = 1\text{ m/s}$ and then varying n to get the best fit.

1-75 A 1.4-m-long, 0.2-cm-diameter electrical wire extends across a room that is maintained at 20°C . Heat is generated in the wire as a result of resistance heating, and the surface temperature of the wire is measured to be 240°C in steady operation. Also, the voltage drop and electric current through the wire are measured to be 110 V and 3 A , respectively. Disregarding any heat transfer by radiation, determine the convection heat transfer coefficient for heat transfer between the outer surface of the wire and the air in the room. *Answer: $170.5\text{ W/m}^2\cdot\text{K}$*

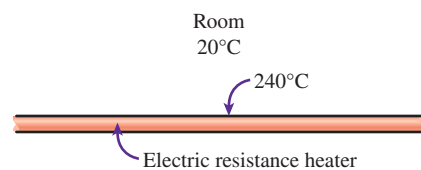


FIGURE P1-75

1-76  Reconsider Prob. 1-75. Using appropriate software, plot the convection heat transfer coefficient as a function of the wire surface temperature in the range of 100°C to 300°C . Discuss the results.

1-77E Using the conversion factors between W and Btu/h , m and ft , and K and R , express the Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8}\text{ W/m}^2\cdot\text{K}^4$ in the English unit $\text{Btu/h}\cdot\text{ft}^2\cdot\text{R}^4$.

1-78 The outer surface of a spacecraft in space has an emissivity of 0.8 and a solar absorptivity of 0.3 . If solar radiation is incident on the spacecraft at a rate of 950 W/m^2 , determine the surface temperature of the spacecraft when the radiation emitted equals the solar energy absorbed.

1-79 Consider a person whose exposed surface area is 1.7 m^2 , emissivity is 0.5 , and surface temperature is 32°C . Determine the rate of heat loss from that person by radiation in a large room having walls at a temperature of (a) 300 K and (b) 280 K . *Answers: (a) 26.7 W , (b) 121 W*

1-80 Consider a sealed 20-cm-high electronic box whose base dimensions are $40\text{ cm} \times 40\text{ cm}$ placed in a vacuum chamber. The emissivity of the outer surface of the box is 0.95 . If the electronic components in the box dissipate a total of 100 W of power and the outer surface temperature of the box is not to exceed 55°C , determine the temperature at which the surrounding surfaces must be kept if this box is to be cooled by radiation alone. Assume the heat transfer from the bottom surface of the box to the stand to be negligible.

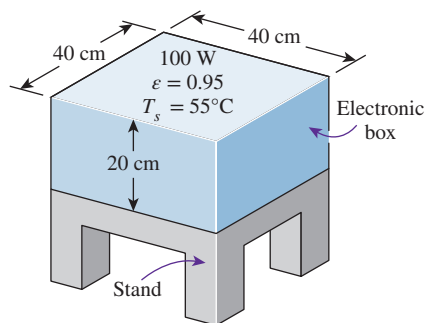


FIGURE P1-80

1-81 Two surfaces, one highly polished and the other heavily oxidized, are found to be emitting the same amount of energy per unit area. The highly polished surface has an emissivity of 0.1 at 1070°C , while the emissivity of the heavily oxidized surface is 0.78. Determine the temperature of the heavily oxidized surface.

1-82 A spherical interplanetary probe with a diameter of 2 m is sent out into the solar system. The probe surface is made of material having an emissivity of 0.9 and an absorptivity of 0.1. Signals from the sensors monitoring the probe surface temperatures are indicating an average value of -40°C for a space temperature of 0 K. If the electronics inside the probe are generating heat at a rate of 100 W/m^3 , determine the incident radiation rate on the probe surface.

1-83 An electronic package in the shape of a sphere with an outer diameter of 100 mm is placed in a large laboratory room. The surface emissivity of the package can assume three different values (0.2, 0.25, and 0.3). The walls of the room are maintained at a constant temperature of 77 K. The electronics in this package can only operate in the surface temperature range of $40^\circ\text{C} \leq T_s \leq 85^\circ\text{C}$. Determine the range of power dissipation ($\dot{W}$) for the electronic package over this temperature range for the three surface emissivity values (ϵ). Plot the results in terms of $\dot{W}(\text{W})$ vs. $T_s(^\circ\text{C})$ for the three different values of emissivity over a surface temperature range of 40 to 85°C with temperature increments of 5°C (total of 10 data points for each ϵ value). Provide a computer-generated graph for the display of your results, and tabulate the data used for the graph. Comment on the results obtained.

Simultaneous Heat Transfer Mechanisms

1-84C Can all three modes of heat transfer occur simultaneously (in parallel) in a medium?

1-85C Can a medium involve (a) conduction and convection, (b) conduction and radiation, or (c) convection and radiation simultaneously? Give examples for the “yes” answers.

1-86C The deep human body temperature of a healthy person remains constant at 37°C while the temperature and the humidity of the environment change with time. Discuss the heat transfer mechanisms between the human body and the environment

in both summer and winter, and explain how a person can keep cooler in summer and warmer in winter.

1-87C We often turn the fan on in summer to help us cool. Explain how a fan makes us feel cooler in the summer. Also explain why some people use ceiling fans also in winter.

1-88 Consider a 20-cm-thick granite wall with a thermal conductivity of $2.79\text{ W/m}\cdot\text{K}$. The temperature of the left surface is held constant at 50°C , whereas the right face is exposed to a flow of 22°C air with a convection heat transfer coefficient of $15\text{ W/m}^2\cdot\text{K}$. Neglecting heat transfer by radiation, find the right wall surface temperature and the heat flux through the wall.

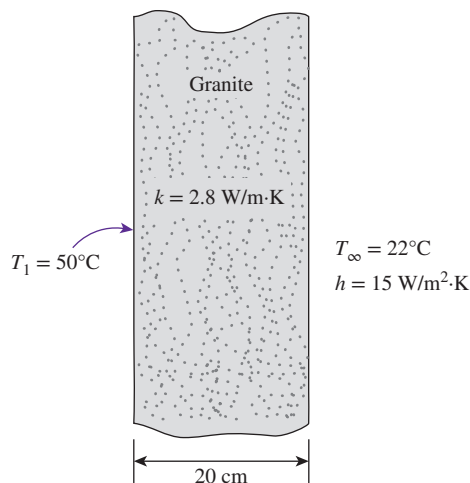


FIGURE P1-88

1-89 A solid plate, with a thickness of 15 cm and a thermal conductivity of $80\text{ W/m}\cdot\text{K}$, is being cooled at the upper surface by air. The air temperature is 10°C , while the temperatures at the upper and lower surfaces of the plate are 50 and 60°C , respectively. Determine the convection heat transfer coefficient of air at the upper surface, and discuss whether the value is reasonable or not for forced convection of air.

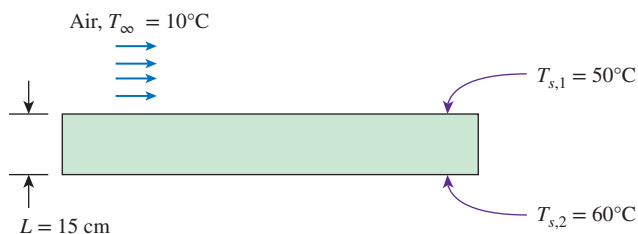


FIGURE P1-89

1-90 Air at 20°C with a convection heat transfer coefficient of $25\text{ W/m}^2\cdot\text{K}$ blows over a horizontal steel hot plate ($k = 43\text{ W/m}\cdot\text{K}$). The surface area of the plate is 0.38 m^2 with a thickness of 2 cm. The plate surface is maintained at a constant temperature

of $T_s = 250^\circ\text{C}$, and the plate loses 300 W from its surface by radiation. Calculate the inside plate temperature (T_i).

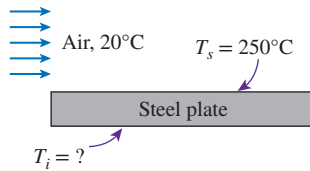


FIGURE P1-90

1-91 An electronic package with a surface area of 1 m^2 placed in an orbiting space station is exposed to space. The electronics in this package dissipate all 1 kW of its power to the space through its exposed surface. The exposed surface has an emissivity of 1.0 and an absorptivity of 0.25. Determine the steady-state exposed surface temperature of the electronic package (a) if the surface is exposed to a solar flux of 750 W/m^2 , and (b) if the surface is not exposed to the sun.

1-92 Consider steady heat transfer between two large parallel plates at constant temperatures of $T_1 = 290\text{ K}$ and $T_2 = 150\text{ K}$ that are $L = 2\text{ cm}$ apart. Assuming the surfaces to be black (emissivity $\varepsilon = 1$), determine the rate of heat transfer between the plates per unit surface area assuming the gap between the plates is (a) filled with atmospheric air, (b) evacuated, (c) filled with fiberglass insulation, and (d) filled with superinsulation having an apparent thermal conductivity of $0.00015\text{ W/m}\cdot\text{K}$.

1-93 Heat treatment is common in processing of semiconductor material. A 200-mm-diameter silicon wafer with thickness of $725\text{ }\mu\text{m}$ is being heat treated in a vacuum chamber by infrared heat. The surrounding walls of the chamber have a uniform temperature of 310 K . The infrared heater provides an incident radiation flux of 200 kW/m^2 on the upper surface of the wafer, and the emissivity and absorptivity of the wafer surface are 0.70. Using a pyrometer, the lower surface temperature of the wafer is measured to be 1000 K . Assuming there is no radiation exchange between the lower surface of the wafer and the surroundings, determine the upper surface temperature of the wafer. (Note: A *pyrometer* is a noncontacting device that intercepts and measures thermal radiation. This device can be used to determine the temperature of an object's surface.)

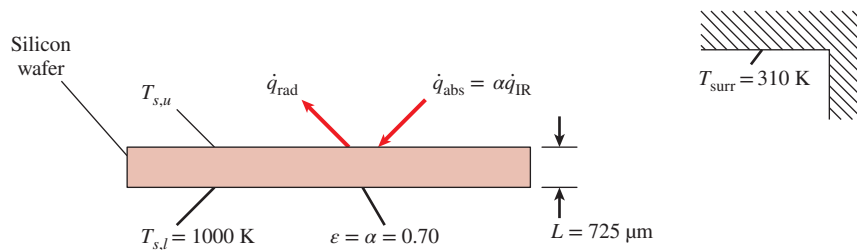


FIGURE P1-93

1-94 Consider a person standing in a room at 23°C . Determine the total rate of heat transfer from this person if the exposed surface area and the skin temperature of the person are 1.7 m^2 and 32°C , respectively, and the convection heat transfer coefficient is $5\text{ W/m}^2\cdot\text{K}$. Take the emissivity of the skin and the clothes to be 0.9, and assume the temperature of the inner surfaces of the room to be the same as the air temperature.

1-95E A 2-in-diameter spherical ball whose surface is maintained at a temperature of 170°F is suspended in the middle of a room at 70°F . If the convection heat transfer coefficient is $15\text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$ and the emissivity of the surface is 0.8, determine the total rate of heat transfer from the ball.

1-96 A 1000-W iron is left on an ironing board with its base exposed to the air at 20°C . The convection heat transfer coefficient between the base surface and the surrounding air is $35\text{ W/m}^2\cdot\text{K}$. If the base has an emissivity of 0.6 and a surface area of 0.02 m^2 , determine the temperature of the base of the iron. *Answer: 674°C*

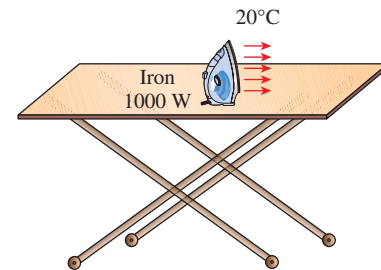


FIGURE P1-96

1-97 C&S The upper surface of a plate is bolted with ASTM B21 naval brass bolts. The bottom surface of the plate is fully insulated. The upper plate surface is exposed to convection at an air temperature of 50°C and a convection heat transfer coefficient of $5\text{ W/m}^2\cdot\text{K}$. Also, thermal radiation exchange occurs between the upper plate surface and the surrounding surface temperature of 300°C . The emissivity of the plate surface and the bolts is 0.3. The ASME Code for Process Piping (ASME B31.3-2014) limits the maximum use

temperature of B21 bolts to 149°C . Determine whether the ASTM B21 bolts on the plate comply with the ASME code.

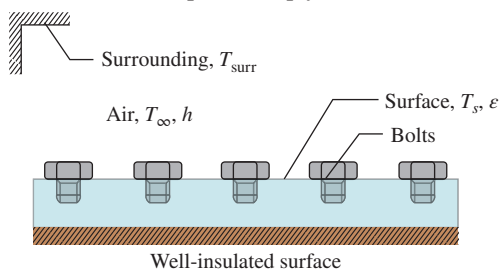


FIGURE P1-97

1-98 A 3-m-internal-diameter spherical tank made of 1-cm-thick stainless steel is used to store iced water at 0°C . The tank is located outdoors at 25°C . Assuming the entire steel tank to be at 0°C and thus the thermal resistance of the tank to be negligible, determine (a) the rate of heat transfer to the iced water in the tank and (b) the amount of ice at 0°C that melts during a 24-h period. The heat of fusion of water at atmospheric pressure is $h_{if} = 333.7 \text{ kJ/kg}$. The emissivity of the outer surface of the tank is 0.75, and the convection heat transfer coefficient on the outer surface can be taken to be $30 \text{ W/m}^2\cdot\text{K}$. Assume the average surrounding surface temperature for radiation exchange to be 15°C . **Answers: (a) 23.1 kW, (b) 5980 kg**

1-99 In the metal processing industry, heat treatment of metals is commonly done using electrically heated draw batch furnaces. Consider a furnace that is situated in a room with surrounding air temperature of 30°C and an average convection heat transfer coefficient of $12 \text{ W/m}^2\cdot\text{K}$. The furnace front is made of a steel plate with thickness of 20 mm and a thermal conductivity of $25 \text{ W/m}\cdot\text{K}$. The outer furnace front surface has an emissivity of 0.23, and the inside surface is subjected to a heat flux of 8 kW/m^2 . Determine the outside surface temperature of the furnace front.

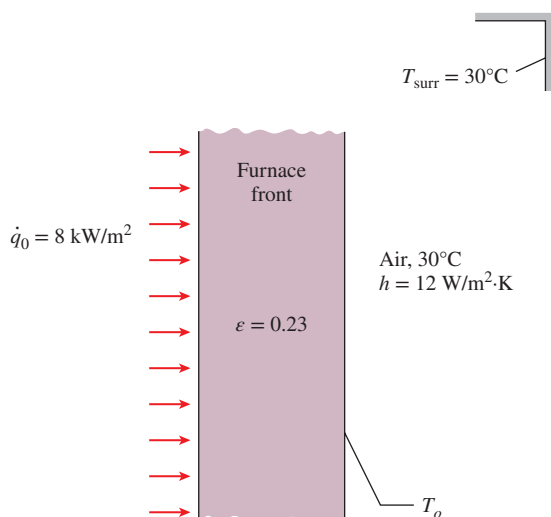


FIGURE P1-99

1-100 Solar radiation is incident on a 5-m^2 solar absorber plate surface at a rate of 800 W/m^2 . Ninety-three percent of the solar radiation is absorbed by the absorber plate, while the remaining 7 percent is reflected away. The solar absorber plate has a surface temperature of 40°C with an emissivity of 0.9 that experiences radiation exchange with the surrounding temperature of -5°C . In addition, convective heat transfer occurs between the absorber plate surface and the ambient air of 20°C with a convection heat transfer coefficient of $7 \text{ W/m}^2\cdot\text{K}$. Determine the efficiency of the solar absorber, which is defined as the ratio of the usable heat collected by the absorber to the incident solar radiation on the absorber.

1-101 A flat-plate solar collector is used to heat water by having water flow through tubes attached at the back of the thin solar absorber plate. The absorber plate has a surface area of 2 m^2 with emissivity and absorptivity of 0.9. The surface temperature of the absorber is 35°C , and solar radiation is incident on the absorber at 500 W/m^2 with a surrounding temperature of 0°C . The convection heat transfer coefficient at the absorber surface is $5 \text{ W/m}^2\cdot\text{K}$, while the ambient temperature is 25°C . Net heat absorbed by the solar collector heats the water from an inlet temperature (T_{in}) to an outlet temperature (T_{out}). If the water flow rate is 5 g/s with a specific heat of $4.2 \text{ kJ/kg}\cdot\text{K}$, determine the temperature rise of the water.

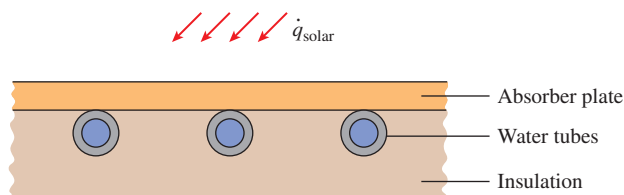


FIGURE P1-101

1-102 **PtD** Heat treatment of metals is commonly done using electrically heated draw batch furnaces. Consider a furnace that is situated in a room with a surrounding air temperature of 30°C and an average convection heat transfer coefficient of $15 \text{ W/m}^2\cdot\text{K}$. The outer furnace front surface has an emissivity of 0.7, and the inside surface is subjected to a heat flux of 5 kW/m^2 . To ensure safety and avoid thermal burns to people working around the furnace, the outer front surface of the furnace should be kept below 50°C . Based on the information given about the furnace, does the furnace front surface require insulation to prevent thermal burns?

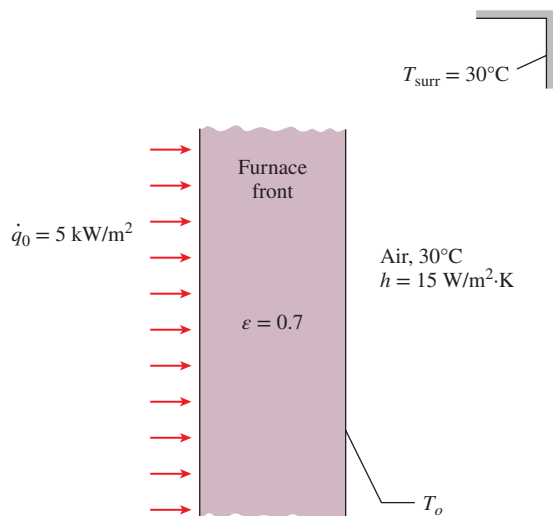


FIGURE P1-102

1-103E Consider a flat-plate solar collector placed horizontally on the flat roof of a house. The collector is 5 ft wide and 15 ft long, and the average temperature of the exposed surface of the collector is 100°F. The emissivity of the exposed surface of the collector is 0.9. Determine the rate of heat loss from the collector by convection and radiation during a calm day when the ambient air temperature is 70°F and the effective sky temperature for radiation exchange is 50°F. Take the convection heat transfer coefficient on the exposed surface to be 2.5 Btu/h·ft²·°F.

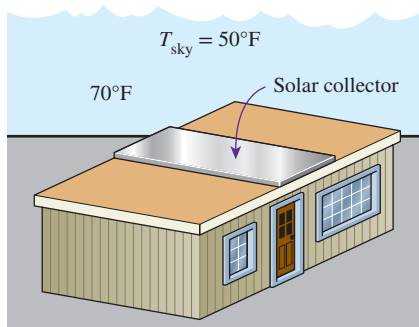


FIGURE P1-103E

1-104 An AISI 304 stainless steel sheet is going through an annealing process inside an electrically heated oven. The ambient air inside the oven is 600°C, while the surrounding surfaces of the oven are at a uniform temperature of 750°C. If the emissivity of the stainless steel sheet is 0.40 and the average convection heat transfer coefficient is 10 W/m²·K, determine the temperature of the stainless steel sheet.

1-105 The roof of a house consists of a 15-cm-thick concrete slab ($k = 2$ W/m·K) that is 15 m wide and 20 m long.

The emissivity of the outer surface of the roof is 0.9, and the convection heat transfer coefficient on that surface is estimated to be 15 W/m²·K. The inner surface of the roof is maintained at 15°C. On a clear winter night, the ambient air is reported to be at 10°C, while the night sky temperature for radiation heat transfer is 255 K. Considering both radiation and convection heat transfer, determine the outer surface temperature and the rate of heat transfer through the roof.

If the house is heated by a furnace burning natural gas with an efficiency of 85 percent, and the unit cost of natural gas is \$0.60/therm (1 therm = 105,500 kJ of energy content), determine the money lost through the roof that night during a 14-h period.

1-106  A nonmetal plate ($k = 0.05$ W/m·K) and an ASME SA-240 stainless steel plate ($k = 15$ W/m·K) are bolted together by ASTM B21 naval brass bolts. Both plates have a thickness of 38 mm, and the brass bolts are bolted through both plates. The bottom steel plate surface (surface 1) is subjected to a uniform heat flux of 200 W/m². The top nonmetal plate surface (surface 2) is exposed to convection at an air temperature of 20°C and a convection heat transfer coefficient of 5 W/m²·K. Also, thermal radiation exchange occurs between the upper plate surface and the surrounding surface temperature of 30°C. The emissivity of the plate surface and the bolts is 0.3. The ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HF-300) limits the maximum use temperature to 260°C for the SA-240 steel plate. The ASME Code for Process Piping (ASME B31.3-2014) limits the maximum use temperature of B21 bolts to 149°C. Determine whether both ASME codes are being complied with under these conditions.

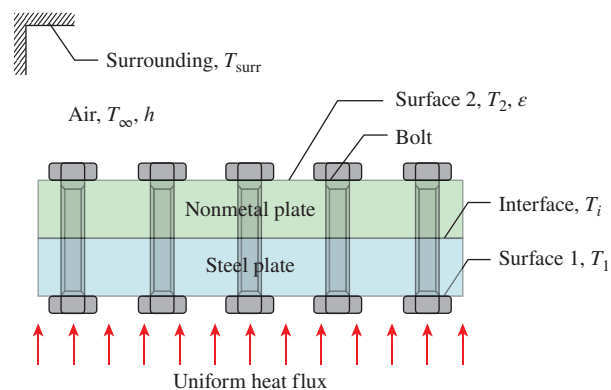


FIGURE P1-106

1-107 The inner and outer surfaces of a 25-cm-thick wall in summer are at 27°C and 44°C, respectively. The outer surface of the wall exchanges heat by radiation with surrounding surfaces at 40°C and by convection with ambient air also at 40°C with a convection heat transfer coefficient of 8 W/m²·K. Solar radiation is

incident on the surface at a rate of 150 W/m^2 . If both the emissivity and the solar absorptivity of the outer surface are 0.8, determine the effective thermal conductivity of the wall.

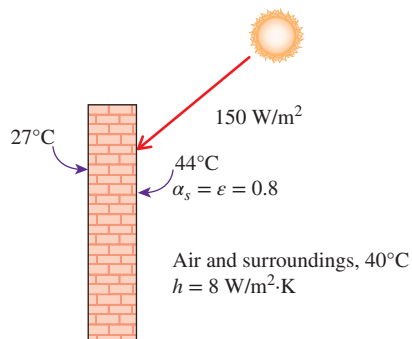


FIGURE P1-107

Solving Engineering Problems

1-108C What is the value of the engineering software packages in (a) engineering education and (b) engineering practice?

1-109  Determine a positive real root of this equation using appropriate software:

$$3.5x^3 - 10x^{0.5} - 3x = -4$$

1-110  Solve this system of two equations with two unknowns using appropriate software:

$$x^3 - y^2 = 10.5$$

$$3xy + y = 4.6$$

1-111  Solve this system of three equations with three unknowns using appropriate software:

$$2x - y + z = 5$$

$$3x^2 + 2y = z + 2$$

$$xy + 2z = 8$$

1-112  Solve this system of three equations with three unknowns using appropriate software:

$$x^2y - z = 1.5$$

$$x - 3y^{0.5} + xz = -2$$

$$x + y - z = 4.2$$

Special Topic: Thermal Comfort

1-113C What is metabolism? What is the range of metabolic rate for an average man? Why are we interested in the metabolic rate of the occupants of a building when we deal with heating and air conditioning?

1-114C Why is the metabolic rate of women, in general, lower than that of men? What is the effect of clothing on the environmental temperature that feels comfortable?

1-115C What is asymmetric thermal radiation? How does it cause thermal discomfort in the occupants of a room?

1-116C How do (a) draft and (b) cold floor surfaces cause discomfort for a room's occupants?

1-117C What is stratification? Is it more likely to occur at places with low ceilings or places with high ceilings? How does it cause thermal discomfort for a room's occupants? How can stratification be prevented?

1-118C Why is it necessary to ventilate buildings? What is the effect of ventilation on energy consumption for heating in winter and for cooling in summer? Is it a good idea to keep the bathroom fans on all the time? Explain.

1-119 Consider a house in Atlanta, Georgia, that is maintained at 22°C and has a total of 20 m^2 of window area. The windows are double-door type with wood frames and metal spacers and have a U -factor of $2.5 \text{ W/m}^2\cdot\text{K}$ (see Prob. 1-120 for the definition of U -factor). The winter average temperature of Atlanta is 11.3°C . Determine the average rate of heat loss through the windows in winter.

1-120 The rate of heat loss through a unit surface area of a window per unit temperature difference between the indoors and the outdoors is called the U -factor. The value of the U -factor ranges from about $1.25 \text{ W/m}^2\cdot\text{K}$ (or $0.22 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$) for low- e coated, argon-filled, quadruple-pane windows to $6.25 \text{ W/m}^2\cdot\text{K}$ (or $1.1 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$) for a single-pane window with aluminum frames. Determine the range for the rate of heat loss through a $1.2\text{-m} \times 1.8\text{-m}$ window of a house that is maintained at 20°C when the outdoor air temperature is -8°C .

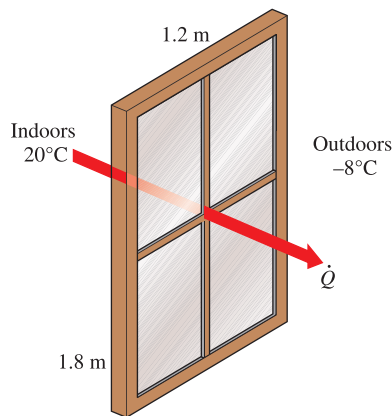


FIGURE P1-120

Review Problems

1-121 A $4\text{-m} \times 5\text{-m} \times 6\text{-m}$ room is to be heated by one ton (1000 kg) of liquid water contained in a tank placed in the room. The room is losing heat to the outside at an average rate of $10,000 \text{ kJ/h}$. The room is initially at 20°C and 100 kPa and

is maintained at an average temperature of 20°C at all times. If the hot water is to meet the heating requirements of this room for a 24-h period, determine the minimum temperature of the water when it is first brought into the room. Assume constant specific heats for both air and water at room temperature. **Answer:** 77.4°C

1-122 Engine valves ($c_p = 440 \text{ J/kg}\cdot\text{K}$ and $\rho = 7840 \text{ kg/m}^3$) are to be heated from 40°C to 800°C in 5 min in the heat treatment section of a valve manufacturing facility. The valves have a cylindrical stem with a diameter of 8 mm and a length of 10 cm. The valve head and the stem may be assumed to be of equal surface area, with a total mass of 0.0788 kg. For a single valve, determine (a) the amount of heat transfer, (b) the average rate of heat transfer, (c) the average heat flux, and (d) the number of valves that can be heat treated per day if the heating section can hold 25 valves and it is used 10 h per day.

1-123 A cylindrical resistor element on a circuit board dissipates 0.8 W of power. The resistor is 2 cm long and has a diameter of 0.4 cm. Assuming heat to be transferred uniformly from all surfaces, determine (a) the amount of heat this resistor dissipates during a 24-h period, (b) the heat flux, and (c) the fraction of heat dissipated from the top and bottom surfaces.

1-124 The heat generated in the circuitry on the surface of a silicon chip ($k = 130 \text{ W/m}\cdot\text{K}$) is conducted to the ceramic substrate to which it is attached. The chip is $6 \text{ mm} \times 6 \text{ mm}$ in size and 0.5 mm thick and dissipates 3 W of power. Disregarding any heat transfer through the 0.5-mm-high side surfaces, determine the temperature difference between the front and back surfaces of the chip in steady operation.

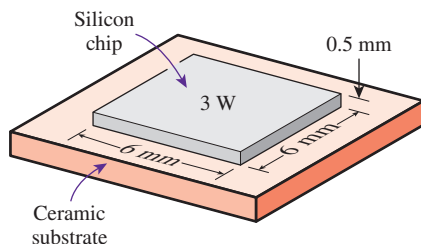


FIGURE P1-124

1-125 A 0.3-cm-thick, 12-cm-high, and 18-cm-long circuit board houses 80 closely spaced logic chips on one side, each dissipating 0.06 W. The board is impregnated with copper fillings and has an effective thermal conductivity of $16 \text{ W/m}\cdot\text{K}$. All the heat generated in the chips is conducted across the circuit board and is dissipated from the back side of the board to the ambient air. Determine the temperature difference between the two sides of the circuit board. **Answer:** 0.042°C

1-126 A 40-cm-long, 800-W electric resistance heating element with diameter 0.5 cm and surface temperature 120°C is immersed in 75 kg of water initially at 20°C . Determine how

long it will take for this heater to raise the water temperature to 80°C . Also, determine the convection heat transfer coefficients at the beginning and at the end of the heating process.

1-127 It is well known that wind makes the cold air feel much colder as a result of the *wind-chill effect* that is due to the increase in the convection heat transfer coefficient with increasing air velocity. The wind-chill effect is usually expressed in terms of the *wind-chill temperature* (WCT), which is the apparent temperature felt by exposed skin. For an outdoor air temperature of 0°C , for example, the wind-chill temperature is -5°C with 20 km/h winds and -9°C with 60 km/h winds. That is, a person exposed to 0°C windy air at 20 km/h will feel as cold as a person exposed to -5°C calm air (air motion under 5 km/h).

For heat transfer purposes, a standing man can be modeled as a 30-cm-diameter, 170-cm-long vertical cylinder with both the top and bottom surfaces insulated and with the side surface at an average temperature of 34°C . For a convection heat transfer coefficient of $15 \text{ W/m}^2\cdot\text{K}$, determine the rate of heat loss from this man by convection in still air at 20°C . What would your answer be if the convection heat transfer coefficient is increased to $30 \text{ W/m}^2\cdot\text{K}$ as a result of winds? What is the wind-chill temperature in this case? **Answers:** 336 W, 672 W, 6°C

1-128 An engine block with a surface area measured to be 0.95 m^2 generates a power output of 50 kW with a net engine efficiency of 35 percent. The engine block operates inside a compartment at 157°C , and the average convection heat transfer coefficient is $50 \text{ W/m}^2\cdot\text{K}$. If convection is the only heat transfer mechanism occurring, determine the engine block surface temperature. **Answer:** 841°C

1-129 Consider an electrical wire submerged in liquid water at atmospheric conditions. The wire has a diameter of 1 mm and a length of 15 cm. The current through the wire is increased until the water reaches a temperature of 100°C . For this situation (boiling water) use an average value of the upper and lower values of the convection heat transfer coefficients given in Table 1-5. The wire surface temperature is to be maintained at 115°C . How much electric power is required for this case?

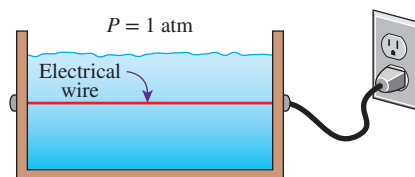


FIGURE P1-129

1-130 A cylindrical fuel rod 2 cm in diameter is encased in a concentric tube and cooled by water. The fuel generates heat uniformly at a rate of 150 MW/m^3 . The convection heat transfer coefficient on the fuel rod is $5000 \text{ W/m}^2\cdot\text{K}$, and the average

temperature of the cooling water, sufficiently far from the fuel rod, is 70°C . Determine the surface temperature of the fuel rod, and discuss whether the value of the given convection heat transfer coefficient on the fuel rod is reasonable.

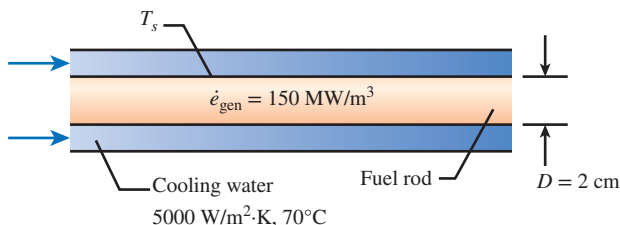


FIGURE P1-130

1-131 **PtD** The critical heat flux (CHF) is a thermal limit at which a boiling crisis occurs whereby an abrupt rise in temperature causes overheating on a fuel rod surface that leads to damage. A cylindrical fuel rod 2 cm in diameter is encased in a concentric tube and cooled by water. The fuel generates heat uniformly at a rate of 150 MW/m^3 . The average temperature of the cooling water, sufficiently far from the fuel rod, is 80°C . The operating pressure of the cooling water is such that the surface temperature of the fuel rod must be kept below 300°C to prevent the cooling water from reaching the critical heat flux. Determine the necessary convection heat transfer coefficient to prevent the critical heat flux from occurring.

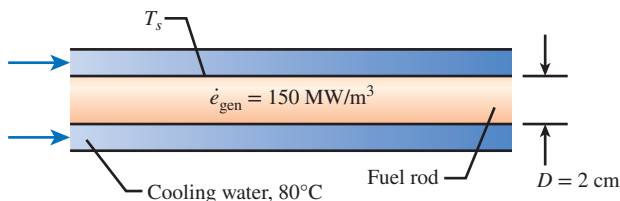


FIGURE P1-131

1-132 Consider a person standing in a room maintained at 20°C at all times. The inner surfaces of the walls, floors, and ceiling of the house are observed to be at an average temperature of 12°C in winter and 23°C in summer. Determine the rates of radiation heat transfer between this person and the surrounding surfaces in both summer and winter if the exposed surface area, emissivity, and the average outer surface temperature of the person are 1.6 m^2 , 0.95 , and 32°C , respectively.

1-133 Consider a $3\text{-m} \times 3\text{-m} \times 3\text{-m}$ cubical furnace whose top and side surfaces closely approximate black surfaces at a temperature of 1200 K . The base surface has an emissivity of

$\varepsilon = 0.7$, and is maintained at 800 K . Determine the net rate of radiation heat transfer to the base surface from the top and side surfaces. **Answer:** 594 kW

1-134 A soldering iron has a cylindrical tip of 2.5 mm in diameter and 20 mm in length. With age and usage, the tip has oxidized and has an emissivity of 0.80 . Assuming that the average convection heat transfer coefficient over the soldering iron tip is $25 \text{ W/m}^2\cdot\text{K}$ and the surrounding air temperature is 20°C , determine the power required to maintain the tip at 400°C .

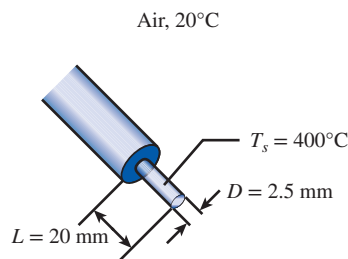


FIGURE P1-134

1-135 A thin metal plate is insulated on the back and exposed to solar radiation on the front surface. The exposed surface of the plate has an absorptivity of 0.7 for solar radiation. If solar radiation is incident on the plate at a rate of 550 W/m^2 and the surrounding air temperature is 10°C , determine the surface temperature of the plate when the heat loss by convection equals the solar energy absorbed by the plate. Take the convection heat transfer coefficient to be $25 \text{ W/m}^2\cdot\text{K}$, and disregard any heat loss by radiation.

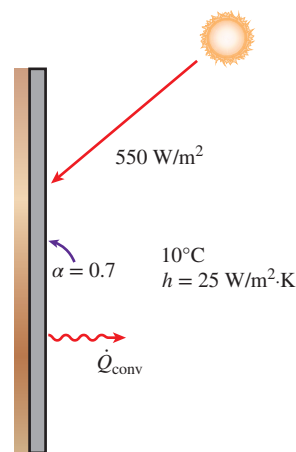


FIGURE P1-135

1-136 Consider a flat-plate solar collector placed on the roof of a house. The temperatures at the inner and outer surfaces of the glass cover are measured to be 28°C and 25°C ,

respectively. The glass cover has a surface area of 2.5 m^2 , a thickness of 0.6 cm , and a thermal conductivity of $0.7 \text{ W/m}\cdot\text{K}$. Heat is lost from the outer surface of the cover by convection and radiation with a convection heat transfer coefficient of $10 \text{ W/m}^2\cdot\text{K}$ and an ambient temperature of 15°C . Determine the fraction of heat lost from the glass cover by radiation.

1-137 An electric heater with the total surface area of 0.25 m^2 and emissivity 0.75 is in a room where the air has a temperature of 20°C and the walls are at 10°C . When the heater consumes 500 W of electric power, its surface has a steady temperature of 120°C . Determine the temperature of the heater surface when it consumes 700 W . Solve the problem (a) assuming negligible radiation and (b) taking radiation into consideration. Based on your results, comment on the assumption made in part (a).

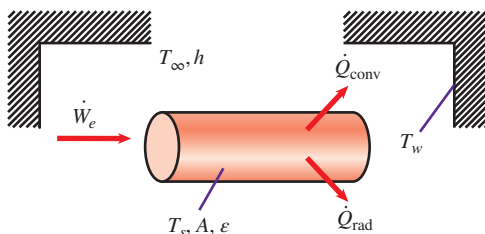


FIGURE P1-137

1-138 An ice skating rink is located in a building where the air is at $T_{\text{air}} = 20^\circ\text{C}$ and the walls are at $T_w = 25^\circ\text{C}$. The convection heat transfer coefficient between the ice and the surrounding air is $h = 10 \text{ W/m}^2\cdot\text{K}$. The emissivity of ice is $\varepsilon = 0.95$. The latent heat of fusion of ice is $h_{if} = 333.7 \text{ kJ/kg}$, and its density is 920 kg/m^3 . (a) Calculate the refrigeration load of the system necessary to maintain the ice at $T_s = 0^\circ\text{C}$ for an ice rink of 12 m by 40 m . (b) How long would it take to melt $\delta = 3 \text{ mm}$ of ice from the surface of the rink if no cooling is supplied and the surface is considered insulated on the back side?

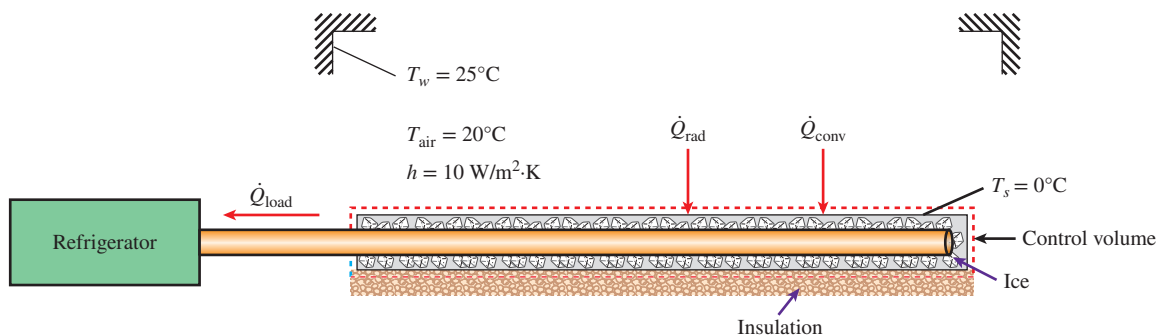


FIGURE P1-138

Fundamentals of Engineering (FE) Exam Problems

1-139 Which expression is used to determine the heat flux for conduction?

- (a) $-kA \frac{dT}{dx}$ (b) $-k \text{ grad } T$ (c) $h(T_2 - T_1)$
(d) $\varepsilon\sigma T^4$ (e) None of them

1-140 Which expression is used to determine the heat flux for convection?

- (a) $-kA \frac{dT}{dx}$ (b) $-k \text{ grad } T$ (c) $h(T_2 - T_1)$
(d) $\varepsilon\sigma T^4$ (e) None of them

1-141 Which expression is used to determine the heat flux emitted by thermal radiation from a surface?

- (a) $-kA \frac{dT}{dx}$ (b) $-k \text{ grad } T$ (c) $h(T_2 - T_1)$
(d) $\varepsilon\sigma T^4$ (e) None of them

1-142 A 1-kW electric resistance heater in a room is turned on and kept on for 50 min. The amount of energy transferred to the room by the heater is

- (a) 1 kJ (b) 50 kJ (c) 3000 kJ
(d) 3600 kJ (e) 6000 kJ

1-143 A 2-kW electric resistance heater submerged in 30-kg water is turned on and kept on for 10 min. During the process, 500 kJ of heat is lost from the water. The temperature rise of the water is

- (a) 5.6°C (b) 9.6°C (c) 13.6°C
(d) 23.3°C (e) 42.5°C

1-144 Eggs with a mass of 0.15 kg per egg and a specific heat of $3.32 \text{ kJ/kg}\cdot^\circ\text{C}$ are cooled from 32°C to 10°C at a rate of 300 eggs per minute. The rate of heat removal from the eggs is

- (a) 11 kW (b) 80 kW (c) 25 kW
(d) 657 kW (e) 55 kW

1-145 A cold bottled drink ($m = 2.5 \text{ kg}$, $c_p = 4200 \text{ J/kg}\cdot\text{K}$) at 5°C is left on a table in a room. The average temperature of

the drink is observed to rise to 15°C in 30 min. The average rate of heat transfer to the drink is

- (a) 23 W (b) 29 W (c) 58 W
(d) 88 W (e) 122 W

1-146 Water enters a pipe at 20°C at a rate of 0.25 kg/s and is heated to 60°C . The rate of heat transfer to the water is

- (a) 10 kW (b) 20.9 kW (c) 41.8 kW
(d) 62.7 kW (e) 167.2 kW

1-147 Air enters a 12-m-long, 7-cm-diameter pipe at 50°C at a rate of 0.06 kg/s. The air is cooled at an average rate of 400 W per square meter surface area of the pipe. The air temperature at the exit of the pipe is

- (a) 4.3°C (b) 17.5°C (c) 32.5°C
(d) 43.4°C (e) 45.8°C

1-148 Heat is lost steadily through a 0.5-cm thick, 2-m $\times$ 3-m window glass whose thermal conductivity is 0.7 W/m $\cdot$ K. The inner and outer surface temperatures of the glass are measured to be 12°C to 9°C . The rate of heat loss by conduction through the glass is

- (a) 420 W (b) 5040 W (c) 17,600 W
(d) 1256 W (e) 2520 W

1-149 Steady heat conduction occurs through a 0.3-m-thick, 9-m $\times$ 3-m composite wall at a rate of 1.2 kW. If the inner and outer surface temperatures of the wall are 15°C and 7°C , the effective thermal conductivity of the wall is

- (a) 0.61 W/m $\cdot$ K (b) 0.83 W/m $\cdot$ K (c) 1.7 W/m $\cdot$ K
(d) 2.2 W/m $\cdot$ K (e) 5.1 W/m $\cdot$ K

1-150 Heat is lost through a brick wall ($k = 0.72$ W/m $\cdot$ K), which is 4 m long, 3 m wide, and 25 cm thick at a rate of 500 W. If the inner surface of the wall is at 22°C , the temperature at the midplane of the wall is

- (a) 0°C (b) 7.5°C (c) 11.0°C
(d) 14.8°C (e) 22°C

1-151 A 10-cm-high and 20-cm-wide circuit board houses on its surface 100 closely spaced chips, each generating heat at a rate of 0.08 W and transferring it by convection and radiation to the surrounding medium at 40°C . Heat transfer from the back surface of the board is negligible. If the combined convection and radiation heat transfer coefficient on the surface of the board is 22 W/m 2 $\cdot$ K, the average surface temperature of the chips is

- (a) 72.4°C (b) 66.5°C (c) 40.4°C
(d) 58.2°C (e) 49.1°C

1-152 A 40-cm-long, 0.4-cm-diameter electric resistance wire submerged in water is used to determine the convection heat transfer coefficient in water during boiling at 1 atm pressure. The surface temperature of the wire is measured to be 114°C when a wattmeter indicates the electric power consumption to be 7.6 kW. The heat transfer coefficient is

- (a) 108 kW/m 2 $\cdot$ K (b) 13.3 kW/m 2 $\cdot$ K
(c) 68.1 kW/m 2 $\cdot$ K (d) 0.76 kW/m 2 $\cdot$ K
(e) 256 kW/m 2 $\cdot$ K

1-153 While driving down a highway early in the evening, the airflow over an automobile establishes an overall heat transfer coefficient of 25 W/m 2 $\cdot$ K. The passenger cabin of this automobile exposes 8 m 2 of surface to the moving ambient air. On a day when the ambient temperature is 33°C , how much cooling must the air-conditioning system supply to maintain a temperature of 20°C in the passenger cabin?

- (a) 0.65 MW (b) 1.4 MW (c) 2.6 MW
(d) 3.5 MW (e) 0.94 MW

1-154 A room is heated by a 1.2-kW electric resistance heater whose wires have a diameter of 4 mm and a total length of 3.4 m. The air in the room is at 23°C , and the interior surfaces of the room are at 17°C . The convection heat transfer coefficient on the surface of the wires is 8 W/m 2 $\cdot$ K. If the rates of heat transfer from the wires to the room by convection and by radiation are equal, the surface temperature of the wire is

- (a) 3534°C (b) 1778°C (c) 1772°C
(d) 98°C (e) 25°C

1-155 Over 90 percent of the energy dissipated by an incandescent lightbulb is in the form of heat, not light. What is the temperature of a vacuum-enclosed tungsten filament with an exposed surface area of 2.03 cm 2 in a 100-W incandescent lightbulb? The emissivity of tungsten at the anticipated high temperatures is about 0.35. Note that the lightbulb consumes 100 W of electrical energy and dissipates all of it by radiation.

- (a) 1870 K (b) 2230 K (c) 2640 K
(d) 3120 K (e) 2980 K

1-156 On a still, clear night, the sky appears to be a blackbody with an equivalent temperature of 250 K. What is the air temperature when a strawberry field cools to 0°C and freezes if the heat transfer coefficient between the plants and air is 6 W/m 2 $\cdot$ K because of a light breeze and the plants have an emissivity of 0.9?

- (a) 14°C (b) 7°C (c) 3°C
(d) 0°C (e) -3°C

1-157 A 30-cm-diameter black ball at 120°C is suspended in air. It is losing heat to the surrounding air at 25°C by convection with a heat transfer coefficient of 12 W/m 2 $\cdot$ K and by radiation to the surrounding surfaces at 15°C . The total rate of heat transfer from the black ball is

- (a) 322 W (b) 595 W (c) 234 W
(d) 472 W (e) 2100 W

1-158 A 3-m 2 black surface at 140°C is losing heat to the surrounding air at 35°C by convection with a heat transfer coefficient of 16 W/m 2 $\cdot$ K and by radiation to the surrounding surfaces at 15°C . The total rate of heat loss from the surface is

- (a) 5105 W (b) 2940 W (c) 3779 W
(d) 8819 W (e) 5040 W

1-159 A person's head can be approximated as a 25-cm-diameter sphere at 35°C with an emissivity of 0.95. Heat is lost from the head to the surrounding air at 25°C by convection with a heat transfer coefficient of 11 W/m 2 $\cdot$ K and by radiation

to the surrounding surfaces at 10°C . Disregarding the neck, determine the total rate of heat loss from the head.

- (a) 22 W (b) 27 W (c) 49 W
(d) 172 W (e) 249 W

1–160 A person standing in a room loses heat to the air in the room by convection and to the surrounding surfaces by radiation. Both the air in the room and the surrounding surfaces are at 20°C . The exposed surface of the person is 1.5 m^2 and has an average temperature of 32°C and an emissivity of 0.90. If the rates of heat transfer from the person by convection and by radiation are equal, the combined heat transfer coefficient is

- (a) $0.008\text{ W/m}^2\cdot\text{K}$ (b) $3.0\text{ W/m}^2\cdot\text{K}$
(c) $5.5\text{ W/m}^2\cdot\text{K}$ (d) $8.3\text{ W/m}^2\cdot\text{K}$
(e) $10.9\text{ W/m}^2\cdot\text{K}$

Design and Essay Problems

1–161 Write an essay on how microwave ovens work, and explain how they cook much faster than conventional ovens. Discuss whether conventional electric or microwave ovens consume more electricity for the same task.

1–162 Using information from the utility bills for the coldest month last year, estimate the average rate of heat loss from your house for that month. In your analysis, consider the contribution of the internal heat sources such as people, lights, and appliances. Identify the primary sources of heat loss from your house, and propose ways of improving the energy efficiency of your house.

1–163 Conduct this experiment to determine the combined heat transfer coefficient between an incandescent lightbulb and the surrounding air and surfaces using a 60-W lightbulb. You will need a thermometer, which can be purchased in a hardware store, and metal glue. You will also need a piece of string and a ruler to calculate the surface area of the lightbulb. First,

measure the air temperature in the room, and then glue the tip of the thermocouple wire of the thermometer to the glass of the lightbulb. Turn the light on and wait until the temperature reading stabilizes. The temperature reading will give the surface temperature of the lightbulb. Assuming 10 percent of the rated power of the bulb is converted to light and is transmitted by the glass, calculate the heat transfer coefficient from Newton's law of cooling.

1–164 It is well known that at the same outdoor air temperature a person is cooled at a faster rate under windy conditions than under calm conditions due to the higher convection heat transfer coefficients associated with windy air. The phrase *wind-chill* is used to relate the rate of heat loss from people under windy conditions to an equivalent air temperature for calm conditions (considered to be a wind or walking speed of 3 mph or 5 km/h). The hypothetical *wind-chill temperature* (WCT), called the *wind-chill temperature index* (WCTI), is an air temperature equivalent to the air temperature needed to produce the same cooling effect under calm conditions. A 2003 report on wind-chill temperature by the U.S. National Weather Service gives the WCTI in metric units as

$$\text{WCTI } (^{\circ}\text{C}) = 13.12 + 0.6215T - 11.37V^{0.16} + 0.3965TV^{0.16}$$

where T is the air temperature in $^{\circ}\text{C}$ and V the wind speed in km/h at 10 m elevation. Show that this relation can be expressed in English units as

$$\text{WCTI } (^{\circ}\text{F}) = 35.74 + 0.6215T - 35.75V^{0.16} + 0.4275TV^{0.16}$$

where T is the air temperature in $^{\circ}\text{F}$ and V the wind speed in mph at 33 ft elevation. Also, prepare a table for WCTI for air temperatures ranging from 10 to -60°C and wind speeds ranging from 10 to 80 km/h. Comment on the magnitude of the cooling effect of the wind and the danger of frostbite.